

# Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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# Motivation

- **Economic Problem:** Selecting best options under costly evaluation
  - Determining each option's true quality costs decision-maker time and effort
  - Low-cost screening of all options  $\Rightarrow$  high-cost evaluation of surviving options
  - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
  - Graduate programs invest time and money into students' development
  - If programs were undervaluing applicants with certain characteristics, future such applicants may receive formerly-withheld opportunities

# Research Questions

- ❶ Which factors in applications affect admission and success in graduate school?
    - Literature measures success with program completion and job placement ranking
  - ❷ Are admissions committees optimizing, or are they making mistakes?
    - Goal is to accept applicants with highest success probabilities who will matriculate
    - Predictors of admission only (success only) may be overvalued (undervalued)
  - ❸ If they are making mistakes, how can they make better admissions decisions?
    - In turn, by changing decision rules, how much better of a cohort can they admit?
- ★ This paper's methods generalize to admissions, hiring, and other settings

# Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
  - Statistical tests of whether committee's decision rule aligns with desired objective
  - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
  - Do estimated success probabilities align with those implied by committee's decisions?
  - Tests reject optimality and counterfactual deviation gains statistically significant when committee misvalues applicant characteristics, but not when it optimizes

# Literature Review

- **Graduate Admissions:** Regress acceptance and success on applicant characteristics
  - **Original Literature:** Attiyeh & Attiyeh (1997); Krueger & Wu (2000); Grove & Wu (2007); Athey, Katz, Krueger, Levitt & Poterba (2007)
  - **Study of 2002 Cohort:** Stock, Finegan & Siegfried (2006–2015)
  - **New Predictors/Methods:** Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
  - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
  - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)
- ★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

# Synthetic Data Protocol (IRB-Approved)

## ① **Administrative office receives data from university departments**

- Algorithmically extracts variables from textual files, such as recommendation letters

## ② **Synthetic data (Patki et al. 2016) used for model development**

- Estimate joint distribution of variables, and simulate synthetic dataset from it
- Rows do not correspond to real subjects, preventing identity disclosure
- Cells with  $< 5$  comparable observations binned to prevent attribute disclosure

## ③ **Administrative office executes final estimation code on actual data**

- Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

# Outline

- 1 Reduced-Form Analysis
- 2 Structural Models
- 3 Structural Estimation
- 4 Optimality Tests

# Data Description

- **Sample:** Applicants to research university's economics, government, history, linguistics, and psychology PhD programs from 2015 to 2023
- **Dependent Variables:** Measures of admission and success
  - **Admission:** Indicators for all stages of admissions process, including waitlist
  - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Independent Variables:** Objective vs. subjective
  - **Objective:** Demographics and performance measures, such as GRE scores
  - **Subjective:** Textual processing of recommendation letters, application essays, CVs, and supplemental forms using dictionaries to construct variables



# Reduced-Form Summary

- **Data:** All applicants to all 5 graduate programs from 2020 to 2023
  - Currently have results for acceptance and objective variables only
  - Implementing synthetic data protocol to complete the empirical work
- **Method:** Logistic regressions with lasso for variable selection
- **Results:** Economics admissions cares most about quantitative GRE and undergraduate GPA, and is most data-driven based on goodness of fit
  - Need success data to see if other programs are making mistakes
  - Prolific recommenders increase acceptance probability  $\Rightarrow$  network effects

# Post-Lasso AMEs: Predicting Acceptance

| Dependent Variable = Accept     | Economics | Government | History   | Linguistics | Psychology |
|---------------------------------|-----------|------------|-----------|-------------|------------|
| GRE Verbal Percentile           | 0.0012*** | 0.0013**   | 0.0032*** |             | 0.0022**   |
| GRE Quantitative Percentile     | 0.0119*** | 0.0006*    | 0.0019*** | 0.0009      |            |
| GRE Analytic Percentile         | 0.0014*** | 0.0009**   |           | 0.0009      |            |
| Undergraduate GPA               | 0.1710*** | 0.0267     | 0.0985**  | 0.0499      |            |
| TOEFL/IELTS                     | 0.0299    | 0.0254     | 0.0758*   | 0.0474      |            |
| Elite U.S. University           | 0.0635*** | 0.0144     | 0.0452*   |             | 0.0346     |
| Elite U.S. Liberal Arts College | 0.0806*** | 0.0247*    | 0.0771**  |             | 0.0604     |
| Elite Foreign University        | 0.0365*   |            | 0.0432    | 0.0535      |            |
| Graduate Degree                 |           |            | 0.0210    |             |            |
| Work Experience                 | 0.0293**  | 0.0157     | 0.0383    |             | 0.0722**   |
| #Professor Recommenders         |           |            | 0.0436**  | 0.0259*     |            |
| #Prolific Recommenders          | 0.0209*** | 0.0115**   | 0.0160    | 0.0184**    | 0.0195     |
| #Math Courses                   | 0.0131*** |            |           |             |            |
| Advanced Math                   | 0.0180    |            |           |             |            |
| Pseudo-R2                       | 0.2671    | 0.1500     | 0.1847    | 0.1722      | 0.1844     |

# Acceptance vs. Success AMEs

| Program = Economics             | Accept    | Success |
|---------------------------------|-----------|---------|
| GRE Verbal Percentile           | 0.0012*** |         |
| GRE Quantitative Percentile     | 0.0119*** |         |
| GRE Analytic Percentile         | 0.0014*** |         |
| Undergraduate GPA               | 0.1710*** |         |
| TOEFL/IELTS                     | 0.0299    |         |
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| Elite U.S. Liberal Arts College | 0.0806*** |         |
| Elite Foreign University        | 0.0365*   |         |
| Work Experience                 | 0.0293**  |         |
| #Prolific Recommenders          | 0.0209*** |         |
| #Math Courses                   | 0.0131*** |         |
| Advanced Math                   | 0.0180    |         |

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality
- Need model of admissions to relate acceptance effects to success effects

► Insufficiency of Comparing AMEs

# Outline

- 1 Reduced-Form Analysis
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# Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
  - Per literature, committee should accept applicants with highest success probabilities
  - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
  - Quantify gains from deviating from actual to counterfactual decision rule

# Model Framework

- **Portfolio Choice Problem:** Accept subset of applicants with highest sum of success probabilities subject to capacity constraint on number of acceptances
- For each applicant  $n \in \{1, \dots, N\}$ , make decision  $d_n \in \{\text{Accept}, \text{Reject}\}$  given:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_A,$$

where  $x$  = objective variables,  $z$  = subjective variables,  $y \in \{\text{Success}, \text{Failure}\}$ ,  
 $N$  = number of applicants, and  $N_A$  = maximum number of acceptances

# Model Framework

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- BEMS (2022) prove that portfolio choice problem reduces to cutoff rule problem
  - Accept applicant if and only if their success probability exceeds cutoff probability
  - Decisions across applicants now independent, so no need to consider all subsets

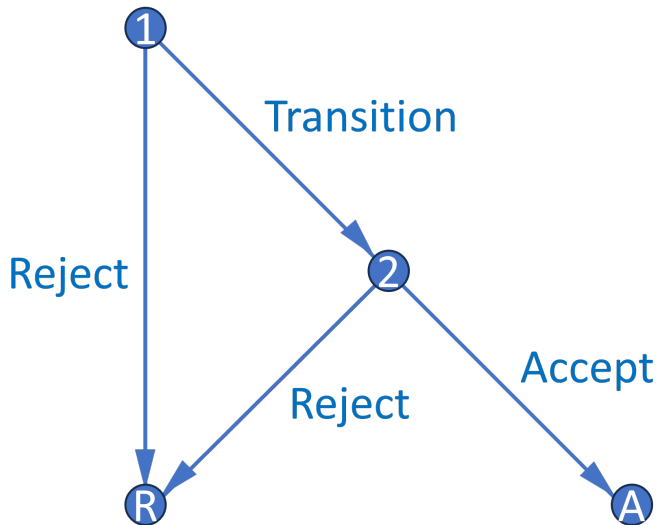
►► Portfolio Choice to Cutoff Rule

# Cutoff Rule Problem

- Start with two-round cutoff rule problem that accounts for costly evaluation:
  - **Round 1:** See only objective variables (e.g. GRE). Can transition application file to Round 2 to see subjective variables (e.g. recommendation letters), or can reject
  - **Round 2:** If chose to transition application file in Round 1, can accept or reject
  - Assume for now committee transitions and accepts optimal number of files
- Along with the baseline dynamic model, there are additional models
  - Static model is simpler and better represents what real-world committees do
  - Matriculation models account for matriculation probabilities in Round 1
  - Waitlist models add a third round and also account for matriculation probabilities



# Decision Graph



# Dynamic Model

- **States:**  $r$  = round,  $x$  = objective variables,  $z$  = subjective variables,  $t$  = year,  $\bar{k}$  = fixed program rank,  $\epsilon_r$  = T1EV preference shock with scale parameter  $\sigma$
- **Controls:**  $d_1 \in \{\text{Transition, Reject}\}$ ,  $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:**  $y \in \{\text{Success, Failure}\}$ ,  $y_r^*$  = marginal outcome

# Dynamic Model

- **States:**  $r = \text{round}$ ,  $x = \text{objective variables}$ ,  $z = \text{subjective variables}$ ,  $t = \text{year}$ ,  $\bar{k} = \text{fixed program rank}$ ,  $\epsilon_r = \text{T1EV preference shock with scale parameter } \sigma$
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- **Outcomes:**  $y \in \{\text{Success, Failure}\}$ ,  $y_r^* = \text{marginal outcome}$

## Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

# Dynamic Model

## Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left( e^{P(y=S|x, \tilde{z}, t, \bar{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma} \right) dF(\tilde{z}|x, t)$$

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## Conditional Choice Probabilities:

$$P(d_r|x, (z), t, \bar{k}) = \frac{\exp(u_r(x, (z), t, \bar{k}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(x, (z), t, \bar{k}, \tilde{d}_r)/\sigma)}$$

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**Round 2:**

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

# Static Model

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- $P(y = S|x, t, \bar{k})$  estimates inconsistent, but fitted values account for  $z$ 's effect on  $y$

# Portfolio Choice with Matriculation

- Accept subset of applicants with highest sum of success times matriculation probabilities subject to capacity constraint on expected number of matriculants
- For each applicant  $n \in \{1, \dots, N\}$ , make decision  $d_n \in \{\text{Accept}, \text{Reject}\}$  given:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where  $(x, z)$  = variables,  $y \in \{\text{Success}, \text{Failure}\}$ ,  $m \in \{\text{Matriculate}, \text{Decline}\}$ ,  
 $N$  = number of applicants, and  $N_M$  = maximum number of matriculants

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such that:  $\sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M$

## Theorem 1 (Proof)

*For large  $N$ , this problem has a cutoff rule solution on **success** probabilities.*

- Likely matriculants more valuable, but also more expensive  $\Rightarrow$  effects cancel
- However, matriculation probabilities matter in Round 1 of a two-round model

# Matriculation-Dynamic Model

## Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

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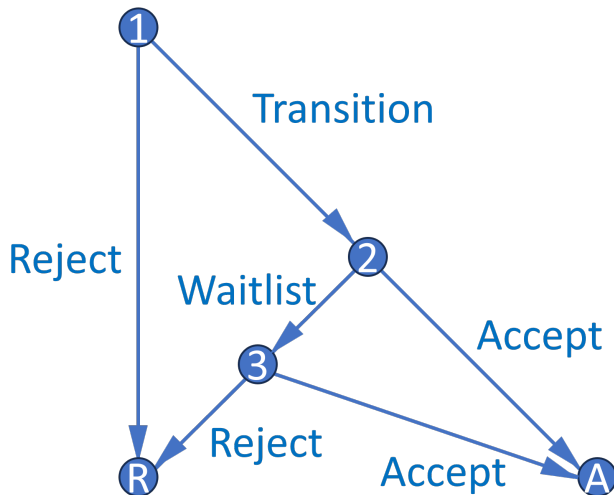
## Round 1:

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$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left( e^{[P(y=S|x, \tilde{z}, t, \bar{k}) - P(y_2^*=S|t)]/\sigma} + 1 \right) P(m=M|x, \tilde{z}, t) dF(\tilde{z} | x, t)$$

# Waitlist Decision Graph



- Three-round model features waitlist and accounts for applicants' decisions
- Dynamic mechanism incentivizes giving early acceptances to likely matriculants

# Waitlist Model, Round 3

- Respecify Round 3 cutoff  $P(y_3^* = S|t)$ , where  $t = \text{year}$ , as  $P(y_3^* = S|\overline{m_{2A}})$ 
  - $\overline{m_{2A}}$  = average early accepted applicant's matriculation probability
    - Enthusiasm variables can affect matriculation independently of success
  - $P(y_3^* = S|\overline{m_{2A}})$  increases in  $\overline{m_{2A}}$  because larger  $\overline{m_{2A}}$  means fewer openings

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$$v_3(x, z, t, \bar{k}, \overline{m}_{2A}, \epsilon_3) = \max_{d_3 \in \{A, R\}} u_3(x, z, t, \bar{k}, \overline{m}_{2A}, d_3) + \epsilon_3(d_3), \text{ such that:}$$

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# Waitlist Model, Round 2

- What information available at start of Round 2 determines  $\overline{m_{2A}}$ ?
  - $\ddot{m}_2$  is defined for all transitioned applicants as  $\overline{m}_2$  with  $m_2$  excluded, where  $m_2$  = applicant's own matriculation probability. Within a year, high  $m_2$  implies low  $\ddot{m}_2$
  - $\ddot{m}_2$  appears when applicant is waitlisted  $\Rightarrow$  represents  $\overline{m}_2$  of applicants committee *can* accept (vs.  $\overline{m_{2A}}$ , which represents  $\overline{m}_2$  of applicants committee *does* accept)

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$$v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2) = \max_{d_2 \in \{A, W\}} u_2(x, z, t, \bar{k}, \ddot{m}_2, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, \ddot{m}_2, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{\beta \mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3) | x, z, t, \bar{k}, \ddot{m}_2] \mathbb{1}(d_2 = W)}_{\text{If waitlist, get discounted } \mathbb{E}[\text{Social Surplus}]}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3) | x, z, t, \bar{k}, \ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log \left( \sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \widetilde{\overline{m_{2A}}}, \tilde{d}_3) / \sigma} \right) dF_{\overline{m_{2A}} | \ddot{m}_2}(\widetilde{\overline{m_{2A}}} | \ddot{m}_2)$$

# Waitlist Dynamics

## Theorem 2 (Proof)

*Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

- Compared to low  $m_2$  applicant, high  $m_2$  applicant has low  $\ddot{m}_2$ , and therefore low  $\bar{F}_{\overline{m_2A}|\ddot{m}_2}$ ,  $\mathbb{E}[P(\text{Cutoff})_3]$ , and  $\mathbb{E}[v_3]$ . This increases relative payoff of accepting them
- Why not accept low  $m_2$  applicant and fill spot with waitlisted high  $m_2$  applicant?
  - ①  $m_2$  declines between Rounds 2 and 3 and has farther to fall for high  $m_2$  applicant
  - ② In general, matriculation probabilities can change unpredictably between rounds

★ **Upshot:** Accepting applicants who want to matriculate benefits the program

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- 4 Optimality Tests

# Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \underbrace{\left[ P_{y_{1,n}}(\theta_{S_1}) P_{y_{2,n}}(\theta_{S_2}) \right]^{\mathbb{1}(y_n=S)}}_{\text{Success probabilities}} \\ \underbrace{\left[ (1 - P_{y_{1,n}}(\theta_{S_1})) (1 - P_{y_{2,n}}(\theta_{S_2})) \right]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T)) P_{y_{1,n}}(\theta_{S_1})^{\mathbb{1}(y_n=S)} (1 - P_{y_{1,n}}(\theta_{S_1}))^{\mathbb{1}(y_n=F)}}_{\text{Conditional choice probabilities}}$$

- In unrestricted likelihood  $\mathcal{L}^U$ , parameterize CCPs with flexible logistic specifications
- In restricted likelihood  $\mathcal{L}^R$ , use model CCPs, which depend on structural parameters
- Do success probabilities implied by outcomes align with those implied by CCPs?

# Cutoff Probability Identification

## Theorem 3 (Proof)

*The maximum likelihood cutoff probability estimate is such that the number of acceptances  $N_A$  equals the expected  $N_A$ . Also, for any cutoff probability and  $N_A$ , the likelihood is highest when the committee accepts the best  $N_A$  applicants.*

- Identify each year's cutoffs by how selective committee is that year
- While selectivity identifies cutoffs, sorting improves restricted likelihood
  - If committee engages in sorting, likelihood-ratio test less likely to reject
  - $LR = 2[\log(\hat{\mathcal{L}}^U) - \log(\hat{\mathcal{L}}^R)] \sim \chi^2_Q$ , such that  $Q = \dim(\theta^U) - \dim(\theta^R)$

# Unrestricted Static AMEs (Optimal)

|                             | Transition            | Accept                 | Success 1              | Success 2              |
|-----------------------------|-----------------------|------------------------|------------------------|------------------------|
| GRE Quantitative Percentile | 0.0240***<br>(0.0018) | 0.0247***<br>(0.0033)  | 0.0155***<br>(0.0015)  | 0.0180***<br>(0.0029)  |
| Gender                      | 0.0440<br>(0.0283)    | 0.0233<br>(0.0451)     | 0.0200<br>(0.0249)     | 0.0197<br>(0.0419)     |
| Demographic 1               | -0.0583<br>(0.0386)   | -0.0476<br>(0.0672)    | -0.0363<br>(0.0338)    | -0.0713<br>(0.0608)    |
| Demographic 2               | 0.0302<br>(0.0431)    | 0.0309<br>(0.0693)     | -0.0169<br>(0.0371)    | -0.0361<br>(0.0672)    |
| Advanced Math               |                       | 0.1396***<br>(0.0443)  |                        | 0.0978**<br>(0.0417)   |
| Committee Score             |                       | 0.0141<br>(0.0108)     |                        | 0.0072<br>(0.0102)     |
| Year Transition More        | 0.1915***<br>(0.0263) | -0.1670***<br>(0.0427) | -0.0085<br>(0.0242)    | -0.0037<br>(0.0430)    |
| Program Rank                |                       |                        | -0.0045***<br>(0.0002) | -0.0044***<br>(0.0003) |

- Data simulated from reduced-form-calibrated data-generating parameters
- Committee's decision rule aligns with maximizing success probabilities

# Unrestricted Static AMEs (Suboptimal)

|                             | Transition             | Accept                 | Success 1              | Success 2              |
|-----------------------------|------------------------|------------------------|------------------------|------------------------|
| GRE Quantitative Percentile | 0.0238***<br>(0.0018)  | 0.0109***<br>(0.0036)  | 0.0158***<br>(0.0015)  | 0.0181***<br>(0.0027)  |
| Gender                      | 0.0500*<br>(0.0283)    | 0.0019<br>(0.0468)     | 0.0195<br>(0.0250)     | 0.0091<br>(0.0416)     |
| Demographic 1               | -0.1610***<br>(0.0371) | -0.0660<br>(0.0669)    | -0.0399<br>(0.0337)    | -0.0804<br>(0.0563)    |
| Demographic 2               | 0.0155<br>(0.0414)     | 0.0348<br>(0.0659)     | -0.0260<br>(0.0369)    | -0.0740<br>(0.0598)    |
| Advanced Math               |                        | 0.0708<br>(0.0465)     |                        | 0.0837**<br>(0.0416)   |
| Committee Score             |                        | 0.0585***<br>(0.0106)  |                        | 0.0096<br>(0.0100)     |
| Year Transition More        | 0.2157***<br>(0.0257)  | -0.1419***<br>(0.0455) | -0.0110<br>(0.0243)    | 0.0205<br>(0.0434)     |
| Program Rank                |                        |                        | -0.0044***<br>(0.0002) | -0.0044***<br>(0.0003) |

- Data simulated from reduced-form-calibrated data-generating parameters
- *Demographic 1* and *Advanced Math* undervalued, *Committee Score* overvalued



# Restricted Static Estimates (Optimal)

|                                   | $P(\text{Cutoff})_1$ | $P(\text{Cutoff})_2$ | Success 1  | Success 2  | Sigma     |
|-----------------------------------|----------------------|----------------------|------------|------------|-----------|
| Year Transition Fewer   Intercept | 66.56%               | 53.82%               | -8.0150*** | -9.7263*** |           |
|                                   | (58.17%, 74.02%)     | (42.08%, 65.15%)     | (1.1629)   | (1.7366)   |           |
| GRE Quantitative Percentile       |                      |                      | 0.1057***  | 0.1158***  |           |
|                                   |                      |                      | (0.0134)   | (0.0193)   |           |
| Gender                            |                      |                      | 0.1987*    | 0.1142     |           |
|                                   |                      |                      | (0.1015)   | (0.1673)   |           |
| Demographic 1                     |                      |                      | -0.2471*   | -0.3338    |           |
|                                   |                      |                      | (0.1346)   | (0.2403)   |           |
| Demographic 2                     |                      |                      | 0.0406     | -0.0047    |           |
|                                   |                      |                      | (0.1504)   | (0.2423)   |           |
| Advanced Math                     |                      |                      |            | 0.6315***  |           |
|                                   |                      |                      |            | (0.1766)   |           |
| Committee Score                   |                      |                      |            | 0.0546     |           |
|                                   |                      |                      |            | (0.0421)   |           |
| Year Transition More              | 46.05%               | 72.10%               | -0.0628    | 0.0285     |           |
|                                   | (39.60%, 52.64%)     | (58.90%, 82.34%)     | (0.1667)   | (0.2701)   |           |
| Program Rank                      |                      |                      | -0.0310*** | -0.0283*** |           |
|                                   |                      |                      | (0.0019)   | (0.0027)   |           |
| Sigma                             |                      |                      |            |            | 0.1913*** |
|                                   |                      |                      |            |            | (0.0252)  |
| LR Statistic   DF   P-Value       | 1.3137               | 9                    | 0.9983     |            |           |

- Likelihood-ratio test does not reject model fitting the data
- In *Year Transition Fewer*, Round 1 cutoff probability  $>$  Round 2 cutoff probability

# Restricted Static Estimates (Suboptimal)

|                                   | $P(\text{Cutoff})_1$       | $P(\text{Cutoff})_2$       | Success 1              | Success 2              | Sigma                 |
|-----------------------------------|----------------------------|----------------------------|------------------------|------------------------|-----------------------|
| Year Transition Fewer   Intercept | 67.57%<br>(58.17%, 75.73%) | 54.20%<br>(41.58%, 66.29%) | -8.0582***<br>(1.2803) | -6.3578***<br>(1.7689) |                       |
| GRE Quantitative Percentile       |                            |                            | 0.1079***<br>(0.0148)  | 0.0727***<br>(0.0189)  |                       |
| Gender                            |                            |                            | 0.2185**<br>(0.1042)   | 0.0409<br>(0.1610)     |                       |
| Demographic 1                     |                            |                            | -0.5505***<br>(0.1395) | -0.3739<br>(0.2380)    |                       |
| Demographic 2                     |                            |                            | -0.0151<br>(0.1505)    | -0.0921<br>(0.2356)    |                       |
| Advanced Math                     |                            |                            |                        | 0.3897**<br>(0.1686)   |                       |
| Committee Score                   |                            |                            |                        | 0.1754***<br>(0.0378)  |                       |
| Year Transition More              | 43.47%<br>(37.12%, 50.05%) | 68.55%<br>(55.51%, 79.20%) | -0.0772<br>(0.1671)    | 0.0376<br>(0.2807)     |                       |
| Program Rank                      |                            |                            | -0.0305***<br>(0.0019) | -0.0281***<br>(0.0027) |                       |
| Sigma                             |                            |                            |                        |                        | 0.2001***<br>(0.0325) |
| LR Statistic   DF   P-Value       | 25.7951                    | 9                          | 0.0022                 |                        |                       |

- Likelihood-ratio test rejects model fitting the data at 1% level
- In *Year Transition Fewer*, Round 1 cutoff probability > Round 2 cutoff probability

# Unrestricted Matriculation-Static AMEs (Optimal)

|                                  | Transition            | Accept                 | Success 1              | Success 2              | Matriculate 1         | Matriculate 2          |
|----------------------------------|-----------------------|------------------------|------------------------|------------------------|-----------------------|------------------------|
| GRE Quantitative Percentile      | 0.0241***<br>(0.0018) | 0.0248***<br>(0.0033)  | 0.0155***<br>(0.0015)  | 0.0181***<br>(0.0029)  | 0.0124*<br>(0.0071)   | 0.0079<br>(0.0061)     |
| Gender                           | 0.0441<br>(0.0283)    | 0.0233<br>(0.0451)     | 0.0201<br>(0.0249)     | 0.0197<br>(0.0419)     | 0.0810<br>(0.0703)    | 0.0649<br>(0.0657)     |
| Demographic 1                    | -0.0587<br>(0.0386)   | -0.0476<br>(0.0672)    | -0.0363<br>(0.0338)    | -0.0717<br>(0.0608)    | 0.0553<br>(0.0907)    | 0.0477<br>(0.1001)     |
| Demographic 2                    | 0.0299<br>(0.0431)    | 0.0308<br>(0.0693)     | -0.0169<br>(0.0371)    | -0.0364<br>(0.0672)    | -0.0260<br>(0.1119)   | 0.0244<br>(0.1069)     |
| Advanced Math                    |                       | 0.1396***<br>(0.0443)  |                        | 0.0977**<br>(0.0417)   |                       | 0.1078<br>(0.0733)     |
| Committee Score                  |                       | 0.0141<br>(0.0108)     |                        | 0.0072<br>(0.0102)     |                       | -0.0985***<br>(0.0235) |
| Year Transition/Matriculate More | 0.1915***<br>(0.0263) | -0.1669***<br>(0.0427) | -0.0084<br>(0.0242)    | -0.0036<br>(0.0430)    | 0.4617***<br>(0.0280) | 0.4427***<br>(0.0331)  |
| Program Rank                     |                       |                        | -0.0045***<br>(0.0002) | -0.0044***<br>(0.0003) |                       |                        |

- Approximate transition payoff with  $P(y = S|x, t, \bar{k})P(m = M|x, t)$

# Unrestricted Matriculation-Static AMEs (Suboptimal)

|                                  | Transition             | Accept                 | Success 1              | Success 2              | Matriculate 1         | Matriculate 2          |
|----------------------------------|------------------------|------------------------|------------------------|------------------------|-----------------------|------------------------|
| GRE Quantitative Percentile      | 0.0238***<br>(0.0018)  | 0.0109***<br>(0.0036)  | 0.0158***<br>(0.0015)  | 0.0181***<br>(0.0027)  | 0.0066<br>(0.0059)    | 0.0017<br>(0.0051)     |
| Gender                           | 0.0500*<br>(0.0283)    | 0.0019<br>(0.0468)     | 0.0195<br>(0.0250)     | 0.0091<br>(0.0416)     | 0.0710<br>(0.0717)    | 0.0169<br>(0.0608)     |
| Demographic 1                    | -0.1610***<br>(0.0371) | -0.0660<br>(0.0669)    | -0.0399<br>(0.0337)    | -0.0804<br>(0.0563)    | 0.0402<br>(0.0870)    | 0.0822<br>(0.0823)     |
| Demographic 2                    | 0.0155<br>(0.0414)     | 0.0348<br>(0.0659)     | -0.0260<br>(0.0369)    | -0.0740<br>(0.0598)    | -0.1047<br>(0.0966)   | -0.0289<br>(0.0880)    |
| Advanced Math                    |                        | 0.0707<br>(0.0465)     |                        | 0.0837**<br>(0.0416)   |                       | 0.0763<br>(0.0674)     |
| Committee Score                  |                        | 0.0585***<br>(0.0106)  |                        | 0.0096<br>(0.0100)     |                       | -0.0791***<br>(0.0176) |
| Year Transition/Matriculate More | 0.2156***<br>(0.0257)  | -0.1420***<br>(0.0455) | -0.0110<br>(0.0243)    | 0.0205<br>(0.0434)     | 0.4572***<br>(0.0414) | 0.4526***<br>(0.0349)  |
| Program Rank                     |                        |                        | -0.0044***<br>(0.0002) | -0.0044***<br>(0.0003) |                       |                        |

- Approximate transition payoff with  $P(y = S|x, t, \bar{k})P(m = M|x, t)$

# Restricted Matriculation-Static Estimates (Optimal)

|                                               | $P(\text{Cutoff})_1$ | $P(\text{Cutoff})_2$ | Success 1  | Success 2  | Matriculate 1 | Matriculate 2 | Sigma     |
|-----------------------------------------------|----------------------|----------------------|------------|------------|---------------|---------------|-----------|
| Year Transition/Matriculate Fewer   Intercept | 26.50%               | 53.69%               | -8.0229*** | -8.3237*** | -8.2894***    | -4.6066       |           |
|                                               | (20.54%, 33.46%)     | (42.32%, 64.69%)     | (1.2042)   | (1.7684)   | (2.5032)      | (4.6480)      |           |
| GRE Quantitative Percentile                   |                      |                      | 0.1071***  | 0.1006***  | 0.0788***     | 0.0593        |           |
|                                               |                      |                      | (0.0140)   | (0.0194)   | (0.0269)      | (0.0459)      |           |
| Gender                                        |                      |                      | 0.1039     | 0.1026     | 0.3893*       | 0.4885        |           |
|                                               |                      |                      | (0.1522)   | (0.1470)   | (0.2210)      | (0.5064)      |           |
| Demographic 1                                 |                      |                      | -0.2437    | -0.2769    | -0.0899       | 0.3587        |           |
|                                               |                      |                      | (0.2094)   | (0.2145)   | (0.3273)      | (0.7459)      |           |
| Demographic 2                                 |                      |                      | -0.0480    | 0.0110     | 0.0296        | 0.1834        |           |
|                                               |                      |                      | (0.2258)   | (0.2131)   | (0.3847)      | (0.7962)      |           |
| Advanced Math                                 |                      |                      |            | 0.5492***  |               | 0.8106        |           |
|                                               |                      |                      |            | (0.1679)   |               | (0.5773)      |           |
| Committee Score                               |                      |                      |            | 0.0492     |               | -0.7402***    |           |
|                                               |                      |                      |            | (0.0368)   |               | (0.2064)      |           |
| Year Transition/Matriculate More              | 29.16%               | 68.54%               | -0.1369    | 0.0137     | 1.8740***     | 3.3280***     |           |
|                                               | (21.96%, 37.57%)     | (55.89%, 78.93%)     | (0.1686)   | (0.2730)   | (0.4069)      | (0.5674)      |           |
| Program Rank                                  |                      |                      | -0.0311*** | -0.0280*** |               |               |           |
|                                               |                      |                      | (0.0019)   | (0.0026)   |               |               |           |
| Sigma                                         |                      |                      |            |            |               |               | 0.1554*** |
|                                               |                      |                      |            |            |               |               | (0.0253)  |
| LR Statistic   DF   P-Value                   | 19.8266              | 9                    | 0.0190     |            |               |               |           |

- Likelihood-ratio test rejects model fitting the data at 5% level

# Restricted Static Estimates (Suboptimal)

|                                               | $P(\text{Cutoff})_1$ | $P(\text{Cutoff})_2$ | Success 1  | Success 2  | Matriculate 1 | Matriculate 2 | Sigma     |
|-----------------------------------------------|----------------------|----------------------|------------|------------|---------------|---------------|-----------|
| Year Transition/Matriculate Fewer   Intercept | 23.30%               | 53.07%               | -8.0768*** | -3.7273**  | -6.1199***    | -1.0400       |           |
|                                               | (16.72%, 31.48%)     | (41.42%, 64.40%)     | (1.2382)   | (1.6988)   | (2.2306)      | (3.6084)      |           |
| GRE Quantitative Percentile                   |                      |                      | 0.1084***  | 0.0455***  | 0.0569**      | 0.0129        |           |
|                                               |                      |                      | (0.0144)   | (0.0175)   | (0.0248)      | (0.0391)      |           |
| Gender                                        |                      |                      | 0.1111     | 0.0275     | 0.3411        | 0.1298        |           |
|                                               |                      |                      | (0.1581)   | (0.1164)   | (0.2134)      | (0.4717)      |           |
| Demographic 1                                 |                      |                      | -0.3744*   | -0.2437    | -0.5399*      | 0.6329        |           |
|                                               |                      |                      | (0.2131)   | (0.1781)   | (0.2762)      | (0.6281)      |           |
| Demographic 2                                 |                      |                      | -0.0814    | -0.0173    | -0.0874       | -0.2225       |           |
|                                               |                      |                      | (0.2176)   | (0.1697)   | (0.3016)      | (0.6864)      |           |
| Advanced Math                                 |                      |                      |            | 0.2559*    |               | 0.5874        |           |
|                                               |                      |                      |            | (0.1400)   |               | (0.5320)      |           |
| Committee Score                               |                      |                      |            | 0.1375***  |               | -0.6092***    |           |
|                                               |                      |                      |            | (0.0346)   |               | (0.1637)      |           |
| Year Transition/Matriculate More              | 22.74%               | 62.05%               | -0.1520    | 0.0048     | 1.4674***     | 3.4846***     |           |
|                                               | (16.19%, 30.97%)     | (50.06%, 72.73%)     | (0.1700)   | (0.2775)   | (0.4102)      | (0.6047)      |           |
| Program Rank                                  |                      |                      | -0.0305*** | -0.0281*** |               |               |           |
|                                               |                      |                      | (0.0019)   | (0.0026)   |               |               |           |
| Sigma                                         |                      |                      |            |            |               |               | 0.1279*** |
|                                               |                      |                      |            |            |               |               | (0.0330)  |
| LR Statistic   DF   P-Value                   | 52.5157              | 9                    | 0.0000     |            |               |               |           |

- Likelihood-ratio test rejects model fitting the data at 1% level

# Unrestricted Waitlist-Hybrid AMEs (Optimal)

|                                  | $\overline{m}_{2A}$   | Transition            | Accept 2              | Accept 3               | Success 1              | Success 3              | Matriculate            |
|----------------------------------|-----------------------|-----------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|
| GRE Quantitative Percentile      |                       | 0.0241***<br>(0.0018) | 0.0148***<br>(0.0034) | 0.0191***<br>(0.0034)  | 0.0168***<br>(0.0017)  | 0.0181***<br>(0.0029)  | 0.0096<br>(0.0075)     |
| Gender                           |                       | 0.0441<br>(0.0283)    | 0.0031<br>(0.0392)    | 0.0418<br>(0.0451)     | 0.0217<br>(0.0270)     | 0.0197<br>(0.0419)     | 0.0538<br>(0.1052)     |
| Demographic 1                    |                       | -0.0587<br>(0.0386)   | 0.0747<br>(0.0665)    | -0.1215**<br>(0.0619)  | -0.0394<br>(0.0366)    | -0.0717<br>(0.0608)    | -0.0304<br>(0.0880)    |
| Demographic 2                    |                       | 0.0299<br>(0.0431)    | 0.0693<br>(0.0699)    | -0.0072<br>(0.0623)    | -0.0183<br>(0.0401)    | -0.0364<br>(0.0672)    | -0.0459<br>(0.1393)    |
| Advanced Math                    |                       |                       | 0.0684*<br>(0.0390)   | 0.1186***<br>(0.0451)  |                        | 0.0976**<br>(0.0417)   | 0.1318<br>(0.1080)     |
| Committee Score                  |                       |                       | 0.0143<br>(0.0093)    | 0.0052<br>(0.0092)     |                        | 0.0072<br>(0.0102)     | -0.1055***<br>(0.0317) |
| Year Transition/Matriculate More |                       | 0.1916***<br>(0.0263) |                       |                        | -0.0091<br>(0.0262)    | -0.0036<br>(0.0430)    | 0.3922***<br>(0.0697)  |
| Program Rank                     |                       |                       |                       |                        | -0.0048***<br>(0.0002) | -0.0044***<br>(0.0003) |                        |
| $\ddot{m}_2$                     | 1.0892***<br>(0.0851) |                       | -0.2160**<br>(0.1049) |                        |                        |                        |                        |
| $\overline{m}_{2A}$              |                       |                       |                       | -0.2774***<br>(0.1033) |                        |                        |                        |

- Waitlist-hybrid model: Round 1 is static, Rounds 2 and 3 are dynamic
- AME signs as expected for  $\ddot{m}_2$  on  $\overline{m}_{2A}$ , and for  $\overline{m}_{2A}$  on *Accept 3*

# Unrestricted Waitlist-Hybrid AMEs (Suboptimal)

|                                  | $\overline{m}_{2A}$   | Transition             | Accept 2              | Accept 3               | Success 1              | Success 3              | Matriculate            |
|----------------------------------|-----------------------|------------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|
| GRE Quantitative Percentile      |                       | 0.0238***<br>(0.0018)  | 0.0061*<br>(0.0034)   | 0.0083**<br>(0.0033)   | 0.0170***<br>(0.0017)  | 0.0181***<br>(0.0027)  | 0.0051<br>(0.0063)     |
| Gender                           |                       | 0.0500*<br>(0.0283)    | -0.0438<br>(0.0431)   | 0.0476<br>(0.0394)     | 0.0210<br>(0.0270)     | 0.0091<br>(0.0416)     | -0.0513<br>(0.0788)    |
| Demographic 1                    |                       | -0.1610***<br>(0.0371) | 0.0407<br>(0.0594)    | -0.1341**<br>(0.0581)  | -0.0431<br>(0.0364)    | -0.0804<br>(0.0563)    | 0.0482<br>(0.1030)     |
| Demographic 2                    |                       | 0.0155<br>(0.0414)     | 0.0850<br>(0.0605)    | -0.0429<br>(0.0539)    | -0.0281<br>(0.0399)    | -0.0740<br>(0.0598)    | -0.1100<br>(0.1089)    |
| Advanced Math                    |                       |                        | 0.0374<br>(0.0417)    | 0.0565<br>(0.0417)     |                        | 0.0837**<br>(0.0416)   | 0.0576<br>(0.0922)     |
| Committee Score                  |                       |                        | 0.0592***<br>(0.0096) | 0.0076<br>(0.0098)     |                        | 0.0096<br>(0.0100)     | -0.0973***<br>(0.0217) |
| Year Transition/Matriculate More |                       | 0.2157***<br>(0.0257)  |                       |                        | -0.0119<br>(0.0262)    | 0.0206<br>(0.0434)     | 0.4256***<br>(0.0578)  |
| Program Rank                     |                       |                        |                       |                        | -0.0047***<br>(0.0002) | -0.0044***<br>(0.0003) |                        |
| $\ddot{m}_2$                     | 0.8883***<br>(0.0489) |                        | -0.0870<br>(0.0900)   |                        |                        |                        |                        |
| $\overline{m}_{2A}$              |                       |                        |                       | -0.3393***<br>(0.1141) |                        |                        |                        |

- Waitlist-hybrid model: Round 1 is static, Rounds 2 and 3 are dynamic
- AME signs as expected for  $\ddot{m}_2$  on  $\overline{m}_{2A}$ , and for  $\overline{m}_{2A}$  on *Accept 3*



# Restricted Waitlist-Hybrid Estimates (Optimal)

|                                               | $\bar{m}_{2A}$        | $P(\text{Cutoff})_1$       | $P(\text{Cutoff})_3$       | Success 1              | Success 3              | Matriculate           | Beta                  | Sigma                 |
|-----------------------------------------------|-----------------------|----------------------------|----------------------------|------------------------|------------------------|-----------------------|-----------------------|-----------------------|
| Year Transition/Matriculate Fewer   Intercept | -0.0252<br>(0.0518)   | 66.58%<br>(58.19%, 74.03%) | 69.29%<br>(54.94%, 80.68%) | -8.0491***<br>(1.1651) | -9.6599***<br>(1.7228) | -5.8313<br>(6.4970)   |                       |                       |
| GRE Quantitative Percentile                   |                       |                            |                            | 0.1060***<br>(0.0135)  | 0.1151***<br>(0.0191)  | 0.0834<br>(0.0644)    |                       |                       |
| Gender                                        |                       |                            |                            | 0.1997**<br>(0.1017)   | 0.1149<br>(0.1664)     | 0.4650<br>(0.9003)    |                       |                       |
| Demographic 1                                 |                       |                            |                            | -0.2477*<br>(0.1349)   | -0.3286<br>(0.2390)    | -0.2627<br>(0.7744)   |                       |                       |
| Demographic 2                                 |                       |                            |                            | 0.0405<br>(0.1507)     | -0.0036<br>(0.2410)    | -0.3966<br>(1.2105)   |                       |                       |
| Advanced Math                                 |                       |                            |                            |                        | 0.6284***<br>(0.1758)  | 1.1399<br>(0.9839)    |                       |                       |
| Committee Score                               |                       |                            |                            |                        | 0.0548<br>(0.0419)     | -0.9123**<br>(0.3787) |                       |                       |
| Year Transition/Matriculate More              |                       | 46.08%<br>(39.62%, 52.67%) | 87.77%<br>(63.97%, 96.67%) | -0.0609<br>(0.1668)    | 0.0204<br>(0.2685)     | 3.3933***<br>(0.7682) |                       |                       |
| Program Rank                                  |                       |                            |                            | -0.0310***<br>(0.0019) | -0.0283***<br>(0.0027) |                       |                       |                       |
| $\bar{m}_2$                                   | 1.0892***<br>(0.0851) |                            |                            |                        |                        |                       |                       |                       |
| Beta                                          |                       |                            |                            |                        |                        |                       | 0.9493***<br>(0.0396) |                       |
| Sigma                                         |                       |                            |                            |                        |                        |                       |                       | 0.1917***<br>(0.0251) |
| LR Statistic   DF   P-Value                   | 8.9955                | 16                         | 0.9136                     |                        |                        |                       |                       |                       |

- Likelihood-ratio test does not reject model fitting the data
- In *Year Transition Fewer*, Round 1 cutoff probability > Round 3 cutoff probability

# Restricted Waitlist-Hybrid Estimates (Suboptimal)

|                                               | $\bar{m}_{2A}$        | $P(\text{Cutoff})_1$       | $P(\text{Cutoff})_3$       | Success 1              | Success 3              | Matriculate            | Beta                  | Sigma                 |
|-----------------------------------------------|-----------------------|----------------------------|----------------------------|------------------------|------------------------|------------------------|-----------------------|-----------------------|
| Year Transition/Matriculate Fewer   Intercept | -0.0498*<br>(0.0260)  | 67.76%<br>(58.29%, 75.97%) | 74.61%<br>(55.80%, 87.24%) | -8.0749***<br>(1.2691) | -6.3299***<br>(1.7373) | -1.4373<br>(3.8091)    |                       |                       |
| GRE Quantitative Percentile                   |                       |                            |                            | 0.1082***<br>(0.0147)  | 0.0718***<br>(0.0186)  | 0.0355<br>(0.0435)     |                       |                       |
| Gender                                        |                       |                            |                            | 0.2189**<br>(0.1045)   | 0.0369<br>(0.1594)     | -0.3545<br>(0.5436)    |                       |                       |
| Demographic 1                                 |                       |                            |                            | -0.5509***<br>(0.1397) | -0.3579<br>(0.2337)    | 0.3334<br>(0.7088)     |                       |                       |
| Demographic 2                                 |                       |                            |                            | -0.0156<br>(0.1510)    | -0.0774<br>(0.2321)    | -0.7602<br>(0.7927)    |                       |                       |
| Advanced Math                                 |                       |                            |                            |                        | 0.3857**<br>(0.1669)   | 0.3982<br>(0.6445)     |                       |                       |
| Committee Score                               |                       |                            |                            |                        | 0.1748***<br>(0.0374)  | -0.6725***<br>(0.1981) |                       |                       |
| Year Transition/Matriculate More              |                       | 43.47%<br>(37.12%, 50.04%) | 93.27%<br>(41.58%, 99.63%) | -0.0797<br>(0.1668)    | 0.1064<br>(0.2766)     | 2.9410***<br>(0.6722)  |                       |                       |
| Program Rank                                  |                       |                            |                            | -0.0306***<br>(0.0019) | -0.0281***<br>(0.0027) |                        |                       |                       |
| $\bar{m}_2$                                   | 0.8883***<br>(0.0489) |                            |                            |                        |                        |                        |                       |                       |
| Beta                                          |                       |                            |                            |                        |                        |                        | 0.8280***<br>(0.0422) |                       |
| Sigma                                         |                       |                            |                            |                        |                        |                        |                       | 0.2012***<br>(0.0325) |
| LR Statistic   DF   P-Value                   | 43.3371               | 16                         | 0.0002                     |                        |                        |                        |                       |                       |

- Likelihood-ratio test rejects model fitting the data at 1% level
- In *Year Transition Fewer*, Round 1 cutoff probability > Round 3 cutoff probability

# Outline

- 1 Reduced-Form Analysis
- 2 Structural Models
- 3 Structural Estimation
- 4 Optimality Tests

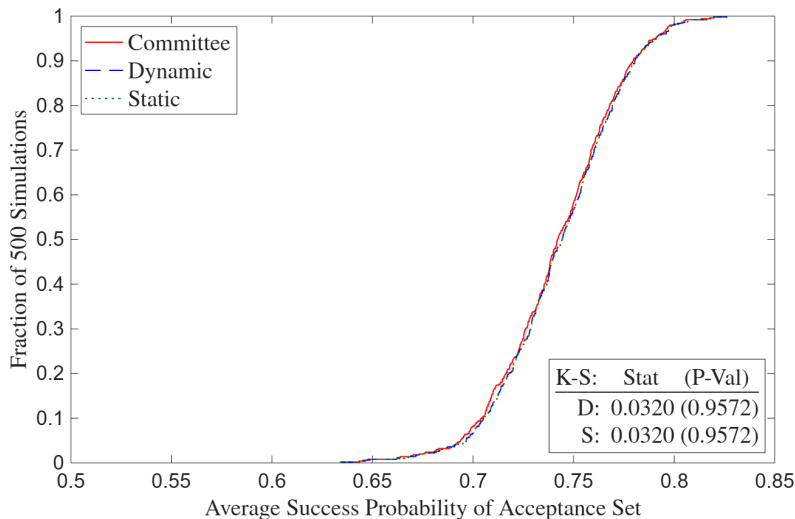
# Likelihood-Ratio and Deviation Gains Tests

- **Likelihood-Ratio:** Determines whether model fits data generated by committee's decision rule, but not whether committee can do better by changing decision rule
- **Deviation Gains:** Suppose committee adopts model's decision rule. Will average success probability of acceptance/matriculation set increase, and if so, by how much?
  - Solve model with  $\sigma = 0$  in CCPs, compute average yearly  $P(\text{Success})$  of model's acceptance/matriculation set, and compare to that of actual decision rule's set
  - $\hat{\theta}_S^R$  may be biased to rationalize suboptimal behavior, so plug in  $\hat{\theta}_S^U$
  - **Example:** If committee puts too little weight on *Advanced Math*, then its restricted success probability parameter estimate will be biased down

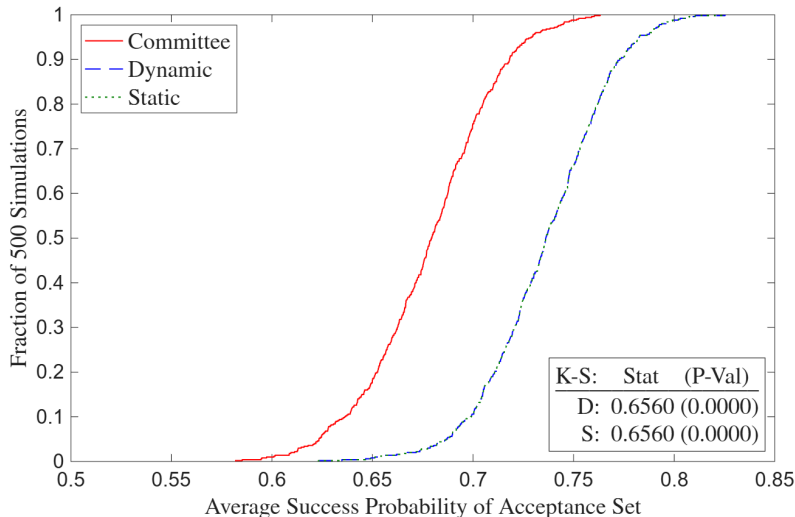
# Deviation Gains Interpretation

- Suppose there are meaningfully large deviation gains. Then committee may have incorrect beliefs about which types of applicants are most likely to succeed
  - Matriculation rate of model's acceptance set may be less than that of actual decision rule's acceptance set, making estimated gains an upper bound
  - In matriculation and waitlist counterfactuals, compare success probabilities of matriculation sets by simulating accepted applicants' matriculation decisions
- ★ **Robustness:** Given too few observations for holdout set and infeasibility of large  $k$ -fold cross-validation due to runtime, implement randomization test
  - Compute average success probabilities with perturbations of  $\hat{\theta}_S^U$

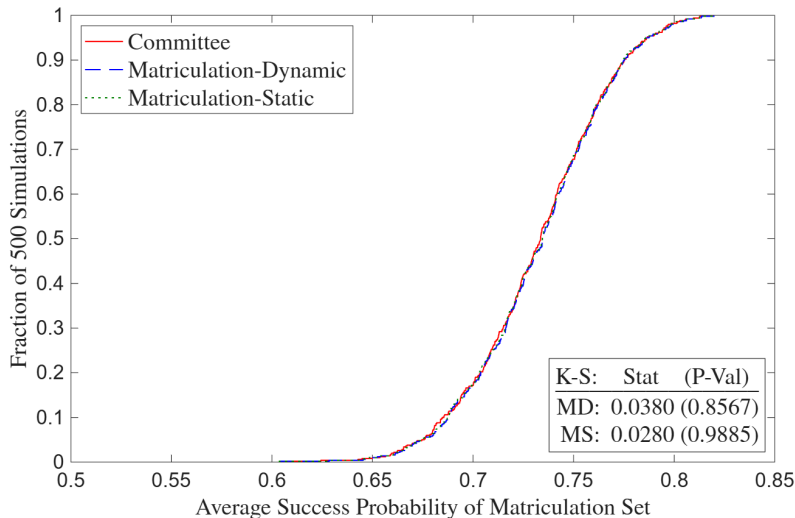
# Deviation Gains (Optimal)



# Deviation Gains (Suboptimal)

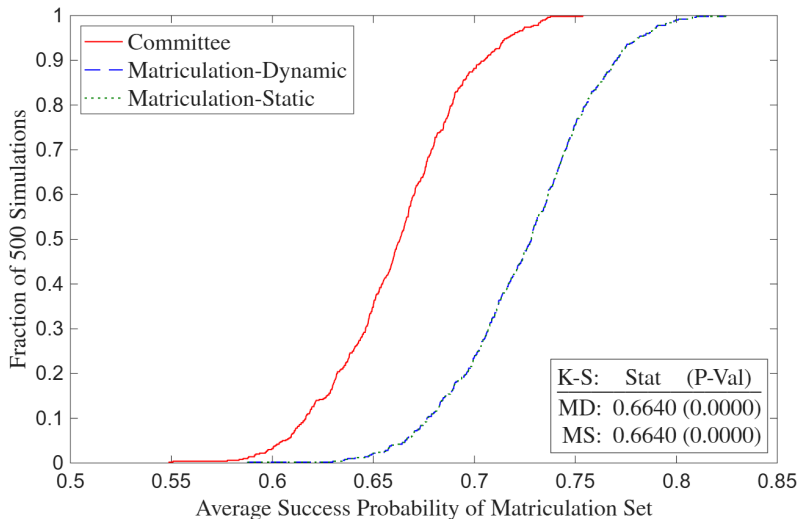


# Matriculation Deviation Gains (Optimal)

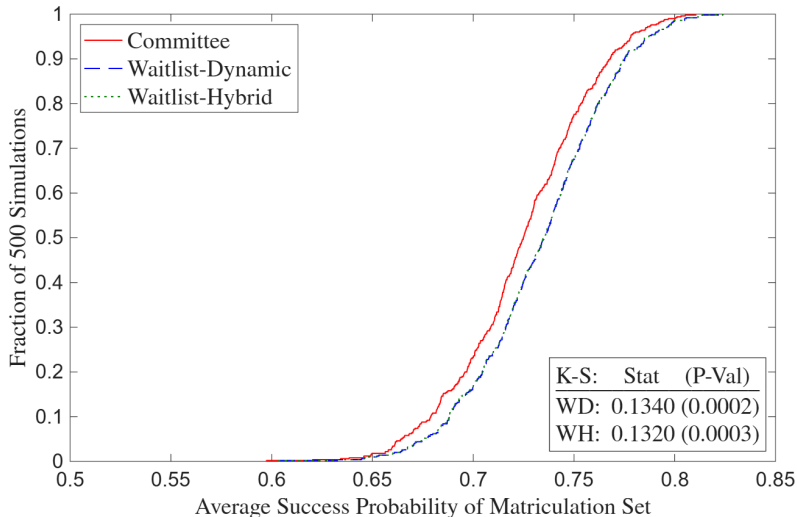




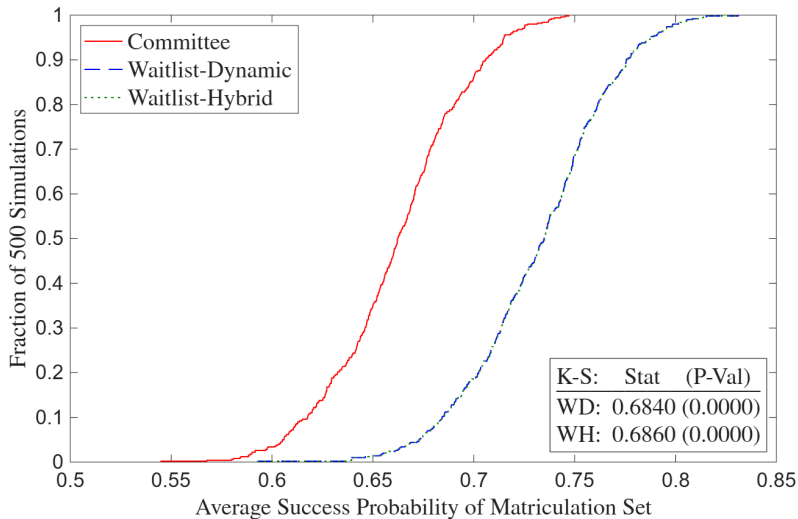
# Matriculation Deviation Gains (Suboptimal)



# Waitlist Deviation Gains (Optimal)



# Waitlist Deviation Gains (Suboptimal)



# Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
  - ① Economics admissions more data-driven than other social science programs
  - ② Tradeoff between marginal cost of reading file vs. quality of marginal acceptance
    - Exploit tradeoff to calculate, in dollars, how much success is worth to committee
    - Alternatively, what if committee uses AI to read every applicant's file?
- **Future Research:** Two-sided matching model across multiple universities

Thank You!

# A1: Synthetic Rows $\neq$ Real Subjects

| Transition | Waitlist | Accept | Matriculate | Success | GRE Quantitative Percentile | Gender | Demographic 1 | Demographic 2 | Advanced Math | Committee Score | Program Rank |
|------------|----------|--------|-------------|---------|-----------------------------|--------|---------------|---------------|---------------|-----------------|--------------|
| 1          | 1        | 0      | 0           | 0       | 89                          | 0      | 0             | 0             | 0             | 5               | 150          |
| 1          | 0        | 1      | 0           | 1       | 96                          | 0      | 1             | 0             | 0             | 6               | 23           |
| 0          | 0        | 0      | 0           | 0       | 88                          | 0      | 1             | 0             | .             | .               | 150          |
| 0          | 0        | 0      | 0           | 0       | 80                          | 1      | 1             | 0             | .             | .               | 150          |
| 0          | 0        | 0      | 0           | 1       | 99                          | 0      | 1             | 0             | .             | .               | 29           |
| 0          | 0        | 0      | 0           | 0       | 81                          | 0      | 0             | 1             | .             | .               | 23           |
| 1          | 0        | 1      | 1           | 1       | 92                          | 0      | 1             | 0             | 0             | 3               | 37           |
| 1          | 1        | 1      | 1           | 1       | 93                          | 0      | 0             | 0             | 1             | 3               | 37           |
| 0          | 0        | 0      | 0           | 0       | 83                          | 1      | 0             | 1             | .             | .               | 48           |
| 1          | 1        | 0      | 0           | 0       | 87                          | 0      | 1             | 0             | 1             | 5               | 34           |



| Transition | Waitlist | Accept | Matriculate | Success | GRE Quantitative Percentile | Gender | Demographic 1 | Demographic 2 | Advanced Math | Committee Score | Program Rank |
|------------|----------|--------|-------------|---------|-----------------------------|--------|---------------|---------------|---------------|-----------------|--------------|
| 1          | 1        | 0      | 0           | 1       | 94                          | 1      | 1             | 0             | 0             | 4               | 29           |
| 1          | 1        | 1      | 1           | 0       | 90                          | 1      | 0             | 1             | 0             | 7               | 37           |
| 0          | 0        | 0      | 0           | 0       | 93                          | 1      | 0             | 1             | .             | .               | 150          |
| 0          | 0        | 0      | 0           | 0       | 75                          | 1      | 1             | 0             | .             | .               | 150          |
| 0          | 0        | 0      | 0           | 0       | 81                          | 1      | 0             | 0             | .             | .               | 150          |
| 0          | 0        | 0      | 0           | 0       | 74                          | 1      | 0             | 0             | .             | .               | 150          |
| 0          | 0        | 0      | 0           | 0       | 84                          | 0      | 0             | 1             | .             | .               | 150          |
| 1          | 1        | 0      | 0           | 0       | 87                          | 1      | 0             | 1             | 0             | 7               | 14           |
| 1          | 1        | 0      | 0           | 0       | 77                          | 0      | 1             | 0             | 0             | 4               | 28           |
| 1          | 0        | 1      | 0           | 0       | 90                          | 0      | 1             | 0             | 0             | 4               | 11           |

# A1: Dependent Variables

Table 1a: Dependent Variables

| Admission                         | Success                         |
|-----------------------------------|---------------------------------|
| <b>1(Transition)</b>              | <b>1(Complete PhD Program)</b>  |
| <i>1(Accept)</i>                  | <i>1(Academic Placement)</i>    |
| <b>1(Waitlist)</b>                | <b>1(Government Placement)</b>  |
| <b>1(Open House)</b>              | <b>1(Consulting Placement)</b>  |
| <i>1(Matriculate)</i>             | <i>1(Tech Placement)</i>        |
| <b>1(Attend PhD Program)</b>      | <b>Placement U.S. Ranking</b>   |
| <b>PhD Program U.S. Ranking</b>   | <b>Placement Global Ranking</b> |
| <b>PhD Program Global Ranking</b> | <b>1(Good Placement)</b>        |

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
  - For non-tenure track, use weighted average of university's and maximum rankings
  - For non-academia, use average of ChatGPT, Claude, and Gemini's rankings

# A1: Objective Variables

Table 1b: Objective Independent Variables

| Demographic                     | Performance                          | Economics-Only Performance              |
|---------------------------------|--------------------------------------|-----------------------------------------|
| 1(Covid Year)                   | GRE Verbal Percentile                | <b>Masters GPA</b>                      |
| Winsorized Age                  | GRE Quantitative Percentile          | 1( <b>Missing Masters GPA</b> )         |
| 1(Female)                       | GRE Analytic Percentile              | 1( <b>Elite U.S. Masters</b> )          |
| 1(U.S. African/Hispanic/Native) | 1(Missing GRE Percentiles)           | 1( <b>Elite International Masters</b> ) |
| 1(U.S. Asian)                   | Undergraduate GPA                    | 1( <b>Other International Masters</b> ) |
| 1(International Asian)          | 1(Missing Undergraduate GPA)         | 1( <b>Research Experience</b> )         |
| 1(International Other)          | TOEFL/IELTS Score                    | <b>Objective Score</b>                  |
|                                 | 1(Missing TOEFL/IELTS Score)         | <b>Relative Committee Score</b>         |
|                                 | 1(Elite U.S. University)             |                                         |
|                                 | 1(Elite U.S. Liberal Arts College)   |                                         |
|                                 | 1(Elite International University)    |                                         |
|                                 | 1(Other International University)    |                                         |
|                                 | 1(Graduate Degree)                   |                                         |
|                                 | 1(Work Experience)                   |                                         |
|                                 | Number of Professor Recommenders     |                                         |
|                                 | Number of Prolific Recommenders      |                                         |
|                                 | <i>Concentration Dummy Variables</i> |                                         |



# A1: Subjective Variables

**Table 1c:** Subjective Independent Variables

| Recommendation Letter        | Essay/CV/Supplemental Form                          |
|------------------------------|-----------------------------------------------------|
| Relative Letter Length       | Essay Length (Essay)                                |
| Relative Letter Features     | $\mathbb{1}(\text{Specific Topic})$ (Essay)         |
| Positive Words (YS 2012)     | $\mathbb{1}(\text{Specific Professor})$ (Essay)     |
| Negative Words (YS 2012)     | $\mathbb{1}(\text{Economics Work Experience})$ (CV) |
| Standout Words (BEMS 2022)   | $\mathbb{1}(\text{Working Paper})$ (CV)             |
| Ability Words (BEMS 2022)    | $\mathbb{1}(\text{Publication})$ (CV)               |
| Research Words (BEMS 2022)   | $\mathbb{1}(\text{Coding})$ (CV)                    |
| Grindstone Words (BEMS 2022) | Number of Math Courses (Supplemental)               |
| Teaching Words (BEMS 2022)   | $\mathbb{1}(\text{Advanced Math})$ (Supplemental)   |
| Communal Words (BEMS 2022)   | $\mathbb{1}(\text{Theory Field})$ (Essay/CV)        |
| Agentic Words (MHM 2009)     | $\mathbb{1}(\text{Applied Field})$ (Essay/CV)       |

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- #MathCourses and  $\mathbb{1}(\text{AdvMath})$  objective after 2020

# A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

# A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

# A1: Dictionaries

- **Recommendation Letters:** Porter stemmer reduces words to their roots

- **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop

- **Essays and CVs:**

- **Specific Topic:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban
- **Economics Work Experience:** assistant, fed, imf, international monetary fund, predoc, ra, research, treasury, world bank
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape

# A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
  - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
  - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for “grindstone terms” (e.g. depend\*, diligen\*)
- Research questions emphasize prediction over inference
  - Mullainathan & Spiess (2017) distinguish  $\hat{\beta}$  vs.  $\hat{y}$  problems
  - If GRE predicts success, committee should use it regardless of why

# A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if  $P(\text{Success}|\text{Skill})$  is an S-curve, then  $\hat{\beta}$  will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
  - Let  $y_n^* = x_n\beta + u_n$ ,  $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$ , and  $y_n = d_n y_n^*$
  - **Step 1:** Run probit of  $d$  on  $z$  and compute inverse Mills  $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
  - **Step 2:** Run OLS or heckprobit of  $y$  on  $(x, \hat{\lambda})$  to get  $\hat{\beta}$
  - **Instruments:** Variables that predict matriculation but not the outcome

# A1: Insufficiency of Comparing AMEs

- Cannot test optimality by comparing AMEs across regressions. Suppose:
  - Math skills have large positive effect on success, coding skills smaller positive effect
  - 25% of applicants have both skills, 25% math only, 25% coding only, 25% neither
  - Committee can accept half of applicants, and other half attend different university
- Optimizing committee accepts half with math and ignores coding
  - **Acceptance Regression:**  $AME_{\text{Math}} = 1, AME_{\text{Coding}} = 0$
  - **Success Regression:**  $1 > AME_{\text{Math}} > AME_{\text{Coding}} > 0$
  - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

# A1: Wald Tests

- Test if admission and success AMEs are same across programs
  - **Example:** Does elite undergraduate university predict admission and/or success more or less for economics than for other programs?
  - **Data:** Simulated from reduced-form-calibrated data-generating parameters
- Can I test if acceptance AMEs equal success AMEs within programs?
  - Predictors of acceptance only (success only) may be overvalued (undervalued), provided committee wants to select based on a “success” model
  - **Problem:** Not a test if admissions decisions are optimal
  - Need structural model of admissions to conclude about optimality



# A1: Logistic Wald Test Across Programs

Assume  $(Y^E \perp\!\!\!\perp Y^{NE})|X$ . Let  $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1 + \exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$ ,

and  $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1 + \exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$ . Finally, let  $N = N_E + N_{NE}$ . Then:

$$\mathcal{L}_N^U(y|x; \theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} [1 - P(Y_n^E = 1|x_n^E; \theta^E)]^{1-Y_n^E} \\ \prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} [1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let  $g(\hat{\theta}) = \frac{1}{N_E} \sum_{n=1}^{N_E} \hat{P}_n^E (1 - \hat{P}_n^E) \hat{\theta}_1^E - \frac{1}{N_{NE}} \sum_{n=1}^{N_{NE}} \hat{P}_n^{NE} (1 - \hat{P}_n^{NE}) \hat{\theta}_1^{NE}$ . Then:  
 $W = g(\hat{\theta})' [\nabla_{\theta} g(\hat{\theta}) \hat{V}(\hat{\theta}) \nabla_{\theta} g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$ . Compare AMEs per Mood (2010)

# A1: Linear Wald Test Across Programs

Assume  $(Y^E \perp\!\!\!\perp Y^{NE})|X$ , i.e. (Economics  $\perp\!\!\!\perp$  Non-Economics) $|$ Predictors.

Let  $f(y_n^E|x_n^E; \beta^E, \sigma_E^2) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E)^2\right]$ , and

$f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE})^2\right]$ . Then:

$$\mathcal{L}_N^U(y|x; \beta, \sigma^2) = \prod_{n=1}^{N_E} f(y_n^E|x_n^E; \beta^E, \sigma_E^2) \prod_{n=1}^{N_{NE}} f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2).$$

- $Y \in \{\mathbb{1}(\text{Standardized Committee Score}), \mathbb{1}(\text{Standardized Placement Rank})\}$
- **Test:** Let  $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$ . (Can also do likelihood-ratio test.)  
Then  $W = g(\hat{\beta}_1)'[\nabla_{\beta, \sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta, \sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e.  $\forall n, \sigma_n^2 = \sigma^2$ )

# A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
  - Need different skills at different stages in the program, such as grades vs. placement
  - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
  - **Admission:** May have different preparation and/or interests
  - **Completion:** Depending on country, may have fewer outside options
  - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
  - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
  - For economics, increased importance of elite predocs and Fed RAs

## A2.1: Portfolio Choice to Cutoff Rule (BEMS 2022)

### Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] + \lambda(N_A - N)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda^* \end{cases}$$

**Upshot:**  $\lambda^*$  is cutoff probability. Denote it hereafter as  $P(y^* = S)$

- $P(y^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

## A2.1: Portfolio Choice to Cutoff Rule with Shocks

### Portfolio Choice Problem:

$$\begin{aligned} \max_{(d_1, \dots, d_N) \in \{A, R\}^N} & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + \epsilon_n(R) \mathbb{1}(d_n = R) \}, \\ \text{such that: } & \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \dots, d_N; \lambda) = \\ & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + [\lambda + \epsilon_n(R)] \mathbb{1}(d_n = R) \} + \lambda(N_A - N) \end{aligned}$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n, \epsilon_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

### Theorem 4 (Proof)

For large  $N$ , there is a cutoff rule solution with logistic CCPs.

► Model Framework

## A2.1: Models of Admissions

- ① **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
  - ① **Static Model:** In Round 1, rank applicants based only on objective variables
  - ② **Myopic Model:** Like dynamic, but distribution of  $z$  not conditional on  $x$
- ② **Matriculation-Dynamic/Static Models:** In Round 1, rank applicants based on combination of success and matriculation probabilities
- ③ **Waitlist-Dynamic/Hybrid Models:** In Round 1, rank applicants based on success probabilities. After Round 1, accept or waitlist applicants in Round 2. After acceptances matriculate or decline, accept or reject waitlisted applicants in Round 3. Favor likely matriculants in Round 2 to increase selectivity in Round 3

## A2.1: Description of Models

- **Round 2:** If accept applicant, get success probability  $P(y = S|x, z, t, \bar{k})$ , where Success =  $\mathbb{1}(\text{Complete Program})$  or  $\mathbb{1}(\text{Good Placement})$ . If reject applicant, get  $P(y_2^* = S|t) = P(\text{Success})$  of year  $t$ 's marginal accepted applicant
- **Round 1:** If reject applicant, get  $P(y_1^* = S|t) = P(\text{Success})$  of year  $t$ 's marginal transitioned applicant. Can be less or greater than  $P(y_2^* = S|t)$  depending on value placed on committee's time vs. admitting a successful cohort
  - **Dynamic:** If transition applicant, get  $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] =$  expected value of Round 2, where  $z$  is integrated out conditional on  $(x, t)$  based on  $F(z|x, t)$
  - **Static:** If transition applicant, get  $P(y = S|x, t, \bar{k})$ , or  $P(\text{Success})$  unconditional on  $z$ . It may differ from  $P(y = S|x, z, t, \bar{k})$ , though it still accounts for  $z$ 's effect on  $y$
  - **Myopic:** In  $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ , replace  $F(z|x, t)$  with  $F(z|t)$

## A2.1: Portfolio Choice to Cutoff Rule, Round 2

### Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] + \lambda_2 (N_A - N_T)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(x_n, z_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda_2^* \end{cases}$$

**Upshot:**  $\lambda_2^*$  is cutoff probability. Denote it hereafter as  $P(y_2^* = S)$

- $P(y_2^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$



## A2.1: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

### Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n; \bar{\lambda}_2^*) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{z|x}(\tilde{z}_n | x_n), \text{ and: } \bar{\lambda}_2^* = \mathbb{E}[\lambda_2^*(\mathbf{x}, \mathbf{z}) | d_1(\mathbf{x})]$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*, \bar{\lambda}_2^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n; \bar{\lambda}_2^*) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n; \bar{\lambda}_2^*) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n; \bar{\lambda}_2^*) < \lambda_1^* \end{cases}$$

**Upshot:**  $\lambda_1^*$  is cutoff level. Denote it hereafter as  $L(y_1^* = S) \in [EV_{2,N_T+1}, EV_{2,N_T}]$

## A2.1: Dynamic Parameterization 1

### Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t; \theta_A) = \frac{\exp(f_A(x, z, t)\theta_A)}{1 + \exp(f_A(x, z, t)\theta_A)}$$

### Success Probability:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}$$

### Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P(y_2^* = S | t; \theta_{S_2^*}) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

## A2.1: Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[ \cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t)\end{aligned}$$

- **Continuous:**  $f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma_{F_{z|x,t,j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x,t,j}}^2}} \exp\left[\frac{-1}{2\sigma_{F_{z|x,t,j}}^2}(z_j - g(x, t)\beta_{F_{z|x,t,j}})^2\right]$
- **Binary:**  $P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}})}{1 + \exp(g(x, t)\theta_{F_{z|x,t,j}})}$
- **Multinomial:**  $P(z_j = (\ell \neq L)|x, t; \theta_{F_{z|x,t,j}}^\ell) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}}^\ell)}{1 + \sum_{i=1}^{L-1} \exp(g(x, t)\theta_{F_{z|x,t,j}}^i)}$

## A2.1: Portfolio Choice to Cutoff Rule, Round 1 (Static)

### Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1 (N_T - N)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S | x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S | x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S | x_n) < \lambda_1^* \end{cases}$$

**Upshot:**  $\lambda_1^*$  is cutoff probability. Denote it hereafter as  $P(y_1^* = S)$

- $P(y_1^* = S) \in [P(y_{N_T+1} = S | x_{N_T+1}), P(y_{N_T} = S | x_{N_T})]$

## A2.1: Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f(z_n|x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}}$$
$$\underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Outcome probabilities}} \prod_{n=N_T+1}^N \underbrace{f(z_n|x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}}$$

- **Problem:** Success may not be observable for recent years
  - **Solution:** Separate block for recent years without success subblocks
  - **Assumption:** Older years representative of recent years
- If years are similar, can make  $(\theta_{S_1^*}, \theta_{S_2^*})$  in  $\mathcal{L}^R$  intercept only

## A2.1: Heckman-Corrected Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F)}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probability}} \prod_{n=N_T+1}^N \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}}
 \end{aligned}$$

- $z$  may be unobservable for applicants who didn't survive Round 1
- Estimate Heckman (1979)-corrected full-information likelihood

## A2.1: Empirical Considerations

- **Selection Effects:** Track success for all applicants, including non-matriculants
  - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
  - In decision blocks containing success, fix *Program Rank* at this university's rank
  - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
  - **Example:** Recommendation letter has nuances that algorithms can't detect
  - Include subjective committee score in  $z$  as a proxy for this information

## A2.1: Cutoff Rule with Matriculation

### Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^N P(m_n = M | w_n) \mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | w_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] P(m_n = M | w_n) + \lambda(N_M - \overline{N_M})$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(w_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda^* \\ R & \text{if } P(y_n = S | w_n) < \lambda^* \end{cases}$$



## A2.1: Cutoff Rule with Matriculation (Large $N$ )

### Portfolio Choice Problem:

$$\max_{\delta \in \{\delta | \delta: \mathbb{R}^K \rightarrow [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w)f(w)dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} [P(y = S|w)\delta(w) + \lambda(1 - \delta(w))]P(m = M|w)f(w)dw + \lambda(N_M - \overline{N}_M)$$

### Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0, 1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

**Upshot:** Point-identified cutoff probability  $\lambda^*$  is with respect to success probabilities

## A2.1: Cutoff Rule with Matriculation, Round 2

### Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] P(m_n = M | w_n) + \lambda_2 (N_M - \overline{N_M})$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(w_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | w_n) < \lambda_2^* \end{cases}$$

## A2.1: Cutoff Rule with Matriculation, Round 1

### Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n; \bar{\lambda}_2^*) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n) - \bar{\lambda}_2^*, 0\} P(m_n = M | x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*, \bar{\lambda}_2^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n; \bar{\lambda}_2^*) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n; \bar{\lambda}_2^*) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n; \bar{\lambda}_2^*) < \lambda_1^* \end{cases}$$

## A2.1: Matriculation-Dynamic Likelihood

### Matriculation Probability:

$$P_m = P(m = M | x, z, t; \theta_M) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$$

$$\mathcal{L}_N^U(\theta) = \underbrace{\prod_{n=1}^{N_T} \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t}}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) \left[ P_{d_{2,n}}(\theta_A) P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)} (1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)} \right]^{\mathbb{1}(d_{2,n}=A)}}_{\text{Conditional choice and matriculation probabilities}}$$

$$\underbrace{(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional Choice Probability}} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Outcome probabilities}} \prod_{n=N_T+1}^N \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional Choice Probability}}$$

## A2.1: Waitlist Model, Round 1

### Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2) | x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in \{A, W\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t), \tilde{d}_2) / \sigma} \right) dF_{z|x, t}(\tilde{z} | x, t)$$

### Conditional Choice Probabilities:

$$P(d_r | \text{states}) = \frac{\exp(u_r(\text{states}, d_r) / \sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(\text{states}, \tilde{d}_r) / \sigma)}$$

## A2.1: Waitlist Model, Round 1

### Round 1 (Hybrid):

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

### Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(\text{states}, \tilde{d}_r)/\sigma)}$$

# A2.1: Waitlist-Dynamic Parameterization 1

## Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, \ddot{m}_2; \theta_{A_2}) = \frac{\exp(f_{A_2}(x, z, \ddot{m}_2)\theta_{A_2})}{1 + \exp(f_{A_2}(x, z, \ddot{m}_2)\theta_{A_2})}$$

$$P_{d_3} = P(d_3 = A | x, z, \overline{m_{2A}}; \theta_{A_3}) = \frac{\exp(f_{A_3}(x, z, \overline{m_{2A}})\theta_{A_3})}{1 + \exp(f_{A_3}(x, z, \overline{m_{2A}})\theta_{A_3})}$$

## Success/Matriculation Probabilities:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}, P_{m_2} = P(m_2 = M | x, z, t; \theta_{M_2}) = \frac{\exp(f_{M_2}(x, z, t)\theta_{M_2})}{1 + \exp(f_{M_2}(x, z, t)\theta_{M_2})}$$

## Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P(y_3^* = S | \overline{m_{2A}}; \theta_{S_3^*}) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}$$

## A2.1: Waitlist-Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t), \tilde{d}_2)/\sigma} \right) dF_{z|x, t}(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_{J_z}} \left[ \cdot \right] dF_{z|x, t, 1}(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_{J_z}, x, t) \cdots dF_{z|x, t, J_z}(\tilde{z}_{J_z}|x, t)\end{aligned}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3)|x, z, t, \bar{k}, \ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log \left( \sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \widetilde{\overline{m_{2A}}}, \tilde{d}_3)/\sigma} \right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

- **Continuous:**  $f_{z|x, t}(z_j|x, t; \beta_{F_{z|x, t, j}}, \sigma_{F_{z|x, t, j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x, t, j}}^2}} \exp \left[ \frac{-1}{2\sigma_{F_{z|x, t, j}}^2} (z_j - g(x, t)\beta_{F_{z|x, t, j}})^2 \right]$
- **Binary:**  $P(z_j = 1|x, t; \theta_{F_{z|x, t, j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x, t, j}})}{1 + \exp(g(x, t)\theta_{F_{z|x, t, j}})}$
- **Multinomial:**  $P(z_j = (\ell \neq L)|x, t; \theta_{F_{z|x, j}}^\ell) = \frac{\exp(g(x, t)\theta_{F_{z|x, j}}^\ell)}{1 + \sum_{i=1}^{L-1} \exp(g(x, t)\theta_{F_{z|x, j}}^i)}$



## A2.1: Waitlist-Dynamic Likelihood

$$\mathcal{L}_{N_1}^U(\theta) = \prod_{n=1}^{N_W} f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_{\overline{m_2A}|\ddot{m}_2}(\overline{m_2A,n}|\ddot{m}_2,n; \theta_{F_{\overline{m_2A}|\ddot{m}_2}}) P_{d_1,n}(\theta_T) (1 - P_{d_2,n}(\theta_{A_2})) P_{d_3,n}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}$$

$$(1 - P_{d_3,n}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_W+1}^{N_T} f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_{\overline{m_2A}|\ddot{m}_2}(\overline{m_2A,n}|\ddot{m}_2,n; \theta_{F_{\overline{m_2A}|\ddot{m}_2}})$$

$$P_{d_1,n}(\theta_T) P_{d_2,n}(\theta_{A_2}) P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^N f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) (1 - P_{d_1,n}(\theta_T))$$

- Estimate  $\theta_{M_2}$  with 1st-stage likelihood to get  $m_{2,n}$  = matriculation probability  $P_{m_{2,n}}$ :

$$\mathcal{L}_{N_T - N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)} (1 - P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$$

- Use Murphy-Topel SEs, and for  $\mathcal{L}_R$ , use  $(\theta_F^U, \theta_S^U)$  as initial guess of  $(\theta_F^R, \theta_S^R)$

## A2.1: Waitlist Notes

- **Estimation:** Estimate  $f_{\overline{m}_{2A}|\ddot{m}_2}$  on all transitioned applicants; slope parameter's expected sign is positive from variation in  $\overline{m}_2$  across years (e.g. due to Covid)
  - **Assumption:** Effect of  $\ddot{m}_2$  on  $\overline{m}_{2A}$  is stationary across years, and therefore can be used to forecast changes in  $\overline{m}_{2A}$  based on changes in  $\ddot{m}_2$  from changes in  $d_2$
- **Dynamics:** What if committee doesn't think dynamically?
  - If  $\ddot{m}_2$  increasing increases  $\overline{F}_{\overline{m}_{2A}|\ddot{m}_2}$ , which increases  $\mathbb{E}[P(\text{Cutoff})_3]$ , then it should
  - Don't need success data for all years to identify these two effects
- **Over/Undershooting:** Committee may wish to avoid both
  - To Round 3's  $P(\text{Cutoff})$ , add target cohort size minus number of matriculants
  - (Negative) parameter may be unidentifiable, so may have to calibrate it

# A2.1: Symmetric Waitlist Model 1

## Round 3:

$$v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, \epsilon_3) = \max_{d_3 \in \{A, R\}} u_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, d_3) = P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_3 = A) + P(y_3^* = S | \overline{m_{2A}}, \overline{m_3}) \mathbb{1}(d_3 = R)$$

## Round 2:

$$v_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, \epsilon_2) = \max_{d_2 \in \{A, W\}} u_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, d_2) =$$

$$P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A) + \beta \mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, \epsilon_3) | x, z, t, \bar{k}, \ddot{m}_2, m_3] \mathbb{1}(d_2 = W), \text{ such that:}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \widetilde{\overline{m_3}}, \epsilon_3) | x, z, t, \bar{k}, \ddot{m}_2, m_3] =$$

$$\int_{\widetilde{\overline{m_{2A}}}} \int_{\widetilde{\overline{m_3}}} \sigma \log \left( \sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \widetilde{\overline{m_{2A}}}, \widetilde{\overline{m_3}}, \tilde{d}_3) / \sigma} \right) dF_{\widetilde{\overline{m_3}} | m_3}(\widetilde{\overline{m_3}} | m_3) dF_{\overline{m_{2A}} | \ddot{m}_2}(\widetilde{\overline{m_{2A}}} | \ddot{m}_2)$$

## A2.1: Symmetric Waitlist Model 2

### Round 1:

$$v_1(x, t, \bar{k}) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

- **Hybrid:**  $u_1(x, t, \bar{k}, d_1) = P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R)$

- **Dynamic:**

$$u_1(x, t, \bar{k}, d_1) = \mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, \epsilon_2)|x, t, \bar{k}]\mathbb{1}(d_1 = T) + L(y_1^* = S|t)\mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t), m_3(x, \tilde{z}, t), \tilde{d}_2)/\sigma} \right) dF_{z|x, t}(\tilde{z}|x, t)$$

### Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(\text{states}, \tilde{d}_r)/\sigma)}$$

# A2.1: Symmetric Waitlist-Dynamic Parameterization 1

## Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, \ddot{m}_2, m_3; \theta_{A_2}) = \frac{\exp(f_{A_2}(x, z, \ddot{m}_2, m_3)\theta_{A_2})}{1 + \exp(f_{A_2}(x, z, \ddot{m}_2, m_3)\theta_{A_2})}$$

$$P_{d_3} = P(d_3 = A | x, z, \overline{m_{2A}}, \overline{m_3}; \theta_{A_3}) = \frac{\exp(f_{A_3}(x, z, \overline{m_{2A}}, \overline{m_3})\theta_{A_3})}{1 + \exp(f_{A_3}(x, z, \overline{m_{2A}}, \overline{m_3})\theta_{A_3})}$$

## Success/Matriculation Probabilities:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}, P_{m_3} = P(m_3 = M | x, z, t; \theta_{M_3}) = \frac{\exp(f_{M_3}(x, z, t)\theta_{M_3})}{1 + \exp(f_{M_3}(x, z, t)\theta_{M_3})}$$

## Cutoff Levels/Probabilities:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P(y_3^* = S | \overline{m_{2A}}, \overline{m_3}; \theta_{S_3^*}) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}}, \overline{m_3})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}}, \overline{m_3})\theta_{S_3^*})}$$

## A2.1: Symmetric Waitlist-Dynamic Parameterization 2

★ Assume  $(\overline{m_{2A}}, \overline{m_3})$  are conditionally independent:

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in \{A, R\}} e^{v_2(x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t), m_3(x, \tilde{z}, t), \tilde{d}_2)/\sigma} \right) dF_{z|x, t}(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_{J_z}} [\cdot] dF_{z|x, t, 1}(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_{J_z}, x, t) \cdots dF_{z|x, t, J_z}(\tilde{z}_{J_z}|x, t)\end{aligned}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, \epsilon_3)|x, z, t, \bar{k}, \ddot{m}_2, m_3] = \int_{\widetilde{\overline{m_{2A}}}} \int_{\widetilde{\overline{m_3}}} \sigma \log \left( \sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \widetilde{\overline{m_{2A}}}, \widetilde{\overline{m_3}}, \tilde{d}_3)/\sigma} \right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

- **Continuous:**  $f_{z|x, t}(z_j|x, t; \beta_{F_{z|x, t, j}}, \sigma_{F_{z|x, t, j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x, t, j}}^2}} \exp \left[ \frac{-1}{2\sigma_{F_{z|x, t, j}}^2} (z_j - g(x, t)\beta_{F_{z|x, t, j}})^2 \right]$
- **Binary:**  $P(z_j = 1|x, t; \theta_{F_{z|x, t, j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x, t, j}})}{1 + \exp(g(x, t)\theta_{F_{z|x, t, j}})}$
- **Multinomial:**  $P(z_j = (\ell \neq L)|x, t; \theta_{F_{z|x, t, j}}^\ell) = \frac{\exp(g(x, t)\theta_{F_{z|x, t, j}}^\ell)}{1 + \sum_{i=1}^{L-1} \exp(g(x, t)\theta_{F_{z|x, t, j}}^i)}$

## A2.1: Symmetric Waitlist-Dynamic Likelihood

$$\mathcal{L}_{N_1}^U(\theta) = \prod_{n=1}^{N_W} f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_{\overline{m_{2A}}|\ddot{m}_2}(\overline{m_{2A,n}}|\ddot{m}_{2,n}; \theta_{F_{\overline{m_{2A}}|\ddot{m}_2}}) f_{\overline{m_3}|m_3}(\overline{m_{3,n}}|m_{3,n}; \theta_{F_{\overline{m_3}|m_3}}) P_{d_{1,n}}(\theta_T)(1-P_{d_{2,n}}(\theta_{A_2}))P_{d_{3,n}}(\theta_{A_3}) \\ \mathbb{1}(d_{3,n}=A)(1-P_{d_{3,n}}(\theta_{A_3}))\mathbb{1}(d_{3,n}=R)P_{y_n}(\theta_S)\mathbb{1}(y_n=S)(1-P_{y_n}(\theta_S))\mathbb{1}(y_n=F) \prod_{n=N_W+1}^{N_T} f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_{\overline{m_{2A}}|\ddot{m}_2}(\overline{m_{2A,n}}|\ddot{m}_{2,n}; \theta_{F_{\overline{m_{2A}}|\ddot{m}_2}}) \\ f_{\overline{m_3}|m_3}(\overline{m_{3,n}}|m_{3,n}; \theta_{F_{\overline{m_3}|m_3}}) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_{A_2}) P_{y_n}(\theta_S) \mathbb{1}(y_n=S)(1-P_{y_n}(\theta_S))\mathbb{1}(y_n=F) \prod_{n=N_T+1}^N f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}})(1-P_{d_{1,n}}(\theta_T))$$

- Estimate  $\theta_{M_3}$  with 1st-stage likelihood to get  $m_{3,n}$ =matriculation probability  $P_{m_{3,n}}$ :  

$$\mathcal{L}_{N_W}^{M_3}(\theta_{M_3}) = \prod_{n=1}^{N_W} P_{m_{3,n}}(\theta_{M_3}) \mathbb{1}(d_{3,n}=A, m_{3,n}=M) (1 - P_{m_{3,n}}(\theta_{M_3})) \mathbb{1}(d_{3,n}=A, m_{3,n}=D)$$
- Use Murphy-Topel SEs, and for  $\mathcal{L}_R$ , use  $(\theta_F^U, \theta_S^U)$  as initial guess of  $(\theta_F^R, \theta_S^R)$

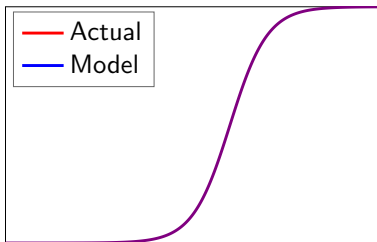
## A2.1: Optimality Tests Notes

- ❶ **Likelihood-Ratio:** Restricted model not nested in unrestricted model (Vuong 1989)
  - **Program Fit:** Does model fit program's behavior?
    - $d_1 = \mathbb{1}(\text{Transition})$ ,  $d_2 = \mathbb{1}(\text{Accept})$
    - If not, could program be behaving suboptimally (Simon 1956)?
  - **Profession Fit:** Does model fit profession's behavior?
    - $d_2 = \mathbb{1}(\text{Attend Program})$  but  $d_1$  unobservable, so must estimate one-stage model
- ❷ **Deviation Gains:** Adjust  $\theta_{S^*}$  so transition and acceptance rates are unchanged
  - **Static:** Rank all applicants by  $P(y = S|x, t, \bar{k})$ , set  $\theta_{S_1^*}$  to match transition rate, rank survivors by  $P(y = S|x, z, t, \bar{k})$ , and set  $\theta_{S_2^*}$  to match acceptance rate
  - **Dynamic:** Use MSM to estimate new  $(\theta_{S_1^*}, \theta_{S_2^*})$ ; moments are each year's transition and acceptance rates. But counterfactual does not require  $(\hat{\theta}_{S_1^*}, \hat{\theta}_{S_2^*})$

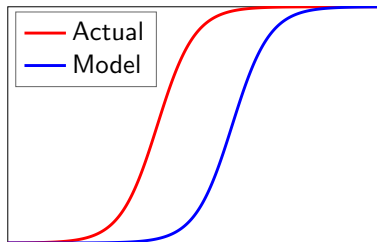


## A2.1: How Deviation Gains Test Works

- 1 Write Bellman equations for committee's optimization problem
- 2 Estimate parameters. Rationality assumption distorts estimates
- 3 Re-estimate parameters without rationality, and use to solve model
- 4 Compare selections from model's decision rule to actual decision rule
- 5 To prevent overfitting, use perturbations of estimates for comparison



Optimizing Committee



Non-Optimizing Committee

# A2.1: Deviation Gains Algorithm 1

## Dynamic/Static/Myopic:

- 1 Rank each  $t$ 's applicants by  $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$  for dynamic and myopic, or  $P(y = S|x, t, \bar{k})$  for static. Transition top  $N_T$  applicants, and reject the rest
- 2 Rank surviving applicants by  $P(y = S|x, z, t, \bar{k})$ , and accept top  $N_A$  survivors.  
Compute  $\overline{P(y = S|x, z, t, \bar{k})}$  of acceptance set, and compare to committee's set

## Matriculation-Dynamic/Static:

- 1 Rank each  $t$ 's applicants by  $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$  for dynamic, or  $P(y = S|x, t, \bar{k})P(m = M|x, t)$  for static. Transition top  $N_T$ , and reject the rest
- 2 Rank surviving applicants by  $P(y = S|x, z, t, \bar{k})$  and accept them in descending order, each matriculating with probability  $P(m = M|x, z, t)$ , until  $N_M$  matriculate.  
Compute  $\overline{P(y = S|x, z, t, \bar{k})}$  of matriculation set, and compare to committee's set

## A2.1: Deviation Gains Algorithm 2

### Waitlist-Dynamic/Hybrid:

- 1 Rank each  $t$ 's applicants by  $\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2)|x, t, \bar{k}]$  for dynamic, or  $P(y = S|x, t, \bar{k})$  for hybrid. Transition top  $N_T$ , and reject the rest
- 2 Rank surviving applicants by  $P(y = S|x, z, t, \bar{k}) - \beta \mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3)|x, z, t, \bar{k}, \ddot{m}_2]$ . Accept top  $N_{2A}$ , each matriculating with probability  $P(m_2 = M|x, z, t)$ , and waitlist the rest
- 3 Rank waitlisted applicants by  $P(y = S|x, z, t, \bar{k})$  and accept them in descending order, each matriculating with probability  $P(m_3 = M|x, z, t)$ , until  $N_M$  matriculate. Compute  $\overline{P(y = S|x, z, t, \bar{k})}$  of matriculation set, and compare to committee's set

## A2.1: Randomization Tests 1

### Theorem 5 (Proof)

Let  $d^R$  be the model's decision rule, and  $d^U$  the actual decision rule. By optimality:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^R[P_{y_n}(\hat{\theta}_S^U)] = A\} \\ \geq \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^U[P_{y_n}(\hat{\theta}_A^U)] = A\} \end{aligned}$$

- But for perturbation  $\widetilde{\theta}_{S,t}^U$  of  $\hat{\theta}_S^U$ 's asymptotic distribution, can have:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta}_{S,t}^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^R[P_{y_n}(\hat{\theta}_S^U)] = A\} \\ < \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta}_{S,t}^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^U[P_{y_n}(\hat{\theta}_A^U)] = A\} \end{aligned}$$

- For matriculation and waitlist models, compare matriculants, not acceptances

## A2.1: Randomization Tests 2

- **Question:** Are deviation gains robust to estimation error?
- **Test 1:** Across  $T$  perturbations, plot CDFs of  $\overline{P(\text{Success})}$  of both acceptance sets. Does the model's CDF stochastically dominate the actual decision rule's CDF?
- **Test 2:** Two-sample Kolmogorov-Smirnov test of equality of CDFs:
  - Let  $X = \overline{P_y[d(\hat{\theta}^U), \widetilde{\theta}_S^U]}$ , and  $F_T(x)$  be  $X$ 's CDF over  $T$  perturbations
  - Then  $D_T = \sup_x |F_T^R(x) - F_T^U(x)|$ . Reject equality if  $D_T > \sqrt{\frac{-\log(\alpha/2)}{T}}$
- **Open Question:** Would deviation gains be borne out in practice?

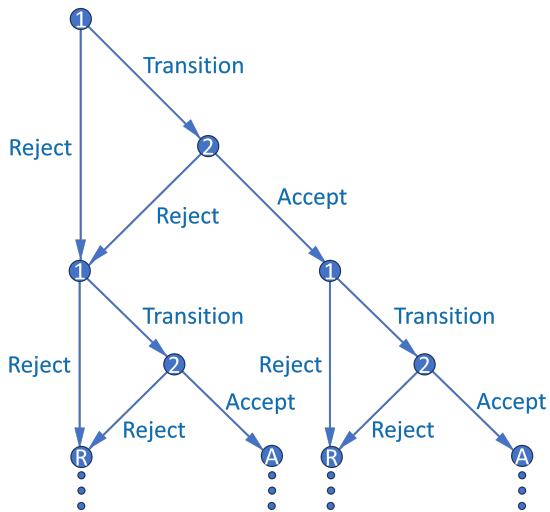
## A2.1: Summary of Results

- **Results:** Estimation results are well behaved
  - **Dynamic:** Likelihood-ratio test always rejects, but deviation gains significant only for simulated dataset where committee misvalues applicant characteristics
  - **Static:** Likelihood-ratio test rejects and gains significant only for suboptimal dataset
  - **Matriculation-Dynamic/Static:** Same as above
  - **Waitlist-Dynamic/Hybrid:** Matriculation estimates are as expected
- Example of cutoff probability reversal if committee transitions few files in Round 1:
  - Three applicants: A, B, and C. Committee transitions two, accepts one
  - A is best and C is worst overall, but A is worst in objective variables
  - Committee rejects A in Round 1 and accepts B, who is worse than A
  - By assuming  $h$  hours to read application file and  $w$  hourly wage, and calculating model's  $\overline{P(\text{Success})}$  induced by all possible  $N_T$ , can put dollar value on success

## A2.1: Extensions

- **Extension 1:** Create matriculation probability uncorrelated with  $P(\text{Success})$ 
  - For accepted applicants, use matriculation predictors in essays/rec letters
  - Can help committee find “people that we can afford” (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes  $P(\text{Success})$ 
  - Instead maximize convex combination of  $P(\text{Success})$  and other objectives
  - **Example:** Committee may want diverse cohort (e.g. across fields):  
 $\alpha P(\text{Success}) + (1 - \alpha)\text{Diversity}$ , where  $\alpha \in [0, 1]$  can be estimated
- **Extension 3:** Multiyear models (see Appendix 2.2)
  - Two-round models repeat over infinite horizon with discounting
  - States contain info about past years/rounds, and committee values future years/rounds (memory includes matriculation probability and diversity)

## A2.2: Multiyear Decision Graph





## A2.2: Multiyear Memory (Matriculation)

- Suppose committee maximizes discounted “lifetime” cohort success
- Respecify  $P(y_2^* = S|t)$ , where  $t = \text{year}$ , as  $L(y_2^* = S|c_{-1})$ 
  - $c_{-1} = \text{previous year's cohort size}$ .  $L(\text{Cutoff}_2)$  increases in  $c_{-1}$  because committee must keep total number of students in program relatively constant across years
- What information available at start of Round 2 determines  $c$ ?
  - **If and only if  $d_2 = A$ :**  $m = \text{applicant's matriculation probability}$
  - For simplicity, suppose matriculation probabilities are exogenous, and success probabilities depend only on applicant characteristics (i.e. not program rank)

## A2.2: Multiyear Dynamics (Matriculation)

### Theorem 6 (Proof)

*Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

- With memory, optimal dynamic decision may not equal optimal myopic one
  - High  $m$  means high  $\bar{F}_{c|m}$ , which means high  $\mathbb{E}[L(\text{Cutoff}_2)']$  and future EV
  - Accepting likely matriculants above this year's margin means that next year in expectation, committee won't have to accept as many marginal applicants
- Harder to estimate since multiyear problem has no closed-form solution
  - Principal component or factor analysis can help reduce dimensionality

## A2.2: Multiyear Model (Matriculation), Round 2

$$v_2(x, z, t, m, c_{-1}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, m, c_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, m, c_{-1}, d_2) = \{P(y = S|x, z, t) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1')|m]\} \mathbb{1}(d_2 = A) \\ + \{L(y_2^* = S|c_{-1}) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1')]\} \mathbb{1}(d_2 = R), \text{ such that:}$$

$$\mathbb{E}[v_1(x', t', c, \epsilon_1')|(m)] = \int_c \int_{t'} \int_{x'} \sigma \log \left( \sum_{d_1' \in \{T, R\}} e^{u_1(x', t', c, d_1')/\sigma} \right) dF_x(x') dF_t(t') dF_{c|(m)}(c|(m))$$

### Conditional Choice Probabilities:

$$P(d_2|x, z, t, m, c_{-1}) = \frac{\exp(u_2(x, z, t, m, c_{-1}, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(x, z, t, m, c_{-1}, \tilde{d}_2)/\sigma)}$$

## A2.2: Multiyear Model (Matriculation), Round 1

$$v_1(x, t, c_{-1}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, c_{-1}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, c_{-1}, d_1) = \mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2) | x, t, c_{-1}] \mathbb{1}(d_1 = T) \\ + \{L(y_1^* = S | t) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1')]\} \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2) | x, t, c_{-1}] = \\ \int_{\tilde{m}} \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \tilde{m}, c_{-1}, \tilde{d}_2) / \sigma} \right) dF_{z|x, t}(\tilde{z} | x, t) dF_{m|x, t}(\tilde{m} | x, t)$$

### Conditional Choice Probabilities:

$$P(d_1 | x, t, c_{-1}) = \frac{\exp(u_1(x, t, c_{-1}, d_1) / \sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(x, t, c_{-1}, \tilde{d}_1) / \sigma)}$$

## A2.2: Multiyear (Matriculation) Parameterization 1

### Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t, c_{-1}; \theta_T) = \frac{\exp(f_T(x, t, c_{-1})\theta_T)}{1 + \exp(f_T(x, t, c_{-1})\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, m, c_{-1}; \theta_A) = \frac{\exp(f_A(x, z, m, c_{-1})\theta_A)}{1 + \exp(f_A(x, z, m, c_{-1})\theta_A)}$$

### Success Probability:

$$P_y = P(y = S | x, z, t; \theta_S) = \frac{\exp(f_S(x, z, t)\theta_S)}{1 + \exp(f_S(x, z, t)\theta_S)}$$

### Cutoff Levels:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = f_{S_1^*}(t)\theta_{S_1^*}$$
$$L_{y_2^*} = L(y_2^* = S | c_{-1}; \theta_{S_2^*}) = f_{S_2^*}(c_{-1})\theta_{S_2^*}$$

## A2.2: Multiyear (Matriculation) Parameterization 2

- Assume  $(x, t)$  are independent, and next year is independent of current year:

$$\mathbb{E}[v_1(x', t', c, \epsilon_1') | (m)] = \int_c \int_{t'} \int_{x'} \sigma \log \left( \sum_{d_1' \in \{T, R\}} e^{u_1(x', t', c, d_1') / \sigma} \right) dF_x(x') dF_t(t') dF_{c|(m)}(c | (m)) =$$

$$\int_c \int_{t_1'} \cdots \int_{t_{J_t}'} \int_{x_1'} \cdots \int_{x_{J_x}'} \left[ \cdot \right] dF_{x,1}(x_1' | x_2', \dots, x_{J_x}') \cdots dF_{x,J_x}(x_{J_x}') dF_{t,1}(t_1' | t_2', \dots, t_{J_t}') \cdots dF_{t,J_t}(t_{J_t}') dF_{c|(m)}(c | (m))$$

- Assume  $(z, m)$  are conditionally independent:

$$\mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2) | x, t, c_{-1}] = \int_{\tilde{m}} \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \tilde{m}, c_{-1}, \tilde{d}_2) / \sigma} \right) dF_{z|x,t}(\tilde{z} | x, t) dF_{m|x}(\tilde{m} | x, t) =$$

$$\int_{\tilde{m}} \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_{J_z}} \left[ \cdot \right] dF_{z|x,t,1}(\tilde{z}_1 | \tilde{z}_2, \dots, \tilde{z}_{J_z}, x, t) \cdots dF_{z|x,t,J_z}(\tilde{z}_{J_z} | x, t) dF_{m|x,t}(\tilde{m} | x, t)$$

## A2.2: Multiyear (Matriculation) Likelihood

$$\begin{aligned} \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_{m|x,t}(m_n|x_n, t_n; \theta_{F_{m|x,t}}) f_{c|m}(c_n|m_n; \theta_{F_{c|m}}) \\ & P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \\ & \prod_{n=N_T+1}^N f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_{m|x,t}(m_n|x_n, t_n; \theta_{F_{m|x,t}}) (1 - P_{d_{1,n}}(\theta_T)) \end{aligned}$$

- As with waitlist-dynamic model, for  $\mathcal{L}_R$ , use  $(\theta_F^U, \theta_S^U)$  as initial guess of  $(\theta_F^R, \theta_S^R)$

## A2.2: Multiyear Memory and Dynamics (Diversity)

- Let additional state  $a$  = any diversity variable affecting success
  - For example,  $a = \mathbb{1}(\text{TheoryField})$ ; theory is less common than applied
  - Let  $\bar{a}_{-1}$  = percentage of last year's acceptance set that's theory
  - If  $|a - \bar{a}_{-1}|$  is small,  $P(\text{Success})$  may decrease because future advisor may be too busy (effect may be unidentifiable if program never gets too imbalanced)

### Theorem 7 (Proof)

*Consider any two applicants, one with  $a = 1$  and one with  $a = 0$ , but with equal success probabilities and therefore different values of  $x$  or  $z$ . Suppose  $P(a' = 1) = p > 0$ . Then for all sufficiently small  $p$ , the  $a = 1$  applicant has a strictly greater Round 2 acceptance probability.*

- Optimal dynamic decision may not equal optimal myopic one. May accept diverse applicant with marginally lower  $P(\text{Success})$  to increase  $P(\text{Success})$  of next year's pool
  - Relative to  $a = 0$ ,  $a = 1 \Rightarrow$  higher  $\bar{F}_{\bar{a}|a} \Rightarrow$  higher  $\mathbb{E}[P(\text{Success})']$  and future EV



## A2.2: Multiyear Model (Diversity), Round 2

$$v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, a, \bar{a}_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, a, \bar{a}_{-1}, d_2) = \{P(y = S|x, z, t, |a - \bar{a}_{-1}|) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a]\} \mathbb{1}(d_2 = A) \\ + \{L(y_2^* = S|t) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)]\} \mathbb{1}(d_2 = R), \text{ such that:}$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a] = \int_{\tilde{\bar{a}}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log \left( \sum_{d'_1 \in \{T, R\}} e^{u_1(\tilde{x}', \tilde{\bar{a}}, \tilde{t}', d'_1)/\sigma} \right) dF_{x'}(\tilde{x}') dF_{t'}(\tilde{t}') dF_{\bar{a}|a}(\tilde{\bar{a}}|a)$$

### Conditional Choice Probabilities:

$$P(d_2|x, z, t, a, \bar{a}_{-1}) = \frac{\exp(u_2(x, z, t, a, \bar{a}_{-1}, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(x, z, t, a, \bar{a}_{-1}, \tilde{d}_2)/\sigma)}$$

## A2.2: Multiyear Model (Diversity), Round 1

$$v_1(x, t, \bar{a}_{-1}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{a}_{-1}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{a}_{-1}, d_1) = \mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) | x, t, \bar{a}_{-1}] \mathbb{1}(d_1 = T) \\ + \{L(y_1^* = S | t) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)]\} \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) | x, t, \bar{a}_{-1}] = \int_{\tilde{a}} \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in A, R} e^{u_2(x, \tilde{z}, t, \tilde{a}, \bar{a}_{-1}, \tilde{d}_2) / \sigma} \right) dF_{z|x, t}(\tilde{z} | x, t) dF_a(\tilde{a})$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)] = \int_{\tilde{a}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log \left( \sum_{d'_1 \in \{T, R\}} e^{u_1(\tilde{x}', \tilde{t}', \tilde{a}, d'_1) / \sigma} \right) dF_{x'}(\tilde{x}') dF_{t'}(\tilde{t}') dF_{\bar{a}}(\tilde{a})$$

**Conditional Choice Probabilities:**

$$P(d_1 | x, t, \bar{a}_{-1}) = \frac{\exp(u_1(x, t, \bar{a}_{-1}, d_1) / \sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(x, t, \bar{a}_{-1}, \tilde{d}_1) / \sigma)}$$

## A2.2: Multiyears (Diversity) Parameterization 1

### Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t, \bar{a}_{-1}; \theta_T) = \frac{\exp(f_T(x, t, \bar{a}_{-1})\theta_T)}{1 + \exp(f_T(x, t, \bar{a}_{-1})\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t, a, \bar{a}_{-1}; \theta_A) = \frac{\exp(f_A(x, z, t, a, \bar{a}_{-1})\theta_A)}{1 + \exp(f_A(x, z, t, a, \bar{a}_{-1})\theta_A)}$$

### Success Probability:

$$P_y = P(y = S | x, z, t, |a - \bar{a}_{-1}|; \theta_S) = \frac{\exp(f_S(x, z, t, |a - \bar{a}_{-1}|)\theta_S)}{1 + \exp(f_S(x, z, t, |a - \bar{a}_{-1}|)\theta_S)}$$

### Cutoff Levels:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = f_{S_1^*}(t)\theta_{S_1^*}$$

$$L_{y_2^*} = L(y_2^* = S | t; \theta_{S_2^*}) = f_{S_2^*}(t)\theta_{S_2^*}$$

## A2.2: Multiyear (Diversity) Parameterization 2

- Assume  $(x, t, \bar{a})$  mutually independent, such that  $\bar{a}|a$ :

$$\begin{aligned} \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a] &= \int_{\tilde{\bar{a}}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log \left( \sum_{d'_1 \in \{T, R\}} e^{u_1(\tilde{x}', \tilde{t}', \tilde{\bar{a}}, d'_1)/\sigma} \right) dF_{x'}(\tilde{x}') dF_{t'}(\tilde{t}') dF_{\bar{a}|a}(\tilde{\bar{a}}|a) = \\ &\int_{\tilde{\bar{a}}} \int_{\tilde{t}'_1} \cdots \int_{\tilde{t}'_{J_t}} \int_{\tilde{x}'_1} \cdots \int_{\tilde{x}'_{J_x}} \left[ \cdot \right] F_{x,1}(x'_1|x'_2, \dots, x'_{J_x}) \cdots dF_{x,J_x}(x'_{J_x}) dF_{t,1}(t'_1|t'_2, \dots, t'_{J_t}) \cdots dF_{t,J_t}(t'_{J_t}) dF_{\bar{a}|a}(\tilde{\bar{a}}|a) \end{aligned}$$

- Assume  $(z, a)$  independent, such that  $z|x$ :

$$\begin{aligned} \mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2)|x, t, \bar{a}_{-1}] &= \int_{\tilde{\bar{a}}} \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in A, R} e^{u_2(x, \tilde{z}, t, \tilde{\bar{a}}, \tilde{a}_{-1}, \tilde{d}_2)/\sigma} \right) dF_{z|x,t}(\tilde{z}|x, t) dF_a(\tilde{\bar{a}}) \\ &= \int_{\tilde{\bar{a}}} \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[ \cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t) dF_a(\tilde{\bar{a}}) \end{aligned}$$

## A2.2: Multiyear (Diversity) Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_a(a) f_{\bar{a}}(\bar{a}) f_{\bar{a}|a}(\bar{a}|a) P_{d_{2,n}}$$

$$P_{y_n}^{\mathbb{1}(y_n=S)} (1 - P_{y_n})^{\mathbb{1}(y_n=F)} \prod_{n=N_A+1}^{N_{T,R}} f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_a(a) f_{\bar{a}}(\bar{a}) (1 - P_{d_{2,n}})$$

$$P_{y_n}^{\mathbb{1}(y_n=S)} (1 - P_{y_n})^{\mathbb{1}(y_n=F)} \prod_{n=N_{T,R}+1}^N f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_a(a) f_{\bar{a}}(\bar{a}) (1 - P_{d_{1,n}})$$

- Estimate  $f_{\bar{a}|a}(\bar{a}|a)$  on  $N_A$  accepted applicants only
- As with waitlist-dynamic model, for  $\mathcal{L}_R$ , use  $(\theta_F^U, \theta_S^U)$  as initial guess of  $(\theta_F^R, \theta_S^R)$

# A3: Full Post-Lasso Logistic Regression Results

Table 2a: Post-Lasso Logistic Regression Results

| Dependent Variable = Accept     | Economics              | Government            | History               | Linguistics          | Psychology            |
|---------------------------------|------------------------|-----------------------|-----------------------|----------------------|-----------------------|
| Covid Year                      | -0.3960**<br>(0.1996)  | -0.4101<br>(0.2609)   | -0.5330<br>(0.3435)   | -0.5737<br>(0.3824)  |                       |
| Winsorized Age                  | -0.0977***<br>(0.0333) | 0.0473*<br>(0.0248)   | 0.0440<br>(0.0325)    |                      |                       |
| Female                          | 0.4109**<br>(0.1679)   | 0.6570***<br>(0.2281) | 0.6797**<br>(0.2832)  |                      | -0.3304<br>(0.4968)   |
| U.S. African/Hispanic/Native    | 0.4355<br>(0.5232)     | 0.9687***<br>(0.3521) | 1.1594***<br>(0.4187) |                      |                       |
| U.S. Asian                      | -0.5139<br>(0.4149)    | -0.5533<br>(0.4435)   |                       | 0.6970<br>(0.6906)   | 0.7934<br>(0.6164)    |
| International Asian             | -0.7960***<br>(0.2665) | -0.0979<br>(0.3808)   | 0.3796<br>(0.4458)    |                      |                       |
| International Other             | 0.5490*<br>(0.2838)    | 0.3749<br>(0.3238)    | 1.1102**<br>(0.4378)  | -0.2036<br>(0.4436)  |                       |
| GRE Verbal Percentile           | 0.0155***<br>(0.0059)  | 0.0335*<br>(0.0180)   | 0.0423**<br>(0.0188)  |                      | 0.0408***<br>(0.0185) |
| GRE Quantitative Percentile     | 0.1549***<br>(0.0184)  | 0.0156*<br>(0.0091)   | 0.0255**<br>(0.0106)  | 0.0157<br>(0.0120)   |                       |
| GRE Analytic Percentile         | 0.0177***<br>(0.0041)  | 0.0232**<br>(0.0101)  |                       | 0.0159<br>(0.0121)   |                       |
| Undergraduate GPA               | 2.2336***<br>(0.6626)  | 0.7026<br>(0.5465)    | 1.3004***<br>(0.5090) | 0.9096<br>(0.9856)   |                       |
| TOEFL/IELTS                     | 0.3900<br>(0.2878)     | 0.6701<br>(0.4545)    | 1.0010*<br>(0.5503)   | 0.8625*<br>(0.5186)  |                       |
| Elite U.S. University           | 0.8292***<br>(0.2312)  | 0.3801<br>(0.2728)    | 0.5964*<br>(0.3401)   |                      | 0.7688<br>(0.4721)    |
| Elite U.S. Liberal Arts College | 1.0529***<br>(0.3373)  | 0.6501*<br>(0.3553)   | 1.0178**<br>(0.5124)  |                      | 1.3415<br>(0.8540)    |
| Elite Foreign University        | 0.4772*<br>(0.2702)    |                       | 0.5710<br>(0.5770)    | 0.9748<br>(0.5983)   |                       |
| Graduate Degree                 |                        |                       | 0.2774<br>(0.3193)    |                      |                       |
| Work Experience                 | 0.3826**<br>(0.1881)   | 0.4139<br>(0.2987)    | 0.5055<br>(0.3290)    | 1.6031**<br>(0.6992) |                       |
| #Professor Recommenders         |                        |                       | 0.5752**<br>(0.2507)  | 0.4712*<br>(0.2697)  |                       |
| #Prolific Recommenders          | 0.2727***<br>(0.0840)  | 0.3028**<br>(0.1229)  | 0.2113<br>(0.1577)    | 0.3359**<br>(0.1456) | 0.4341<br>(0.2885)    |
| #Math Courses                   | 0.1715***<br>(0.6593)  |                       |                       |                      |                       |
| Advanced Math                   | 0.2350<br>(0.1777)     |                       |                       |                      |                       |
| Missing GRE                     | 0.1827<br>(0.7660)     |                       |                       | -0.0009<br>(0.3941)  | -0.8985<br>(0.5569)   |
| Missing TOEFL/IELTS             | -0.3983*<br>(0.2266)   | 0.5313<br>(0.4700)    | 0.5689<br>(0.5289)    |                      |                       |
| Concentration FE                | No                     | Yes                   | Yes                   | Yes                  | Yes                   |
| Observations                    | 2078                   | 2231                  | 690                   | 737                  | 486                   |
| Pseudo-R <sup>2</sup>           | 0.2671                 | 0.1500                | 0.1847                | 0.1722               | 0.1844                |
| CV Pseudo-R <sup>2</sup>        | 0.2168                 | 0.0895                | 0.0715                | 0.0104               | 0.0445                |

Robust standard errors in parentheses \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

## • Economics Positive ( $p < .05$ ):

Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Work Experience, #Prolific Recommenders, #Math Courses

## • Economics Negative ( $p < .05$ ):

Covid Year, Winsorized Age, International Asian

# A3: Full Post-Lasso Logistic Regression AMEs

Table 2b: Post-Lasso Logistic Regression AMEs

| Dependent Variable – Accept     | Economics              | Government            | History               | Linguistics          | Psychology           |
|---------------------------------|------------------------|-----------------------|-----------------------|----------------------|----------------------|
| Covid Year                      | -0.0303**<br>(0.0152)  | -0.0159<br>(0.0100)   | -0.0404<br>(0.0261)   | -0.0315<br>(0.0212)  |                      |
| Winsorized Age                  | -0.0075***<br>(0.0025) | 0.0018*<br>(0.0010)   | 0.0033<br>(0.0025)    |                      |                      |
| Female                          | 0.0314**<br>(0.0128)   | 0.0249***<br>(0.0087) | 0.0515**<br>(0.0220)  |                      | -0.0144<br>(0.0222)  |
| U.S. African/Hispanic/Native    | 0.0333<br>(0.0400)     | 0.0367***<br>(0.0136) | 0.0678***<br>(0.0318) |                      |                      |
| U.S. Asian                      | -0.0394<br>(0.0318)    | -0.0210<br>(0.0245)   |                       | 0.0383<br>(0.0382)   | 0.0357<br>(0.0270)   |
| International Asian             | -0.0610***<br>(0.0203) | -0.0037<br>(0.0144)   |                       | 0.0208<br>(0.0245)   |                      |
| International Other             | 0.0420*<br>(0.0217)    | 0.0142<br>(0.0123)    | 0.0841**<br>(0.0333)  | -0.0112<br>(0.0245)  |                      |
| GRE Verbal Percentile           | 0.0012***<br>(0.0005)  | 0.0013*<br>(0.0007)   | 0.0032**<br>(0.0015)  |                      | 0.0022**<br>(0.0009) |
| GRE Quantitative Percentile     | 0.0119***<br>(0.0014)  | 0.0006*<br>(0.0004)   | 0.0019**<br>(0.0008)  | 0.0009<br>(0.0007)   |                      |
| GRE Analytic Percentile         | 0.0014***<br>(0.0003)  | 0.0009**<br>(0.0004)  |                       | 0.0009<br>(0.0007)   |                      |
| Undergraduate GPA               | 0.1710***<br>(0.0506)  | 0.0267<br>(0.0208)    | 0.0985**<br>(0.0397)  | 0.0490<br>(0.0542)   |                      |
| TOEFL/IELTS                     | 0.0299<br>(0.0219)     | 0.0254<br>(0.0173)    | 0.0758*<br>(0.0420)   | 0.0474<br>(0.0290)   |                      |
| Elite U.S. University           | 0.0635***<br>(0.0176)  | 0.0144<br>(0.0104)    | 0.0452**<br>(0.0262)  |                      | 0.0346<br>(0.0214)   |
| Elite U.S. Liberal Arts College | 0.0806***<br>(0.0255)  | 0.0247*<br>(0.0136)   | 0.0771**<br>(0.0387)  |                      | 0.0604<br>(0.0398)   |
| Elite Foreign University        | 0.0365*<br>(0.0207)    |                       | 0.0432<br>(0.0434)    | 0.0535<br>(0.0331)   |                      |
| Graduate Degree                 |                        |                       | 0.0210<br>(0.0243)    |                      |                      |
| Work Experience                 | 0.0293**<br>(0.0143)   | 0.0157<br>(0.0113)    | 0.0383<br>(0.0250)    |                      | 0.0722**<br>(0.0335) |
| #Professor Recommenders         |                        |                       | 0.0436**<br>(0.0195)  | 0.0259*<br>(0.0147)  |                      |
| #Prolific Recommenders          | 0.0209***<br>(0.0064)  | 0.0115**<br>(0.0047)  | 0.0160<br>(0.0119)    | 0.0184**<br>(0.0080) | 0.0195<br>(0.0127)   |
| #Math Courses                   | 0.0131***<br>(0.0045)  |                       |                       |                      |                      |
| Advanced Math                   | 0.0180<br>(0.0136)     |                       |                       |                      |                      |
| Missing GRE                     | 0.0140<br>(0.0587)     |                       |                       | -0.0050<br>(0.0216)  | -0.0404<br>(0.0253)  |
| Missing TOEFL/IELTS             | -0.0395*<br>(0.0175)   | 0.0202<br>(0.0179)    | 0.0431<br>(0.0399)    |                      |                      |
| Concentration FE                | No                     | Yes                   | Yes                   | Yes                  | Yes                  |
| Observations                    | 2078                   | 2231                  | 690                   | 737                  | 486                  |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

## • Economics Positive ( $p < .05$ ):

Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Work Experience, #Prolific Recommenders, #Math Courses

## • Economics Negative ( $p < .05$ ):

Covid Year, Winsorized Age, International Asian

# A3: Means by Program

Table 3a: Means by Program

|                                 | Economics | Government | History | Linguistics | Psychology |
|---------------------------------|-----------|------------|---------|-------------|------------|
| Accept                          | 0.1121    | 0.0430     | 0.0986  | 0.0678      | 0.0535     |
| Covid Year                      | 0.2478    | 0.2792     | 0.3029  | 0.3460      | 0.3004     |
| Winsorized Age                  | 26.2782   | 27.9633    | 26.8478 | 28.0692     | 25.8230    |
| Female                          | 0.4201    | 0.4182     | 0.4232  | 0.6133      | 0.7490     |
| U.S. African/Hispanic/Native    | 0.0289    | 0.0766     | 0.1159  | 0.0665      | 0.1728     |
| U.S. Asian                      | 0.0313    | 0.0421     | 0.0304  | 0.0353      | 0.0576     |
| International Asian             | 0.5322    | 0.2510     | 0.1058  | 0.2714      | 0.1523     |
| International Other             | 0.2517    | 0.2967     | 0.2203  | 0.3256      | 0.1584     |
| GRE Verbal Percentile           | 71.8677   | 81.3486    | 82.6054 | 73.5451     | 75.6192    |
| GRE Quantitative Percentile     | 86.2163   | 63.3428    | 49.3295 | 57.8455     | 58.0000    |
| GRE Analytic Percentile         | 52.6351   | 72.1905    | 75.0460 | 63.6824     | 69.2077    |
| Undergraduate GPA               | 3.6534    | 3.5675     | 3.6531  | 3.6567      | 3.5695     |
| TOEFL/IELTS                     | 7.5232    | 7.5422     | 7.4252  | 7.5444      | 7.3158     |
| Elite U.S. University           | 0.1516    | 0.1847     | 0.2565  | 0.1954      | 0.2840     |
| Elite U.S. Liberal Arts College | 0.0423    | 0.0672     | 0.0696  | 0.0231      | 0.0535     |
| Elite Foreign University        | 0.0717    | 0.0578     | 0.0478  | 0.0488      | 0.0206     |
| Other Foreign University        | 0.5736    | 0.3953     | 0.2333  | 0.5102      | 0.1852     |
| Graduate Degree                 | 0.7478    | 0.7553     | 0.6652  | 0.7544      | 0.4259     |
| Work Experience                 | 0.5938    | 0.7463     | 0.6638  | 0.7544      | 0.7593     |
| #Professor Recommenders         | 2.5173    | 2.3469     | 2.6261  | 2.4355      | 2.1831     |
| #Prolific Recommenders          | 1.0760    | 0.6912     | 0.5043  | 0.7069      | 0.3128     |
| #Math Courses                   | 6.6396    |            |         |             |            |
| Advanced Math                   | 0.4827    |            |         |             |            |
| Missing GRE                     | 0.0255    | 0.4612     | 0.6217  | 0.6839      | 0.4650     |
| Missing Undergraduate GPA       | 0.6607    | 0.4827     | 0.3072  | 0.5726      | 0.2490     |
| Missing TOEFL/IELTS             | 0.5958    | 0.7714     | 0.8449  | 0.7707      | 0.8827     |
| Observations                    | 2078      | 2231       | 690     | 737         | 486        |

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS



# A3: Means by Program, Accept = 1

Table 3b: Means by Program, Accept = 1

|                                 | Economics | Government | History | Linguistics | Psychology |
|---------------------------------|-----------|------------|---------|-------------|------------|
| Covid Year                      | 0.1888    | 0.2188     | 0.2059  | 0.2200      | 0.2308     |
| Winsorized Age                  | 25.0815   | 27.7292    | 27.2794 | 27.1000     | 25.7692    |
| Female                          | 0.4850    | 0.5000     | 0.5588  | 0.5800      | 0.6154     |
| U.S. African/Hispanic/Native    | 0.0386    | 0.1458     | 0.1765  | 0.0400      | 0.0769     |
| U.S. Asian                      | 0.0429    | 0.0312     | 0.0147  | 0.0600      | 0.1154     |
| International Asian             | 0.4163    | 0.1667     | 0.0882  | 0.3600      | 0.0769     |
| International Other             | 0.2575    | 0.2292     | 0.3235  | 0.2000      | 0.1538     |
| GRE Verbal Percentile           | 84.3980   | 88.8725    | 86.6328 | 77.9161     | 85.7729    |
| GRE Quantitative Percentile     | 91.9289   | 68.3708    | 54.8113 | 64.5504     | 62.3462    |
| GRE Analytic Percentile         | 69.0355   | 82.7222    | 78.0879 | 68.8958     | 74.6169    |
| Undergraduate GPA               | 3.7336    | 3.6404     | 3.7204  | 3.7004      | 3.6341     |
| TOEFL/IELTS                     | 7.6419    | 7.6050     | 7.4730  | 7.6237      | 7.3178     |
| Elite U.S. University           | 0.2833    | 0.3125     | 0.3382  | 0.2200      | 0.5385     |
| Elite U.S. Liberal Arts College | 0.1030    | 0.1458     | 0.1176  | 0.0200      | 0.1154     |
| Elite Foreign University        | 0.1030    | 0.0521     | 0.0588  | 0.1000      | 0.0385     |
| Other Foreign University        | 0.3863    | 0.2396     | 0.2647  | 0.4000      | 0.1154     |
| Graduate Degree                 | 0.5837    | 0.6771     | 0.7206  | 0.7400      | 0.4615     |
| Work Experience                 | 0.6309    | 0.8125     | 0.7647  | 0.7000      | 0.9231     |
| #Professor Recommenders         | 2.5494    | 2.3750     | 2.8235  | 2.7200      | 2.2308     |
| #Prolific Recommenders          | 1.2833    | 0.9167     | 0.6765  | 1.0800      | 0.4615     |
| #Math Courses                   | 7.2575    |            |         |             |            |
| Advanced Math                   | 0.6781    |            |         |             |            |
| Missing GRE                     | 0.0086    | 0.2917     | 0.6324  | 0.5800      | 0.1923     |
| Missing Undergraduate GPA       | 0.4893    | 0.3125     | 0.3088  | 0.4800      | 0.1923     |
| Missing TOEFL/IELTS             | 0.6567    | 0.8854     | 0.8529  | 0.7600      | 0.8846     |
| Observations                    | 233       | 96         | 68      | 50          | 26         |

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

# A3: Correlation Matrix

Table 4: Correlation Matrix

|                                 | Covid | Age   | Fem   | USAHN | USAsn | IntAsn | IntOth | GREV  | GREQ  | GREa  | GPA   | T/I   | EUSUni | EUSLA | EFor  | OthFor | Grad  | Work  | Prof  | Pro   | MGRE  | MGPA  | MT/I |
|---------------------------------|-------|-------|-------|-------|-------|--------|--------|-------|-------|-------|-------|-------|--------|-------|-------|--------|-------|-------|-------|-------|-------|-------|------|
| Covid Year                      | 1.00  |       |       |       |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| Winsorized Age                  | 0.01  | 1.00  |       |       |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| Female                          | 0.02  | -0.13 | 1.00  |       |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| U.S. African/Hispanic/Native    | 0.00  | -0.04 | 0.04  | 1.00  |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| U.S. Asian                      | -0.00 | -0.06 | 0.05  | -0.05 | 1.00  |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| International Asian             | -0.05 | -0.07 | 0.02  | -0.19 | -0.14 | 1.00   |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| International Other             | 0.01  | 0.18  | -0.06 | -0.17 | -0.12 | -0.42  | 1.00   |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| GRE Verbal Percentile           | 0.03  | -0.06 | -0.04 | 0.00  | 0.04  | -0.12  | -0.15  | 1.00  |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| GRE Quantitative Percentile     | -0.02 | -0.16 | -0.10 | -0.18 | 0.00  | 0.44   | -0.10  | 0.16  | 1.00  |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| GRE Analytic Percentile         | 0.05  | -0.06 | 0.03  | 0.08  | 0.08  | -0.37  | -0.08  | 0.60  | -0.20 | 1.00  |       |       |        |       |       |        |       |       |       |       |       |       |      |
| Undergraduate GPA               | -0.02 | -0.29 | 0.07  | -0.13 | -0.02 | 0.02   | 0.00   | 0.06  | 0.12  | 0.05  | 1.00  |       |        |       |       |        |       |       |       |       |       |       |      |
| TOEFL/IELTS                     | -0.01 | -0.07 | 0.01  | -0.02 | 0.01  | 0.04   | 0.00   | 0.24  | 0.12  | 0.19  | 0.00  | 1.00  |        |       |       |        |       |       |       |       |       |       |      |
| Elite U.S. University           | 0.01  | -0.12 | 0.02  | 0.12  | 0.14  | -0.12  | -0.22  | 0.16  | -0.00 | 0.19  | -0.01 | -0.02 | 1.00   |       |       |        |       |       |       |       |       |       |      |
| Elite U.S. Liberal Arts College | -0.02 | -0.02 | 0.04  | 0.04  | 0.04  | -0.08  | -0.09  | 0.12  | -0.01 | 0.13  | -0.02 | -0.00 | -0.11  | 1.00  |       |        |       |       |       |       |       |       |      |
| Elite Foreign University        | -0.00 | -0.04 | -0.00 | -0.06 | -0.04 | 0.09   | 0.09   | 0.02  | 0.08  | -0.02 | 0.01  | 0.04  | -0.12  | -0.06 | 1.00  |        |       |       |       |       |       |       |      |
| Other Foreign University        | -0.02 | 0.20  | -0.04 | -0.23 | -0.14 | 0.34   | 0.39   | -0.26 | 0.20  | -0.39 | 0.02  | 0.02  | -0.43  | -0.21 | -0.22 | 1.00   |       |       |       |       |       |       |      |
| Graduate Degree                 | 0.00  | 0.41  | -0.08 | -0.13 | -0.09 | 0.21   | 0.15   | -0.14 | 0.05  | -0.20 | -0.22 | 0.03  | -0.21  | -0.10 | 0.05  | 0.37   | 1.00  |       |       |       |       |       |      |
| Work Experience                 | 0.01  | 0.33  | 0.05  | 0.05  | 0.01  | -0.18  | 0.11   | 0.01  | -0.15 | 0.11  | -0.08 | -0.03 | -0.02  | 0.03  | -0.03 | -0.01  | 0.10  | 1.00  |       |       |       |       |      |
| #Professor Recommenders         | -0.02 | -0.19 | -0.04 | -0.04 | 0.01  | 0.11   | -0.09  | 0.01  | 0.09  | -0.04 | 0.14  | 0.03  | -0.02  | 0.02  | -0.00 | -0.02  | -0.03 | -0.18 | 1.00  |       |       |       |      |
| #Prolific Recommenders          | 0.01  | -0.06 | -0.02 | -0.07 | -0.01 | 0.24   | -0.10  | -0.01 | 0.25  | -0.11 | 0.04  | 0.10  | -0.01  | -0.01 | 0.06  | 0.08   | 0.17  | -0.09 | 0.19  | 1.00  |       |       |      |
| Missing GRE                     | 0.09  | 0.15  | 0.05  | 0.11  | -0.01 | -0.22  | 0.17   | 0.09  | -0.33 | 0.17  | -0.09 | -0.10 | -0.03  | -0.04 | -0.03 | -0.01  | 0.02  | 0.10  | -0.08 | -0.20 | 1.00  |       |      |
| Missing Undergraduate GPA       | -0.01 | 0.18  | -0.04 | -0.24 | -0.15 | 0.36   | 0.42   | -0.24 | 0.21  | -0.38 | 0.02  | 0.04  | -0.46  | -0.23 | 0.23  | 0.83   | 0.37  | -0.02 | -0.03 | 0.10  | -0.01 | 1.00  |      |
| Missing TOEFL/IELTS             | 0.05  | -0.07 | 0.03  | 0.16  | 0.11  | -0.32  | -0.22  | 0.16  | -0.24 | 0.32  | -0.02 | -0.02 | 0.29   | 0.14  | -0.04 | -0.58  | -0.20 | 0.05  | -0.07 | -0.05 | 0.12  | -0.58 | 1.00 |
| Observations                    | 6222  |       |       |       |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |

# A3: Logistic Regression Results

Table 5a: Logistic Regression Results

| Dependent Variable = Accept  | Economics              | Government            | History               | Linguistics          | Psychology            |
|------------------------------|------------------------|-----------------------|-----------------------|----------------------|-----------------------|
| Covid Year                   | -0.3866**<br>(0.2010)  | -0.4332**<br>(0.2612) | -0.5752**<br>(0.3479) | -0.5501<br>(0.4120)  | -0.1766<br>(0.5545)   |
| Winsorized Age               | -0.0995***<br>(0.0356) | 0.0454*<br>(0.0271)   | 0.0380<br>(0.0327)    | 0.0222<br>(0.0532)   | -0.0520<br>(0.0899)   |
| Female                       | 0.4079**<br>(0.1681)   | 0.6702***<br>(0.2299) | 0.6539**<br>(0.2971)  | 0.1311<br>(0.3452)   | -0.3060<br>(0.5606)   |
| U.S. African/Hispanic/Native | 0.4221<br>(0.5258)     | 1.0081***<br>(0.3540) | 1.1601***<br>(0.4425) | 0.0120<br>(0.3411)   | -0.8263<br>(1.1611)   |
| U.S. Asian                   | -0.5394<br>(0.4199)    | -0.5660<br>(0.6504)   | -0.3260<br>(1.2512)   | 0.8638<br>(0.7201)   | 1.1711<br>(0.7276)    |
| International Asian          | -0.8445***<br>(0.2945) | -0.1853<br>(0.4146)   | 0.9405<br>(0.7213)    | 0.4890<br>(0.6729)   | -0.7827<br>(0.8233)   |
| International Other          | 0.5034<br>(0.3172)     | 0.3781<br>(0.4131)    | 1.3529***<br>(0.5086) | 0.1061<br>(0.7483)   | 0.6230<br>(1.1070)    |
| GRE Verbal Percentile        | 0.0135***<br>(0.0060)  | 0.0261*<br>(0.0160)   | 0.0753**<br>(0.0344)  | -0.0062<br>(0.0154)  | 0.0607***<br>(0.0218) |
| GRE Quantitative Percentile  | 0.1547***<br>(0.0186)  | 0.0161*<br>(0.0086)   | 0.0210*<br>(0.0116)   | 0.0211<br>(0.0150)   | -0.0138<br>(0.0159)   |
| GRE Analytic Percentile      | 0.0179***<br>(0.0042)  | 0.0596**<br>(0.0044)  | 0.0030<br>(0.0134)    | 0.0175<br>(0.0127)   | -0.0202<br>(0.0218)   |
| Undergraduate GPA            | 2.2212***<br>(0.7219)  | 0.6968<br>(0.5476)    | 1.3648**<br>(0.5513)  | 1.2645<br>(0.9703)   | 1.7772*<br>(0.9487)   |
| TOEFL/IELTS                  | 0.3878<br>(0.2883)     | 0.6466<br>(0.4708)    | 1.0653**<br>(0.5502)  | 0.9650*<br>(0.5466)  | -0.4656<br>(0.6122)   |
| Elite U.S. University        | 0.8850***<br>(0.2960)  | 0.3943<br>(0.3135)    | 0.6541*<br>(0.3862)   | -0.6057<br>(0.4985)  | 1.4303**<br>(0.6192)  |
| Elite U.S. Liberal Arts      | 1.1173***<br>(0.3513)  | 0.6279*<br>(0.3754)   | 1.8608***<br>(0.5399) | -0.6086<br>(1.4186)  | 2.3161***<br>(1.0642) |
| Elite Foreign University     | 1.0061*<br>(0.6060)    | 0.4759<br>(0.2224)    | 1.7715**<br>(0.7938)  | 1.6195*<br>(0.8587)  | 0.3889<br>(1.9889)    |
| Other Foreign University     | 0.5442<br>(0.5765)     | 0.5777<br>(0.7686)    | 1.9733**<br>(0.6902)  | 0.4168<br>(0.7157)   | -1.0737<br>(1.3016)   |
| Graduate Degree              | 0.0467<br>(0.2238)     | 0.0733<br>(0.2581)    | 0.3128<br>(0.3280)    | 0.4049<br>(0.4896)   | 0.4892<br>(0.6399)    |
| Work Experience              | 0.3952***<br>(0.1889)  | 0.4027<br>(0.2963)    | 0.4802<br>(0.3295)    | 0.4801*<br>(0.3778)  | 0.8261**<br>(0.8564)  |
| #Professor Recommenders      | 0.0340<br>(0.1317)     | -0.0179<br>(0.1339)   | 0.9511**<br>(0.2554)  | 0.4781*<br>(0.2662)  | -0.0981<br>(0.2660)   |
| #Public Recommenders         | 0.2630***<br>(0.0854)  | 0.3825***<br>(0.1262) | 0.1880<br>(0.1576)    | 0.3281**<br>(0.1544) | 0.6601*<br>(0.3145)   |
| #Math Courses                | 0.1689***<br>(0.0596)  |                       |                       |                      |                       |
| Advanced Math                | 0.2319<br>(0.1790)     |                       |                       |                      |                       |
| Missing GRE                  | 0.1889<br>(0.7646)     | -0.3259<br>(0.3100)   | 0.6282<br>(0.4275)    | -0.0831<br>(0.3761)  | -1.1514**<br>(0.5702) |
| Missing Undergraduate GPA    | -0.4745<br>(0.4891)    | -0.4017<br>(0.5440)   | -1.5402**<br>(0.6803) | -1.1091*<br>(0.6481) | 0.4690<br>(1.1831)    |
| Missing TOEFL/IELTS          | -0.3706<br>(0.2452)    | 0.6189<br>(0.5657)    | 0.6745<br>(0.5190)    | -0.0440<br>(0.5099)  | -1.7224*<br>(0.9271)  |
| Concentration FE             | No                     | Yes                   | Yes                   | Yes                  | Yes                   |
| Observations                 | 2078                   | 2231                  | 660                   | 737                  | 486                   |
| Pseudo R <sup>2</sup>        | 0.2877                 | 0.1525                | 0.1982                | 0.1871               | 0.2475                |
| CV Pseudo R <sup>2</sup>     | 0.2302                 | 0.0720                | 0.0604                | -0.0265              | -0.2787               |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- **Economics Positive ( $p < .05$ ):**  
Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ( $p < .05$ ):**  
Covid Year, Winsorized Age, International Asian

# A3: Logistic Regression AMEs

Table 5b: Logistic Regression AMEs

| Dependent Variable = Accept     | Economics              | Government            | History               | Linguistics          | Psychology            | Non-Economics          |
|---------------------------------|------------------------|-----------------------|-----------------------|----------------------|-----------------------|------------------------|
| Covid Year                      | -0.0303**<br>(0.0153)  | -0.0164<br>(0.0300)   | -0.0431**<br>(0.0261) | -0.0298<br>(0.0224)  | -0.0075<br>(0.0238)   | -0.0218***<br>(0.0084) |
| Winsorized Age                  | -0.0076***<br>(0.0027) | 0.0017**<br>(0.0010)  | 0.0028<br>(0.0025)    | 0.0012<br>(0.0020)   | -0.0022<br>(0.0038)   | 0.0016*<br>(0.0009)    |
| Female                          | 0.0312***<br>(0.0129)  | 0.0254***<br>(0.0088) | 0.0400***<br>(0.0220) | 0.0071<br>(0.0187)   | -0.0216<br>(0.0239)   | 0.0201***<br>(0.0075)  |
| U.S. African/Hispanic/Native    | 0.0323<br>(0.0402)     | 0.0381***<br>(0.0137) | 0.0869***<br>(0.0332) | 0.0007<br>(0.0455)   | -0.0352<br>(0.0487)   | 0.0384***<br>(0.0119)  |
| U.S. Asian                      | -0.0413<br>(0.0322)    | -0.0215<br>(0.0247)   | -0.0244<br>(0.0936)   | 0.0468<br>(0.0391)   | 0.0499<br>(0.0313)    | 0.0052<br>(0.0185)     |
| International Asian             | -0.0646***<br>(0.0224) | -0.0070<br>(0.0157)   | 0.0704<br>(0.0543)    | 0.0265<br>(0.0363)   | -0.0334<br>(0.0393)   | 0.0069<br>(0.0132)     |
| International Other             | 0.0385<br>(0.0243)     | 0.0143<br>(0.0157)    | 0.0113***<br>(0.0385) | -0.0057<br>(0.0406)  | 0.0266<br>(0.0471)    | 0.0288*<br>(0.0132)    |
| GRE Verbal Percentile           | 0.0012***<br>(0.0005)  | 0.0011*<br>(0.0006)   | 0.0056**<br>(0.0025)  | -0.0005<br>(0.0008)  | 0.0026***<br>(0.0010) | 0.0015***<br>(0.0005)  |
| GRE Quantitative Percentile     | 0.0118***<br>(0.0014)  | 0.0066*<br>(0.0003)   | 0.0016*<br>(0.0009)   | 0.0011<br>(0.0007)   | -0.0006<br>(0.0007)   | 0.0005**<br>(0.0003)   |
| GRE Analytic Percentile         | 0.0014***<br>(0.0003)  | 0.0008***<br>(0.0004) | 0.0002<br>(0.0010)    | -0.0009<br>(0.0007)  | -0.0009<br>(0.0009)   | 0.0007**<br>(0.0003)   |
| Undergraduate GPA               | 0.1699***<br>(0.0551)  | 0.0264<br>(0.0308)    | 0.0222**<br>(0.0423)  | 0.0984<br>(0.0522)   | 0.0757**<br>(0.0418)  | 0.0468***<br>(0.0176)  |
| TOEFL/IELTS                     | 0.0207<br>(0.0219)     | 0.0044<br>(0.0179)    | 0.0821**<br>(0.0417)  | 0.0523*<br>(0.0301)  | -0.0198<br>(0.0259)   | 0.0326***<br>(0.0123)  |
| Elite U.S. University           | 0.0677***<br>(0.0225)  | 0.0149<br>(0.0119)    | 0.0401**<br>(0.0281)  | 0.0610**<br>(0.0264) | 0.0233***<br>(0.0271) | 0.0409**<br>(0.0098)   |
| Elite U.S. Liberal Arts College | 0.0859***<br>(0.0288)  | 0.0238*<br>(0.0143)   | 0.0801**<br>(0.0402)  | -0.0130<br>(0.0761)  | 0.0667**<br>(0.0480)  | 0.0420***<br>(0.0136)  |
| Elite Foreign University        | 0.0776**<br>(0.0311)   | 0.0700*<br>(0.0350)   | 0.0807**<br>(0.0591)  | 0.0166<br>(0.0481)   | 0.0470***<br>(0.0850) | 0.0409***<br>(0.0241)  |
| Other Foreign University        | 0.0416<br>(0.0441)     | 0.0219<br>(0.0291)    | 0.1028*<br>(0.0527)   | 0.0227<br>(0.0391)   | -0.0458<br>(0.0556)   | 0.0273<br>(0.0221)     |
| Graduate Degree                 | 0.0036<br>(0.0171)     | 0.0028<br>(0.0098)    | 0.0234<br>(0.0247)    | 0.0219<br>(0.0262)   | 0.0208<br>(0.0273)    | 0.0106<br>(0.0089)     |
| Work Experience                 | 0.0301**<br>(0.0144)   | 0.0152<br>(0.0113)    | 0.0372<br>(0.0248)    | 0.0081<br>(0.0205)   | 0.0776**<br>(0.0368)  | 0.0221**<br>(0.0089)   |
| #Professor Recommenders         | 0.0025<br>(0.0101)     | -0.0007<br>(0.0051)   | 0.0209*<br>(0.0190)   | -0.0042<br>(0.0142)  | 0.0263<br>(0.0113)    | 0.0063<br>(0.0047)     |
| #Prolific Recommenders          | 0.0202***<br>(0.0065)  | 0.0114***<br>(0.0048) | 0.0141<br>(0.0118)    | 0.0178**<br>(0.0082) | 0.0251*<br>(0.0130)   | 0.0133***<br>(0.0038)  |
| #Math Courses                   | 0.0129***<br>(0.0045)  |                       |                       |                      |                       |                        |
| Advanced Math                   | 0.0177<br>(0.0137)     |                       |                       |                      |                       |                        |
| Missing GRE                     | 0.0345<br>(0.0585)     | -0.0123<br>(0.0118)   | 0.0470<br>(0.0319)    | -0.0045<br>(0.0204)  | -0.0401*<br>(0.0252)  | -0.0062<br>(0.0088)    |
| Missing Undergraduate GPA       | -0.0363<br>(0.0375)    | -0.0186<br>(0.0206)   | -0.1160**<br>(0.0524) | -0.0601*<br>(0.0355) | 0.0200<br>(0.0605)    | -0.0291<br>(0.0179)    |
| Missing TOEFL/IELTS             | -0.0290<br>(0.0188)    | 0.0234<br>(0.0215)    | 0.0505<br>(0.0388)    | -0.0034<br>(0.0276)  | -0.0734*<br>(0.0417)  | 0.0108<br>(0.0135)     |
| Concentration FE                | No                     | Yes                   | Yes                   | Yes                  | Yes                   | Yes                    |
| Observations                    | 2078                   | 2231                  | 690                   | 737                  | 486                   | 4144                   |
| Wald Statistic / DF / P-Value   | 115.8637               | 23                    | 0.0000                |                      |                       |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- Economics Positive ( $p < .05$ ):**  
 Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Work Experience, #Prolific Recommenders, #Math Courses
- Economics Negative ( $p < .05$ ):**  
 Covid Year, Winsorized Age, International Asian

# A3: Logistic Regression Results (Synthetic)

Table 6a: Logistic Regression Results (Synthetic)

| Dependent Variable = Accept     | Economics             | Government           | History              | Linguistics            | Psychology             |
|---------------------------------|-----------------------|----------------------|----------------------|------------------------|------------------------|
| Covid Year                      | -0.1907<br>(0.1633)   | -0.0323<br>(0.2215)  | 0.2028<br>(0.3078)   | -0.4761<br>(0.3477)    | -0.5946<br>(0.4520)    |
| Winsorized Age                  | -0.0506**<br>(0.0201) | -0.0198<br>(0.0174)  | 0.0576*<br>(0.0307)  | -0.0164<br>(0.0357)    | 0.1203***<br>(0.0400)  |
| Female                          | 0.1941<br>(0.1402)    | 0.0023<br>(0.2004)   | 0.1389<br>(0.2526)   | 0.2596<br>(0.3255)     | 0.2119<br>(0.4310)     |
| U.S. African/Hispanic/Native    | 0.3053<br>(0.3724)    | -0.1457<br>(0.4303)  | -0.2379<br>(0.4670)  | -1.6401<br>(1.0720)    | 0.5413<br>(0.5306)     |
| U.S. Asian                      | 0.5663<br>(0.3642)    | 0.6016<br>(0.4452)   | -0.7934<br>(1.1675)  | -0.4505<br>(0.7936)    | 0.7122<br>(0.8863)     |
| International Asian             | -0.1428<br>(0.2169)   | 0.1630<br>(0.2803)   | 0.5780<br>(0.3740)   | -0.4965<br>(0.4215)    | 0.1868<br>(0.5860)     |
| International Other             | -0.1231<br>(0.2311)   | 0.1395<br>(0.3046)   | 0.1097<br>(0.3237)   | -0.1602<br>(0.3832)    | -1.0807<br>(0.7583)    |
| GRE Verbal Percentile           | 0.0167***<br>(0.0040) | 0.0247**<br>(0.0117) | 0.0154<br>(0.0237)   | -0.0029<br>(0.0100)    | 0.0132<br>(0.0184)     |
| GRE Quantitative Percentile     | 0.0233***<br>(0.0060) | -0.0027<br>(0.0051)  | 0.0026<br>(0.0086)   | 0.0304**<br>(0.0137)   | 0.0097<br>(0.0090)     |
| GRE Analytic Percentile         | 0.0042<br>(0.0031)    | 0.0004<br>(0.0056)   | -0.0073<br>(0.0060)  | -0.0042<br>(0.0101)    | 0.0069<br>(0.0103)     |
| Undergraduate GPA               | 0.2262<br>(0.4745)    | 0.6327<br>(0.4985)   | 1.3552**<br>(0.5796) | -1.3171**<br>(0.5594)  | -0.1002<br>(0.6152)    |
| TOEFL/IELTS                     | -0.4613**<br>(0.2157) | 0.5141<br>(0.4185)   | -0.7648<br>(0.6814)  | 0.2446<br>(0.7480)     | -0.7354<br>(0.8025)    |
| Elite U.S. University           | -0.4850*<br>(0.2560)  | -0.2243<br>(0.3258)  | 0.0572<br>(0.3421)   | 0.4764<br>(0.4788)     | 0.3941<br>(0.4782)     |
| Elite U.S. Liberal Arts College | 0.2429<br>(0.3049)    | 0.2355<br>(0.4042)   | 0.4860<br>(0.4964)   | 0.9652<br>(0.9631)     | 0.4860<br>(0.7361)     |
| Elite Foreign University        | -0.1622<br>(0.3015)   | -1.1029<br>(0.7484)  | 0.7404<br>(0.5686)   | 0.4801<br>(0.7841)     | 3.0530***<br>(0.9673)  |
| Other Foreign University        | -0.1067<br>(0.1679)   | 0.2043<br>(0.2732)   | 0.1721<br>(0.3391)   | 0.2723<br>(0.4292)     | 0.1731<br>(0.5926)     |
| Graduate Degree                 | -0.0198<br>(0.1824)   | 0.1770<br>(0.2439)   | -0.1601<br>(0.2765)  | -0.8698***<br>(0.3320) | -0.2769<br>(0.3814)    |
| Work Experience                 | 0.2525*<br>(0.1441)   | -0.0396<br>(0.2396)  | -0.2850<br>(0.2629)  | -0.1936**<br>(0.3417)  | -0.1936**<br>(0.4805)  |
| #Professor Recommenders         | -0.1059<br>(0.0609)   | 0.0911<br>(0.1138)   | -0.1828<br>(0.1473)  | 0.2181<br>(0.1817)     | -0.4023**<br>(0.1965)  |
| #Profic Recommenders            | 0.0009<br>(0.0656)    | 0.2280**<br>(0.1108) | -0.1743<br>(0.1723)  | 0.0673<br>(0.1642)     | 0.5314**<br>(0.2213)   |
| #Math Courses                   | 0.0366<br>(0.0421)    |                      |                      |                        |                        |
| Advanced Math                   | 0.1961<br>(0.1416)    |                      |                      |                        |                        |
| Missing GRE                     | -0.1963<br>(0.4328)   | -0.2029<br>(0.2195)  | -0.0031<br>(0.2686)  | -0.0433<br>(0.3503)    | -0.6040<br>(0.4017)    |
| Missing Undergraduate GPA       | 0.1531<br>(0.1548)    | 0.0895<br>(0.2291)   | -0.2455<br>(0.3015)  | 0.6298**<br>(0.3125)   | -0.3574<br>(0.4342)    |
| Missing TOEFL/IELTS             | -0.2950*<br>(0.1640)  | 0.4898<br>(0.3480)   | 0.5752<br>(0.3466)   | 0.1346<br>(0.4346)     | -2.0781***<br>(0.7033) |
| Concentration FE                | No                    | Yes                  | Yes                  | Yes                    | Yes                    |
| Observations                    | 2078                  | 2231                 | 690                  | 737                    | 486                    |
| Pseudo-R <sup>2</sup>           | 0.0524                | 0.0396               | 0.0753               | 0.1046                 | 0.2230                 |
| CV Pseudo-R <sup>2</sup>        | 0.0139                | -0.0189              | -0.0867              | 0.0055                 | -0.0014                |

Hobart standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- **Economics Positive ( $p < .05$ ):**  
GRE Verbal Percentile, GRE Quantitative Percentile
- **Economics Negative ( $p < .05$ ):**  
Winsorized Age, TOEFL/IELTS

# A3: Logistic Regression AMEs (Synthetic)

Table 6b: Logistic Regression AMEs (Synthetic)

| Dependent Variable = Accept     | Economics             | Government            | History              | Linguistics            | Psychology             | Non-Economics         |
|---------------------------------|-----------------------|-----------------------|----------------------|------------------------|------------------------|-----------------------|
| Covid Year                      | -0.0204<br>(0.0167)   | -0.0014<br>(0.0090)   | 0.0192<br>(0.0291)   | -0.0298<br>(0.0216)    | -0.0345<br>(0.0262)    | -0.0089<br>(0.0086)   |
| Winsorized Age                  | -0.0524**<br>(0.0021) | -0.0009<br>(0.0008)   | 0.0054*<br>(0.0029)  | -0.0010<br>(0.0022)    | 0.0070***<br>(0.0023)  | 0.0004<br>(0.0008)    |
| Female                          | 0.0198<br>(0.0143)    | 0.0041<br>(0.0089)    | 0.0131<br>(0.0239)   | 0.0162<br>(0.0204)     | 0.0123<br>(0.0251)     | 0.0099<br>(0.0078)    |
| U.S. African/Hispanic/Native    | 0.0312<br>(0.0381)    | -0.0065<br>(0.0192)   | -0.0225<br>(0.0441)  | -0.1021<br>(0.0673)    | 0.0314<br>(0.0306)     | -0.0137<br>(0.0147)   |
| U.S. Asian                      | 0.0579<br>(0.0372)    | 0.0266<br>(0.0199)    | -0.0750<br>(0.1101)  | -0.0266<br>(0.0495)    | 0.0413<br>(0.0516)     | 0.0241<br>(0.0179)    |
| International Asian             | -0.0146<br>(0.0222)   | 0.0073<br>(0.0125)    | 0.0546<br>(0.0353)   | -0.0309<br>(0.0263)    | 0.0108<br>(0.0399)     | 0.0075<br>(0.0112)    |
| International Other             | -0.0126<br>(0.0236)   | 0.0062<br>(0.0136)    | 0.0104<br>(0.0306)   | -0.0100<br>(0.0237)    | -0.0627<br>(0.0443)    | -0.0014<br>(0.0166)   |
| GRE Verbal Percentile           | 0.0017***<br>(0.0005) | 0.0011***<br>(0.0005) | 0.0015<br>(0.0012)   | -0.0002<br>(0.0006)    | 0.0008<br>(0.0010)     | 0.0008***<br>(0.0004) |
| GRE Quantitative Percentile     | 0.0024***<br>(0.0009) | -0.0001<br>(0.0002)   | 0.0003<br>(0.0008)   | 0.0019***<br>(0.0009)  | 0.0006<br>(0.0006)     | 0.0003<br>(0.0002)    |
| GRE Analytic Percentile         | 0.0004<br>(0.0003)    | 0.0000<br>(0.0003)    | -0.0007<br>(0.0009)  | -0.0003<br>(0.0006)    | 0.0004<br>(0.0006)     | 0.0000<br>(0.0002)    |
| Undergraduate GPA               | 0.0234<br>(0.0485)    | 0.0282<br>(0.0223)    | 0.1281**<br>(0.0553) | -0.0820**<br>(0.0358)  | -0.0058<br>(0.0357)    | 0.0130<br>(0.0158)    |
| TOEFL/IELTS                     | -0.0478**<br>(0.0221) | 0.0229<br>(0.0187)    | -0.0723<br>(0.0645)  | 0.0152<br>(0.0466)     | -0.0427<br>(0.0469)    | 0.0050<br>(0.0187)    |
| Elite U.S. University           | -0.0496*<br>(0.0261)  | -0.0100<br>(0.0145)   | 0.0054<br>(0.0324)   | 0.0297<br>(0.0297)     | 0.0229<br>(0.0274)     | 0.0036<br>(0.0108)    |
| Elite U.S. Liberal Arts College | 0.0248<br>(0.0312)    | 0.0078<br>(0.0180)    | 0.0223<br>(0.0469)   | 0.0303<br>(0.0602)     | 0.0502<br>(0.0418)     | 0.0217<br>(0.0160)    |
| Elite Foreign University        | -0.0166<br>(0.0308)   | -0.0532<br>(0.0337)   | 0.0700<br>(0.0537)   | 0.1771***<br>(0.0490)  | 0.0158<br>(0.0484)     | 0.0158<br>(0.0179)    |
| Other Foreign University        | -0.0111<br>(0.0202)   | 0.0091<br>(0.0122)    | 0.0257<br>(0.0321)   | 0.0108<br>(0.0267)     | 0.0175<br>(0.0344)     | 0.0116<br>(0.0103)    |
| Graduate Degree                 | -0.0020<br>(0.0166)   | 0.0079<br>(0.0109)    | -0.0151<br>(0.0261)  | -0.0542***<br>(0.0206) | -0.0161<br>(0.0220)    | -0.0093<br>(0.0085)   |
| Work Experience                 | 0.0258*<br>(0.0148)   | -0.0017<br>(0.0107)   | -0.0070<br>(0.0248)  | -0.0165<br>(0.0213)    | -0.0692**<br>(0.0275)  | -0.0093<br>(0.0086)   |
| #Professor Recommenders         | -0.0108<br>(0.0093)   | 0.0041<br>(0.0051)    | -0.0173<br>(0.0140)  | 0.0136<br>(0.0114)     | -0.0286***<br>(0.0110) | -0.0027<br>(0.0042)   |
| #Profic Recommenders            | 0.0004<br>(0.0067)    | 0.0098*<br>(0.0050)   | -0.0165<br>(0.0164)  | 0.0042<br>(0.0102)     | 0.0268**<br>(0.0128)   | 0.0083*<br>(0.0045)   |
| #Math Courses                   | 0.0037<br>(0.0043)    |                       |                      |                        |                        |                       |
| Advanced Math                   | 0.0201<br>(0.0145)    |                       |                      |                        |                        |                       |
| Missing GRE                     | -0.0204<br>(0.0443)   | -0.0090<br>(0.0098)   | -0.0001<br>(0.0254)  | -0.0027<br>(0.0218)    | -0.0350<br>(0.0239)    | -0.0098<br>(0.0079)   |
| Missing Undergraduate GPA       | 0.0157<br>(0.0158)    | 0.0040<br>(0.0102)    | -0.0232<br>(0.0285)  | 0.0392*<br>(0.0201)    | -0.0207<br>(0.0254)    | 0.0019<br>(0.0083)    |
| Missing TOEFL/IELTS             | -0.0302*<br>(0.0168)  | 0.0218<br>(0.0156)    | 0.0544<br>(0.0517)   | 0.0084<br>(0.0272)     | -0.1205***<br>(0.0434) | 0.0063<br>(0.0130)    |
| Concentration FE                | No                    | Yes                   | Yes                  | Yes                    | Yes                    | Yes                   |
| Observations                    | 2078                  | 2231                  | 690                  | 737                    | 486                    | 4144                  |
| Wald Statistic   DF   P-Value   | 37.4658               | 23                    | 0.0290               |                        |                        |                       |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- **Economics Positive ( $p < .05$ ):**  
GRE Verbal Percentile, GRE Quantitative Percentile
- **Economics Negative ( $p < .05$ ):**  
Winsorized Age, TOEFL/IELTS

# A3: Lasso Logistic Regression Results

Table 7: Lasso Logistic Regression Results

| Dependent Variable = Accept     | Economics | Government | History | Linguistics | Psychology |
|---------------------------------|-----------|------------|---------|-------------|------------|
| Covid Year                      | -0.3330   | -0.3123    | -0.3498 | -0.2378     |            |
| Winsorized Age                  | -0.0788   | 0.0266     | 0.0200  |             |            |
| Female                          | 0.3352    | 0.5096     | 0.5068  |             | -0.0654    |
| U.S. African/Hispanic/Native    | 0.2638    | 0.8041     | 0.7157  |             |            |
| U.S. Asian                      | -0.2802   | -0.2918    |         | 0.0413      | 0.1155     |
| International Asian             | -0.6744   | -0.0726    |         | 0.0465      |            |
| International Other             | 0.4729    | 0.1946     | 0.7240  | -0.1087     |            |
| GRE Verbal Percentile           | 0.0139    | 0.0252     | 0.0272  |             | 0.0343     |
| GRE Quantitative Percentile     | 0.1349    | 0.0119     | 0.0187  | 0.0154      |            |
| GRE Analytic Percentile         | 0.0166    | 0.0208     |         | 0.0080      |            |
| Undergraduate GPA               | 1.9977    | 0.4621     | 0.7414  | 0.1714      |            |
| TOEFL/IELTS                     | 0.4009    | 0.3295     | 0.5199  | 0.5151      |            |
| Elite U.S. University           | 0.6886    | 0.2709     | 0.3267  |             | 0.4606     |
| Elite U.S. Liberal Arts College | 0.8956    | 0.5199     | 0.5985  |             | 0.4470     |
| Elite Foreign University        | 0.3724    |            | 0.0414  | 0.4343      |            |
| Other Foreign University        |           |            |         |             |            |
| Graduate Degree                 |           |            | 0.0978  |             |            |
| Work Experience                 | 0.2835    | 0.3447     | 0.3273  |             | 0.5559     |
| #Professor Recommenders         |           |            | 0.3739  | 0.2322      |            |
| #Prolific Recommenders          | 0.2243    | 0.2362     | 0.1651  | 0.2213      | 0.0632     |
| #Math Courses                   | 0.1493    |            |         |             |            |
| Advanced Math                   | 0.2272    |            |         |             |            |
| Missing GRE                     |           | -0.2304    |         | -0.0527     | -0.5416    |
| Missing Undergraduate GPA       |           |            |         |             |            |
| Missing TOEFL/IELTS             | -0.2364   | 0.2852     | 0.0796  |             |            |
| Concentration FE                | No        | Yes        | Yes     | Yes         | Yes        |
| Observations                    | 2078      | 2231       | 690     | 737         | 486        |
| Pseudo-R <sup>2</sup>           | 0.2642    | 0.1456     | 0.1589  | 0.1486      | 0.1362     |
| CV Pseudo-R <sup>2</sup>        | 0.2297    | 0.0912     | 0.0592  | 0.0850      | 0.0537     |

# A3: Elastic-Net Logistic Regression Results

Table 8: Elastic-Net Logistic Regression Results

| Dependent Variable = Accept     | Economics | Government | History | Linguistics | Psychology |
|---------------------------------|-----------|------------|---------|-------------|------------|
| Covid Year                      | -0.3323   | -0.3013    | -0.3652 | -0.2642     | -0.0428    |
| Winsorized Age                  | -0.0786   | 0.0231     | 0.0235  |             |            |
| Female                          | 0.3316    | 0.4214     | 0.4373  |             | -0.2716    |
| U.S. African/Hispanic/Native    | 0.2622    | 0.7009     | 0.6038  |             | -0.2580    |
| U.S. Asian                      | -0.2741   | -0.4137    | -0.3622 | 0.1731      | 0.4670     |
| International Asian             | -0.6634   | -0.1850    | 0.1309  | 0.1086      | -0.2639    |
| International Other             | 0.4732    | 0.1832     | 0.5739  | -0.1280     | 0.0824     |
| GRE Verbal Percentile           | 0.0140    | 0.0187     | 0.0188  |             | 0.0236     |
| GRE Quantitative Percentile     | 0.1324    | 0.0117     | 0.0166  | 0.0136      |            |
| GRE Analytic Percentile         | 0.0165    | 0.0157     | 0.0055  | 0.0089      |            |
| Undergraduate GPA               | 1.9971    | 0.5294     | 0.7165  | 0.3104      | 0.3833     |
| TOEFL/IELTS                     | 0.4029    | 0.4179     | 0.6717  | 0.5298      |            |
| Elite U.S. University           | 0.6894    | 0.3109     | 0.3740  |             | 0.5669     |
| Elite U.S. Liberal Arts College | 0.8906    | 0.5376     | 0.6277  |             | 0.7887     |
| Elite Foreign University        | 0.3738    | 0.0332     | 0.3675  | 0.5020      | 0.3558     |
| Other Foreign University        |           | 0.0233     | 0.1382  |             |            |
| Graduate Degree                 |           | 0.0315     | 0.1873  |             | 0.1018     |
| Work Experience                 | 0.2822    | 0.3230     | 0.3192  |             | 0.5851     |
| #Professor Recommenders         |           | -0.0179    | 0.3250  | 0.2333      |            |
| #Prolific Recommenders          | 0.2238    | 0.2164     | 0.1665  | 0.2212      | 0.1970     |
| #Math Courses                   | 0.1489    |            |         |             |            |
| Advanced Math                   | 0.2309    |            |         |             |            |
| Missing GRE                     |           | -0.3089    | 0.1123  | -0.0864     | -0.5616    |
| Missing Undergraduate GPA       |           | -0.1040    | -0.1046 |             |            |
| Missing TOEFL/IELTS             | -0.2369   | 0.2950     | 0.2603  |             | -0.1782    |
| Concentration FE                | No        | Yes        | Yes     | Yes         | Yes        |
| Observations                    | 2078      | 2231       | 690     | 737         | 486        |
| Pseudo-R <sup>2</sup>           | 0.2640    | 0.1422     | 0.1626  | 0.1519      | 0.1653     |
| CV Pseudo-R <sup>2</sup>        | 0.2297    | 0.0999     | 0.0786  | 0.0859      | 0.0582     |
| $\alpha^*$                      | 0.9       | 0.0        | 0.0     | 0.4         | 0.1        |



# A3: Economics Means (Matriculation)

Table 9: Economics Means (Matriculation)

| Mean                        | All Observations | Matriculate = 1 |
|-----------------------------|------------------|-----------------|
| Matriculate                 | 0.2814           |                 |
| Covid Year                  | 0.1905           | 0.2000          |
| Winsorized Age              | 25.0303          | 25.1692         |
| Female                      | 0.4848           | 0.4154          |
| International               | 0.6710           | 0.7385          |
| GRE Verbal Percentile       | 84.5065          | 79.7385         |
| GRE Quantitative Percentile | 91.9784          | 91.2308         |
| GRE Analytic Percentile     | 69.1775          | 65.4615         |
| Undergraduate GPA           | 3.7160           | 3.6868          |
| TOEFL/IELTS                 | 7.6394           | 7.5953          |
| Elite Undergraduate         | 0.4892           | 0.4000          |
| Graduate Degree             | 0.5801           | 0.6308          |
| Work Experience             | 0.6320           | 0.5231          |
| #Professor Recommenders     | 2.5498           | 2.5846          |
| #Prolific Recommenders      | 1.2727           | 1.1692          |
| Advanced Math               | 0.6753           | 0.5077          |
| Missing Undergraduate GPA   | 0.4892           | 0.5077          |
| Missing TOEFL/IELTS         | 0.6537           | 0.6000          |
| Observations                | 231              | 65              |

# A3: Correlation Matrix (Matriculation)

Table 10: Correlation Matrix (Matriculation)

|                             | Covid | Age   | Fem   | Int   | GREV  | GREQ  | GREA  | GPA   | T/I   | Elite | Grad  | Work  | Prof | Pro   | Math  | MGPA  | MT/I |
|-----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|-------|-------|------|
| Covid Year                  | 1.00  |       |       |       |       |       |       |       |       |       |       |       |      |       |       |       |      |
| Winsorized Age              | 0.03  | 1.00  |       |       |       |       |       |       |       |       |       |       |      |       |       |       |      |
| Female                      | 0.04  | -0.09 | 1.00  |       |       |       |       |       |       |       |       |       |      |       |       |       |      |
| International               | -0.04 | 0.11  | 0.02  | 1.00  |       |       |       |       |       |       |       |       |      |       |       |       |      |
| GRE Verbal Percentile       | -0.04 | -0.14 | -0.04 | -0.34 | 1.00  |       |       |       |       |       |       |       |      |       |       |       |      |
| GRE Quantitative Percentile | 0.03  | -0.03 | -0.15 | 0.26  | 0.04  | 1.00  |       |       |       |       |       |       |      |       |       |       |      |
| GRE Analytic Percentile     | 0.08  | -0.09 | -0.01 | -0.50 | 0.44  | -0.19 | 1.00  |       |       |       |       |       |      |       |       |       |      |
| Undergraduate GPA           | -0.00 | -0.37 | 0.10  | -0.40 | 0.21  | -0.13 | 0.23  | 1.00  |       |       |       |       |      |       |       |       |      |
| TOEFL/IELTS                 | -0.06 | -0.05 | -0.03 | 0.24  | 0.23  | 0.15  | 0.02  | -0.21 | 1.00  |       |       |       |      |       |       |       |      |
| Elite Undergraduate         | -0.03 | -0.17 | 0.11  | -0.37 | 0.33  | -0.00 | 0.27  | 0.19  | -0.18 | 1.00  |       |       |      |       |       |       |      |
| Graduate Degree             | -0.06 | 0.42  | -0.12 | 0.43  | -0.25 | 0.12  | -0.25 | -0.48 | 0.17  | -0.38 | 1.00  |       |      |       |       |       |      |
| Work Experience             | -0.04 | 0.34  | 0.00  | -0.19 | 0.08  | -0.17 | 0.30  | -0.02 | -0.03 | 0.05  | -0.07 | 1.00  |      |       |       |       |      |
| #Professor Recommenders     | -0.06 | -0.24 | -0.01 | 0.23  | -0.05 | 0.09  | -0.11 | 0.11  | 0.02  | -0.07 | 0.05  | -0.30 | 1.00 |       |       |       |      |
| #Prolific Recommenders      | 0.04  | -0.02 | 0.14  | 0.14  | -0.02 | 0.11  | -0.08 | -0.11 | 0.10  | -0.11 | 0.21  | -0.02 | 0.22 | 1.00  |       |       |      |
| Advanced Math               | 0.01  | -0.17 | 0.01  | -0.21 | 0.24  | 0.06  | 0.17  | 0.15  | -0.12 | 0.22  | -0.25 | 0.01  | 0.04 | 0.14  | 1.00  |       |      |
| Missing Undergraduate GPA   | 0.01  | 0.28  | -0.12 | 0.65  | -0.34 | 0.18  | -0.39 | -0.64 | 0.32  | -0.58 | 0.57  | -0.04 | 0.03 | 0.19  | -0.26 | 1.00  |      |
| Missing TOEFL/IELTS         | 0.05  | -0.20 | 0.12  | -0.51 | 0.27  | -0.08 | 0.38  | 0.45  | -0.46 | 0.49  | -0.36 | 0.12  | 0.01 | -0.00 | 0.23  | -0.71 | 1.00 |
| Observations                | 231   |       |       |       |       |       |       |       |       |       |       |       |      |       |       |       |      |

# A3: Logistic Regression Results (Matriculation)

Table 11a: Logistic Regression Results (Matriculation)

| Dependent Variable = Matriculate | Coefficient            | AME                    | Lasso   | Post-Lasso Coefficient | Post-Lasso AME         |
|----------------------------------|------------------------|------------------------|---------|------------------------|------------------------|
| Covid Year                       | 0.1384<br>(0.4306)     | 0.0232<br>(0.0724)     |         |                        |                        |
| Winsorized Age                   | 0.0261<br>(0.0900)     | 0.0044<br>(0.0152)     |         |                        |                        |
| Female                           | -0.6244*<br>(0.3570)   | -0.1049*<br>(0.0589)   | -0.3914 | -0.6158*<br>(0.3471)   | -0.1037*<br>(0.0573)   |
| International                    | 0.7372<br>(0.5175)     | 0.1239<br>(0.0856)     | 0.1687  | 0.6690<br>(0.4693)     | 0.1127<br>(0.0779)     |
| GRE Verbal Percentile            | -0.0224*<br>(0.0130)   | -0.0038*<br>(0.0021)   | -0.0194 | -0.0215*<br>(0.0123)   | -0.0036*<br>(0.0020)   |
| GRE Quantitative Percentile      | -0.0609*<br>(0.0348)   | -0.0102*<br>(0.0058)   | -0.0425 | -0.0631*<br>(0.0341)   | -0.0106*<br>(0.0056)   |
| GRE Analytic Percentile          | 0.0034<br>(0.0091)     | 0.0006<br>(0.0015)     |         |                        |                        |
| Undergraduate GPA                | -4.1670***<br>(1.6083) | -0.7001***<br>(0.2570) | -2.2674 | -4.1144***<br>(1.4501) | -0.6932***<br>(0.2319) |
| TOEFL/IELTS                      | -0.3597<br>(0.5858)    | -0.0604<br>(0.0984)    | -0.1628 | -0.2931<br>(0.5263)    | -0.0494<br>(0.0887)    |
| Elite Undergraduate              | -0.9465*<br>(0.4954)   | -0.1590*<br>(0.0827)   | -0.3943 | -0.9245*<br>(0.4844)   | -0.1558*<br>(0.0816)   |
| Graduate Degree                  | -0.2158<br>(0.4760)    | -0.0363<br>(0.0799)    |         |                        |                        |
| Work Experience                  | -0.8483**<br>(0.4006)  | -0.1425**<br>(0.0666)  | -0.6164 | -0.7994**<br>(0.3458)  | -0.1347**<br>(0.0572)  |
| #Professor Recommenders          | 0.0296<br>(0.2316)     | 0.0050<br>(0.0389)     |         |                        |                        |
| #Prolific Recommenders           | 0.0268<br>(0.1948)     | 0.0045<br>(0.0328)     |         |                        |                        |
| Advanced Math                    | -1.0856***<br>(0.3627) | -0.1824***<br>(0.0574) | -0.8427 | -1.0371***<br>(0.3463) | -0.1747***<br>(0.0548) |
| Missing Undergraduate GPA        | -2.1263***<br>(0.8113) | -0.3573***<br>(0.1304) | -0.9377 | -2.0496***<br>(0.7114) | -0.3453***<br>(0.1135) |
| Missing TOEFL/IELTS              | -0.1722<br>(0.6110)    | -0.0289<br>(0.1026)    |         |                        |                        |
| Observations                     | 231                    | 231                    | 231     | 231                    | 231                    |
| Pseudo-R <sup>2</sup>            | 0.1469                 |                        | 0.1288  | 0.1445                 |                        |
| CV Pseudo-R <sup>2</sup>         | -0.0928                |                        | 0.0112  | 0.0485                 |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- **Positive ( $p < .05$ ):** None
- **Negative ( $p < .05$ ):** Undergraduate GPA, Work Experience, Advanced Math, Missing Undergraduate GPA

## A3: Simulated Wald Test Summary Statistics (Same)

Table 11b: Simulated Wald Test Summary Statistics (Same)

|                             | Economics |        | Non-Economics 1 |        | Non-Economics 2 |        |
|-----------------------------|-----------|--------|-----------------|--------|-----------------|--------|
|                             | Mean      | SD     | Mean            | SD     | Mean            | SD     |
| Accept                      | 0.1335    | 0.3401 | 0.0900          | 0.2862 | 0.0786          | 0.2691 |
| Success                     | 0.3000    | 0.4583 | 0.4057          | 0.4910 | 0.1157          | 0.3199 |
| GRE Quantitative Percentile | 84.8390   | 7.5405 | 65.1343         | 7.4495 | 65.6114         | 7.6706 |
| Gender                      | 0.3935    | 0.4885 | 0.3952          | 0.4889 | 0.5914          | 0.4916 |
| Demographic 1               | 0.5005    | 0.5000 | 0.3091          | 0.4621 | 0.2600          | 0.4386 |
| Demographic 2               | 0.2505    | 0.4333 | 0.2957          | 0.4563 | 0.3414          | 0.4742 |
| Advanced Math               | 0.5095    | 0.4999 |                 |        |                 |        |
| Committee Score             | 4.9325    | 1.9877 | 6.0191          | 2.0205 | 6.0243          | 2.0038 |
| Observations                | 2000      |        | 2300            |        | 700             |        |

# A3: Simulated Wald Test Summary Statistics (Different)

Table 12a: Simulated Wald Test Summary Statistics (Different)

|                             | Economics |        | Non-Economics 1 |        | Non-Economics 2 |        |
|-----------------------------|-----------|--------|-----------------|--------|-----------------|--------|
|                             | Mean      | SD     | Mean            | SD     | Mean            | SD     |
| Accept                      | 0.1335    | 0.3401 | 0.0452          | 0.2078 | 0.0543          | 0.2266 |
| Success                     | 0.3000    | 0.4583 | 0.3126          | 0.4636 | 0.1314          | 0.3379 |
| GRE Quantitative Percentile | 84.8390   | 7.5405 | 65.1343         | 7.4495 | 65.6114         | 7.6706 |
| Gender                      | 0.3935    | 0.4885 | 0.3952          | 0.4889 | 0.5914          | 0.4916 |
| Demographic 1               | 0.5005    | 0.5000 | 0.3091          | 0.4621 | 0.2600          | 0.4386 |
| Demographic 2               | 0.2505    | 0.4333 | 0.2957          | 0.4563 | 0.3414          | 0.4742 |
| Advanced Math               | 0.5095    | 0.4999 |                 |        |                 |        |
| Committee Score             | 4.9325    | 1.9877 | 6.0191          | 2.0205 | 6.0243          | 2.0038 |
| Observations                | 2000      |        | 2300            |        | 700             |        |

# A3: Simulated Wald Test Correlation Matrix (Same)

Table 12b: Simulated Wald Test Correlation Matrix (Same)

|                             | Acc     | Suc     | GRE     | Gen     | Dem1    | Dem2    | Com    | NEcon1  | NEcon2 |
|-----------------------------|---------|---------|---------|---------|---------|---------|--------|---------|--------|
| Accept                      | 1       |         |         |         |         |         |        |         |        |
| Success                     | 0.1603  | 1       |         |         |         |         |        |         |        |
| GRE Quantitative Percentile | 0.2772  | 0.1665  | 1       |         |         |         |        |         |        |
| Gender                      | 0.0484  | -0.0581 | -0.0795 | 1       |         |         |        |         |        |
| Demographic 1               | -0.0233 | 0.0199  | 0.3377  | -0.0136 | 1       |         |        |         |        |
| Demographic 2               | 0.0487  | 0.0148  | -0.0876 | -0.0209 | -0.4918 | 1       |        |         |        |
| Committee Score             | 0.2045  | 0.1486  | -0.0770 | 0.0980  | -0.1567 | -0.0359 | 1      |         |        |
| Non-Economics 1             | -0.0474 | 0.1636  | -0.6017 | -0.0501 | -0.1326 | 0.0238  | 0.1930 | 1       |        |
| Non-Economics 2             | -0.0357 | -0.1787 | -0.2472 | 0.1384  | -0.0988 | 0.0514  | 0.0854 | -0.3724 | 1      |
| Observations                | 5000    |         |         |         |         |         |        |         |        |

» Acceptance vs. Success AMEs

# A3: Simulated Wald Test Correlation Matrix (Different)

Table 13a: Simulated Wald Test Correlation Matrix (Different)

|                             | Acc     | Suc     | GRE     | Gen     | Dem1    | Dem2    | Com    | NEcon1  | NEcon2 |
|-----------------------------|---------|---------|---------|---------|---------|---------|--------|---------|--------|
| Accept                      | 1       |         |         |         |         |         |        |         |        |
| Success                     | 0.0885  | 1       |         |         |         |         |        |         |        |
| GRE Quantitative Percentile | 0.2376  | 0.1167  | 1       |         |         |         |        |         |        |
| Gender                      | 0.0582  | -0.0220 | -0.0795 | 1       |         |         |        |         |        |
| Demographic 1               | 0.0152  | -0.0004 | 0.3377  | -0.0136 | 1       |         |        |         |        |
| Demographic 2               | 0.0078  | 0.0003  | -0.0876 | -0.0209 | -0.4918 | 1       |        |         |        |
| Committee Score             | 0.1437  | 0.1500  | -0.0770 | 0.0980  | -0.1567 | -0.0359 | 1      |         |        |
| Non-Economics 1             | -0.1232 | 0.0624  | -0.6017 | -0.0501 | -0.1326 | 0.0238  | 0.1930 | 1       |        |
| Non-Economics 2             | -0.0405 | -0.1352 | -0.2472 | 0.1384  | -0.0988 | 0.0514  | 0.0854 | -0.3724 | 1      |
| Observations                | 5000    |         |         |         |         |         |        |         |        |

» Acceptance vs. Success AMEs

# A3: Simulated Wald Test Results (Same)

Table 13b: Simulated Wald Test Results (Same)

|                                      | Dependent Variable = Accept |                        |                         |                        | Dependent Variable = Success |                        |                         |                        |
|--------------------------------------|-----------------------------|------------------------|-------------------------|------------------------|------------------------------|------------------------|-------------------------|------------------------|
|                                      | Economics                   |                        | Non-Economics           |                        | Economics                    |                        | Non-Economics           |                        |
|                                      | Coef                        | AME                    | Coef                    | AME                    | Coef                         | AME                    | Coef                    | AME                    |
| Intercept                            | -16.3848***<br>(1.1315)     | -1.5822***<br>(0.0956) | -28.5524***<br>(1.4963) | -1.3677***<br>(0.0500) | -10.4586***<br>(0.7236)      | -1.9105***<br>(0.1070) | -10.2610***<br>(0.4815) | -1.8514***<br>(0.0628) |
| GRE Quantitative Percentile          | 0.1509***<br>(0.0131)       | 0.0146***<br>(0.0012)  | 0.3245***<br>(0.0179)   | 0.0155***<br>(0.0007)  | 0.1058***<br>(0.0088)        | 0.0193***<br>(0.0014)  | 0.1154***<br>(0.0070)   | 0.0208***<br>(0.0011)  |
| Gender                               | 0.4714***<br>(0.1475)       | 0.0455***<br>(0.0140)  | 0.7880***<br>(0.1696)   | 0.0377***<br>(0.0081)  | -0.1384<br>(0.1095)          | -0.0253<br>(0.0200)    | -0.0497<br>(0.0895)     | -0.0090<br>(0.0161)    |
| Demographic 1                        | -0.6287***<br>(0.2088)      | -0.0607***<br>(0.0200) | -1.8488***<br>(0.2445)  | -0.0886***<br>(0.0115) | -0.3486**<br>(0.1467)        | -0.0637**<br>(0.0267)  | -0.3079***<br>(0.1123)  | -0.0556***<br>(0.0202) |
| Demographic 2                        | 0.1531<br>(0.2072)          | 0.0148<br>(0.0200)     | 0.3703*<br>(0.1929)     | 0.0177*<br>(0.0091)    | 0.1756<br>(0.1558)           | 0.0321<br>(0.0284)     | 0.0282<br>(0.1069)      | 0.0051<br>(0.0193)     |
| Advanced Math                        | 0.2889*<br>(0.1612)         | 0.0279*<br>(0.0156)    |                         |                        | 0.3267***<br>(0.1112)        | 0.0597***<br>(0.0202)  |                         |                        |
| Committee Score                      | 0.2241***<br>(0.0413)       | 0.0216***<br>(0.0039)  | 0.3994***<br>(0.0506)   | 0.0191***<br>(0.0023)  | 0.1026***<br>(0.0287)        | 0.0187***<br>(0.0052)  | 0.0757***<br>(0.0228)   | 0.0137***<br>(0.0041)  |
| Non-Economics 1                      |                             |                        | 0.8376***<br>(0.2297)   | 0.0401***<br>(0.0107)  |                              |                        | 1.9483***<br>(0.1236)   | 0.3515***<br>(0.0205)  |
| Observations   Pseudo-R <sup>2</sup> | 5000                        | 0.3251                 |                         |                        | 5000                         | 0.1431                 |                         |                        |
| LL   Parameters   AIC                | -1130.4                     | 14                     | 2288.8                  |                        | -2691.4                      | 14                     | 5410.8                  |                        |
| Wald Statistic   DF   P-Value        | 3.1018                      | 5                      | 0.6843                  |                        | 3.0042                       | 5                      | 0.6993                  |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.



# A3: Simulated Wald Test Results (Different)

Table 14a: Simulated Wald Test Results (Different)

|                                      | Dependent Variable = Accept |                        |                        |                        | Dependent Variable = Success |                        |                        |                        |
|--------------------------------------|-----------------------------|------------------------|------------------------|------------------------|------------------------------|------------------------|------------------------|------------------------|
|                                      | Economics                   |                        | Non-Economics          |                        | Economics                    |                        | Non-Economics          |                        |
|                                      | Coef                        | AME                    | Coef                   | AME                    | Coef                         | AME                    | Coef                   | AME                    |
| Intercept                            | -16.3291***<br>(1.1274)     | -1.5784***<br>(0.0955) | -8.4753***<br>(0.8673) | -0.3642***<br>(0.0412) | -10.4581***<br>(0.7237)      | -1.9104***<br>(0.1070) | -3.2236***<br>(0.4018) | -0.5997***<br>(0.0728) |
| GRE Quantitative Percentile          | 0.1502***<br>(0.0131)       | 0.0145***<br>(0.0012)  | 0.0402***<br>(0.0128)  | 0.0017***<br>(0.0006)  | 0.1058***<br>(0.0088)        | 0.0193***<br>(0.0014)  | 0.0034<br>(0.0062)     | 0.0006<br>(0.0012)     |
| Gender                               | 0.4702***<br>(0.1473)       | 0.0454***<br>(0.0140)  | 0.5942***<br>(0.1820)  | 0.0255***<br>(0.0079)  | -0.1372<br>(0.1095)          | -0.0251<br>(0.0200)    | 0.0375<br>(0.0868)     | 0.0070<br>(0.0161)     |
| Demographic 1                        | -0.6246***<br>(0.2085)      | -0.0604***<br>(0.0200) | -0.2541<br>(0.2393)    | -0.0109<br>(0.0103)    | -0.3544**<br>(0.1467)        | -0.0647**<br>(0.0267)  | -0.1152<br>(0.1103)    | -0.0214<br>(0.0205)    |
| Demographic 2                        | 0.1555<br>(0.2070)          | 0.0150<br>(0.0200)     | 0.0081<br>(0.2109)     | 0.0003<br>(0.0091)     | 0.1713<br>(0.1557)           | 0.0313<br>(0.0284)     | -0.0562<br>(0.1030)    | -0.0105<br>(0.0192)    |
| Advanced Math                        | 0.2886*<br>(0.1611)         | 0.0279*<br>(0.0156)    |                        |                        | 0.3284***<br>(0.1112)        | 0.0600***<br>(0.0202)  |                        |                        |
| Committee Score                      | 0.2241***<br>(0.0413)       | 0.0217***<br>(0.0039)  | 0.3817***<br>(0.0528)  | 0.0164***<br>(0.0025)  | 0.1026***<br>(0.0287)        | 0.0187***<br>(0.0052)  | 0.1808***<br>(0.0228)  | 0.0336***<br>(0.0041)  |
| Non-Economics 1                      |                             |                        | -0.0452<br>(0.1995)    | -0.0019<br>(0.0086)    |                              |                        | 1.1393***<br>(0.1219)  | 0.2120***<br>(0.0219)  |
| Observations   Pseudo-R <sup>2</sup> | 5000                        | 0.1488                 |                        |                        | 5000                         | 0.0758                 |                        |                        |
| LL   Parameters   AIC                | -1155.7                     | 14                     | 2339.5                 |                        | -2747.2                      | 14                     | 5522.4                 |                        |
| Wald Statistic   DF   P-Value        | 112.5897                    | 5                      | 0.0000                 |                        | 124.7379                     | 5                      | 0.0000                 |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Summary Statistics (Optimal)

Table 15a: Simulated Summary Statistics (Optimal)

|                                  | Mean    | SD      |
|----------------------------------|---------|---------|
| Transition                       | 0.3740  | 0.4839  |
| Waitlist                         | 0.3040  | 0.4600  |
| Accept                           | 0.1260  | 0.3318  |
| Matriculate                      | 0.0540  | 0.2260  |
| Success                          | 0.3080  | 0.4617  |
| GRE Quantitative Percentile      | 84.9310 | 7.5286  |
| Gender                           | 0.3970  | 0.4893  |
| Demographic 1                    | 0.5380  | 0.4986  |
| Demographic 2                    | 0.2580  | 0.4375  |
| Advanced Math                    | 0.5060  | 0.5000  |
| Committee Score                  | 5.0240  | 2.0638  |
| Year Transition/Matriculate More | 0.5000  | 0.5000  |
| Program Rank                     | 77.3340 | 53.8839 |
| Observations                     | 1000    |         |

## A4: Simulated Summary Statistics (Suboptimal)

Table 15b: Simulated Summary Statistics (Suboptimal)

|                                  | Mean    | SD      |
|----------------------------------|---------|---------|
| Transition                       | 0.3850  | 0.4866  |
| Waitlist                         | 0.2960  | 0.4565  |
| Accept                           | 0.1320  | 0.3385  |
| Matriculate                      | 0.0480  | 0.2138  |
| Success                          | 0.3070  | 0.4612  |
| GRE Quantitative Percentile      | 84.9310 | 7.5286  |
| Gender                           | 0.3970  | 0.4893  |
| Demographic 1                    | 0.5380  | 0.4986  |
| Demographic 2                    | 0.2580  | 0.4375  |
| Advanced Math                    | 0.5060  | 0.5000  |
| Committee Score                  | 5.0240  | 2.0638  |
| Year Transition/Matriculate More | 0.5000  | 0.5000  |
| Program Rank                     | 77.1650 | 53.8030 |
| Observations                     | 1000    |         |

# A4: Simulated Correlation Matrix (Optimal)

Table 16a: Simulated Correlation Matrix (Optimal)

|                                  | Trans   | Wait    | Acc     | Mat     | Suc     | GRE     | Gen     | Dem1    | Dem2    | Math    | Com     | Year   | Rank |
|----------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|--------|------|
| Transition                       | 1       |         |         |         |         |         |         |         |         |         |         |        |      |
| Waitlist                         | 0.8550  | 1       |         |         |         |         |         |         |         |         |         |        |      |
| Accept                           | 0.4912  | 0.1159  | 1       |         |         |         |         |         |         |         |         |        |      |
| Matriculate                      | 0.3091  | 0.1018  | 0.6292  | 1       |         |         |         |         |         |         |         |        |      |
| Success                          | 0.0797  | 0.0253  | 0.1514  | 0.1952  | 1       |         |         |         |         |         |         |        |      |
| GRE Quantitative Percentile      | 0.3387  | 0.2134  | 0.3353  | 0.2296  | 0.3433  | 1       |         |         |         |         |         |        |      |
| Gender                           | 0.0233  | 0.0147  | 0.0307  | 0.0141  | 0.0076  | -0.0398 | 1       |         |         |         |         |        |      |
| Demographic 1                    | 0.0157  | -0.0198 | 0.0255  | 0.0262  | 0.0621  | 0.2726  | -0.0147 | 1       |         |         |         |        |      |
| Demographic 2                    | 0.0449  | 0.0475  | 0.0309  | 0.0108  | -0.0171 | -0.0817 | 0.0027  | -0.6363 | 1       |         |         |        |      |
| Advanced Math                    | 0.0238  | -0.0079 | 0.0859  | 0.0856  | 0.1696  | 0.1015  | -0.0608 | 0.0994  | -0.0436 | 1       |         |        |      |
| Committee Score                  | 0.0221  | -0.0024 | 0.0467  | -0.0478 | 0.0898  | 0.0440  | -0.0292 | -0.0670 | 0.0673  | 0.2325  | 1       |        |      |
| Year Transition/Matriculate More | 0.1860  | 0.2044  | -0.0181 | 0.1416  | -0.0477 | -0.0410 | -0.0225 | -0.0361 | 0.0457  | 0.0120  | 0.0329  | 1      |      |
| Program Rank                     | -0.0484 | 0.0114  | -0.1432 | -0.1788 | -0.4839 | -0.2223 | -0.0013 | -0.0508 | 0.0029  | -0.1054 | -0.0424 | 0.0451 | 1    |
| Observations                     | 1000    |         |         |         |         |         |         |         |         |         |         |        |      |

» Dynamic Unrestricted AMEs (Optimal)

# A4: Simulated Correlation Matrix (Suboptimal)

Table 16b: Simulated Correlation Matrix (Suboptimal)

|                                  | Trans   | Wait    | Acc     | Mat     | Suc     | GRE     | Gen     | Dem1    | Dem2    | Math    | Com     | Year   | Rank |
|----------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|--------|------|
| Transition                       | 1       |         |         |         |         |         |         |         |         |         |         |        |      |
| Waitlist                         | 0.8195  | 1       |         |         |         |         |         |         |         |         |         |        |      |
| Accept                           | 0.4929  | 0.0254  | 1       |         |         |         |         |         |         |         |         |        |      |
| Matriculate                      | 0.2838  | -0.0124 | 0.5758  | 1       |         |         |         |         |         |         |         |        |      |
| Success                          | 0.0749  | 0.0386  | 0.0927  | 0.1345  | 1       |         |         |         |         |         |         |        |      |
| GRE Quantitative Percentile      | 0.3032  | 0.2166  | 0.2198  | 0.1468  | 0.3433  | 1       |         |         |         |         |         |        |      |
| Gender                           | 0.0301  | 0.0425  | 0.0157  | -0.0005 | 0.0050  | -0.0398 | 1       |         |         |         |         |        |      |
| Demographic 1                    | -0.0830 | -0.0670 | -0.0712 | 0.0017  | 0.0602  | 0.2726  | -0.0147 | 1       |         |         |         |        |      |
| Demographic 2                    | 0.1018  | 0.0632  | 0.0874  | 0.0066  | -0.0208 | -0.0817 | 0.0027  | -0.6363 | 1       |         |         |        |      |
| Advanced Math                    | 0.0254  | -0.0122 | 0.0662  | 0.0628  | 0.1633  | 0.1015  | -0.0608 | 0.0994  | -0.0436 | 1       |         |        |      |
| Committee Score                  | 0.0446  | -0.0585 | 0.1515  | 0.0133  | 0.0921  | 0.0440  | -0.0292 | -0.0670 | 0.0673  | 0.2325  | 1       |        |      |
| Year Transition/Matriculate More | 0.2158  | 0.1928  | 0.0236  | 0.1684  | -0.0499 | -0.0410 | -0.0225 | -0.0361 | 0.0457  | 0.0120  | 0.0329  | 1      |      |
| Program Rank                     | -0.0608 | -0.0029 | -0.1116 | -0.1676 | -0.4782 | -0.2140 | 0.0057  | -0.0400 | -0.0021 | -0.1029 | -0.0463 | 0.0401 | 1    |
| Observations                     | 1000    |         |         |         |         |         |         |         |         |         |         |        |      |

► Static Unrestricted AMEs (Suboptimal)

# A4: Full Unrestricted Static Estimates (Optimal)

Table 17a: Unrestricted Static Estimates (Optimal)

|                             | Transition              |                        | Accept                  |                        | Success 1              |                        | Success 2              |                        |
|-----------------------------|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -11.5886***<br>(1.0232) | -2.2559***<br>(0.1454) | -13.1494***<br>(2.0072) | -2.3728***<br>(0.2886) | -8.1224***<br>(1.1371) | -1.1660***<br>(0.1344) | -9.5306***<br>(2.0699) | -1.4839***<br>(0.2689) |
| GRE Quantitative Percentile | 0.1235***<br>(0.0118)   | 0.0240***<br>(0.0018)  | 0.1371***<br>(0.0223)   | 0.0247***<br>(0.0033)  | 0.1078***<br>(0.0133)  | 0.0155***<br>(0.0015)  | 0.1158***<br>(0.0227)  | 0.0180***<br>(0.0029)  |
| Gender                      | 0.2262<br>(0.1464)      | 0.0440<br>(0.0283)     | 0.1290<br>(0.2502)      | 0.0233<br>(0.0451)     | 0.1396<br>(0.1738)     | 0.0200<br>(0.0249)     | 0.1262<br>(0.2698)     | 0.0197<br>(0.0419)     |
| Demographic 1               | -0.2995<br>(0.1988)     | -0.0583<br>(0.0386)    | -0.2639<br>(0.3743)     | -0.0476<br>(0.0672)    | -0.2529<br>(0.2358)    | -0.0363<br>(0.0338)    | -0.4577<br>(0.3931)    | -0.0713<br>(0.0608)    |
| Demographic 2               | 0.1552<br>(0.2215)      | 0.0302<br>(0.0431)     | 0.1711<br>(0.3843)      | 0.0309<br>(0.0693)     | -0.1180<br>(0.2580)    | -0.0169<br>(0.0371)    | -0.2318<br>(0.4320)    | -0.0361<br>(0.0672)    |
| Advanced Math               |                         |                        | 0.7733***<br>(0.2534)   | 0.1396***<br>(0.0443)  |                        |                        | 0.6278**<br>(0.2748)   | 0.0978**<br>(0.0417)   |
| Committee Score             |                         |                        | 0.0781<br>(0.0602)      | 0.0141<br>(0.0108)     |                        |                        | 0.0462<br>(0.0655)     | 0.0072<br>(0.0102)     |
| Year Transition More        | 0.9837***<br>(0.1472)   | 0.1915***<br>(0.0263)  | -0.9252***<br>(0.2512)  | -0.1670***<br>(0.0427) | -0.0592<br>(0.1689)    | -0.0085<br>(0.0242)    | -0.0239<br>(0.2762)    | -0.0037<br>(0.0430)    |
| Program Rank                |                         |                        |                         |                        | -0.0310***<br>(0.0019) | -0.0045***<br>(0.0002) | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |
| Observations                | 1000                    |                        |                         |                        |                        |                        |                        |                        |
| LL   Parameters   AIC       | -1375.9                 | 30                     | 2811.7                  |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Full Unrestricted Static Estimates (Suboptimal)

Table 17b: Unrestricted Static Estimates (Suboptimal)

|                             | Transition              |                        | Accept                 |                        | Success 1              |                        | Success 2              |                        |
|-----------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -11.2297***<br>(1.0144) | -2.1801***<br>(0.1474) | -6.7297***<br>(1.7376) | -1.3296***<br>(0.3202) | -8.2833***<br>(1.1393) | -1.1914***<br>(0.1341) | -9.6070***<br>(1.9611) | -1.4986***<br>(0.2508) |
| GRE Quantitative Percentile | 0.1226***<br>(0.0117)   | 0.0238***<br>(0.0018)  | 0.0550***<br>(0.0191)  | 0.0109***<br>(0.0036)  | 0.1097***<br>(0.0133)  | 0.0158***<br>(0.0015)  | 0.1163***<br>(0.0216)  | 0.0181***<br>(0.0027)  |
| Gender                      | 0.2576*<br>(0.1469)     | 0.0500*<br>(0.0283)    | 0.0097<br>(0.2368)     | 0.0019<br>(0.0468)     | 0.1355<br>(0.1741)     | 0.0195<br>(0.0250)     | 0.0584<br>(0.2669)     | 0.0091<br>(0.0416)     |
| Demographic 1               | -0.8291***<br>(0.1966)  | -0.1610***<br>(0.0371) | -0.3339<br>(0.3394)    | -0.0660<br>(0.0669)    | -0.2777<br>(0.2344)    | -0.0399<br>(0.0337)    | -0.5157<br>(0.3648)    | -0.0804<br>(0.0563)    |
| Demographic 2               | 0.0798<br>(0.2135)      | 0.0155<br>(0.0414)     | 0.1762<br>(0.3346)     | 0.0348<br>(0.0659)     | -0.1808<br>(0.2566)    | -0.0260<br>(0.0369)    | -0.4742<br>(0.3857)    | -0.0740<br>(0.0598)    |
| Advanced Math               |                         |                        | 0.3581<br>(0.2375)     | 0.0708<br>(0.0465)     |                        |                        | 0.5369**<br>(0.2704)   | 0.0837**<br>(0.0416)   |
| Committee Score             |                         |                        | 0.2962***<br>(0.0592)  | 0.0585***<br>(0.0106)  |                        |                        | 0.0616<br>(0.0643)     | 0.0096<br>(0.0100)     |
| Year Transition More        | 1.1109***<br>(0.1479)   | 0.2157***<br>(0.0257)  | -0.7185***<br>(0.2397) | -0.1419***<br>(0.0455) | -0.0766<br>(0.1688)    | -0.0110<br>(0.0243)    | 0.1314<br>(0.2795)     | 0.0205<br>(0.0434)     |
| Program Rank                |                         |                        |                        |                        | -0.0304***<br>(0.0019) | -0.0044***<br>(0.0002) | -0.0279***<br>(0.0028) | -0.0044***<br>(0.0003) |
| Observations                | 1000                    |                        |                        |                        |                        |                        |                        |                        |
| LL   Parameters   AIC       | -1404.1                 |                        | 30                     |                        | 2868.1                 |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Full Restricted Static Estimates (Optimal)

Table 18a: Restricted Static Estimates (Optimal)

|                             | Cutoff 1               | $P(\text{Cutoff})_1$       | Cutoff 2             | $P(\text{Cutoff})_2$       | Success 1              | Success 2              | Sigma                 |
|-----------------------------|------------------------|----------------------------|----------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                   | 0.6884***<br>(0.1830)  | 66.56%<br>(58.17%, 74.02%) | 0.1530<br>(0.2411)   | 53.82%<br>(42.08%, 65.15%) | -8.0150***<br>(1.1629) | -9.7263***<br>(1.7366) |                       |
| GRE Quantitative Percentile |                        |                            |                      |                            | 0.1057***<br>(0.0134)  | 0.1158***<br>(0.0193)  |                       |
| Gender                      |                        |                            |                      |                            | 0.1987*<br>(0.1015)    | 0.1142<br>(0.1673)     |                       |
| Demographic 1               |                        |                            |                      |                            | -0.2471*<br>(0.1346)   | -0.3338<br>(0.2403)    |                       |
| Demographic 2               |                        |                            |                      |                            | 0.0406<br>(0.1504)     | -0.0047<br>(0.2423)    |                       |
| Advanced Math               |                        |                            |                      |                            |                        | 0.6315***<br>(0.1766)  |                       |
| Committee Score             |                        |                            |                      |                            |                        | 0.0546<br>(0.0421)     |                       |
| Year Transition More        | -0.8466***<br>(0.2184) | 46.05%<br>(39.60%, 52.64%) | 0.7965**<br>(0.3769) | 72.10%<br>(58.90%, 82.34%) | -0.0628<br>(0.1667)    | 0.0285<br>(0.2701)     |                       |
| Program Rank                |                        |                            |                      |                            | -0.0310***<br>(0.0019) | -0.0283***<br>(0.0027) |                       |
| Sigma                       |                        |                            |                      |                            |                        |                        | 0.1913***<br>(0.0252) |
| Observations                | 1000                   |                            |                      |                            |                        |                        |                       |
| LL   Parameters   AIC       | -1376.5                | 21                         | 2795.0               |                            |                        |                        |                       |
| LR Statistic   DF   P-Value | 1.3137                 | 9                          | 0.9983               |                            |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.



# A4: Full Restricted Static Estimates (Suboptimal)

Table 18b: Restricted Static Estimates (Suboptimal)

|                             | Cutoff 1               | $P(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success 1              | Success 2              | Sigma                 |
|-----------------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                   | 0.7340***<br>(0.2062)  | 67.57%<br>(58.17%, 75.73%) | 0.1682<br>(0.2593) | 54.20%<br>(41.58%, 66.29%) | -8.0582***<br>(1.2803) | -6.3578***<br>(1.7689) |                       |
| GRE Quantitative Percentile |                        |                            |                    |                            | 0.1079***<br>(0.0148)  | 0.0727***<br>(0.0189)  |                       |
| Gender                      |                        |                            |                    |                            | 0.2185**<br>(0.1042)   | 0.0409<br>(0.1610)     |                       |
| Demographic 1               |                        |                            |                    |                            | -0.5505***<br>(0.1395) | -0.3739<br>(0.2380)    |                       |
| Demographic 2               |                        |                            |                    |                            | -0.0151<br>(0.1505)    | -0.0921<br>(0.2356)    |                       |
| Advanced Math               |                        |                            |                    |                            |                        | 0.3897**<br>(0.1686)   |                       |
| Committee Score             |                        |                            |                    |                            |                        | 0.1754***<br>(0.0378)  |                       |
| Year Transition More        | -0.9965***<br>(0.2396) | 43.47%<br>(37.12%, 50.05%) | 0.6110<br>(0.3768) | 68.55%<br>(55.51%, 79.20%) | -0.0772<br>(0.1671)    | 0.0376<br>(0.2807)     |                       |
| Program Rank                |                        |                            |                    |                            | -0.0305***<br>(0.0019) | -0.0281***<br>(0.0027) |                       |
| Sigma                       |                        |                            |                    |                            |                        |                        | 0.2001***<br>(0.0325) |
| Observations                | 1000                   |                            |                    |                            |                        |                        |                       |
| LL   Parameters   AIC       | -1417.0                | 21                         | 2875.9             |                            |                        |                        |                       |
| LR Statistic   DF   P-Value | 25.7951                | 9                          | 0.0022             |                            |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Unrestricted Dynamic Estimates (Optimal)

Table 19a: Unrestricted Dynamic Estimates (Optimal)

|                             | Advanced Math         |                       | Committee Score       |  | Transition              |                        | Accept                  |                        | Success                |                        |
|-----------------------------|-----------------------|-----------------------|-----------------------|--|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient           | AME                   | Coefficient           |  | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    |
| Intercept                   | -1.9035**<br>(0.7564) | -0.4665**<br>(0.1830) | 3.5851***<br>(0.7240) |  | -11.5883***<br>(1.0232) | -2.2558***<br>(0.1454) | -13.1511***<br>(2.0073) | -2.3731***<br>(0.2886) | -9.5511***<br>(2.0717) | -1.4866***<br>(0.2688) |
| GRE Quantitative Percentile | 0.0208**<br>(0.0090)  | 0.0051**<br>(0.0022)  | 0.0126<br>(0.0086)    |  | 0.1236***<br>(0.0118)   | 0.0241***<br>(0.0018)  | 0.1371***<br>(0.0223)   | 0.0247***<br>(0.0033)  | 0.1160***<br>(0.0227)  | 0.0181***<br>(0.0029)  |
| Gender                      | -0.2334*<br>(0.1309)  | -0.0572*<br>(0.0319)  | -0.0581<br>(0.1315)   |  | 0.2266<br>(0.1464)      | 0.0441<br>(0.0283)     | 0.1288<br>(0.2502)      | 0.0232<br>(0.0451)     | 0.1262<br>(0.2699)     | 0.0196<br>(0.0419)     |
| Demographic 1               | 0.3691**<br>(0.1732)  | 0.0905**<br>(0.0421)  | -0.3561**<br>(0.1714) |  | -0.3013<br>(0.1988)     | -0.0586<br>(0.0386)    | -0.2639<br>(0.3743)     | -0.0476<br>(0.0672)    | -0.4604<br>(0.3932)    | -0.0717<br>(0.0608)    |
| Demographic 2               | 0.0924<br>(0.1911)    | 0.0227<br>(0.0468)    | 0.1200<br>(0.1890)    |  | 0.1538<br>(0.2215)      | 0.0299<br>(0.0431)     | 0.1707<br>(0.3843)      | 0.0308<br>(0.0693)     | -0.2341<br>(0.4322)    | -0.0364<br>(0.0672)    |
| Advanced Math               |                       |                       | 0.9753***<br>(0.1270) |  |                         |                        | 0.7736***<br>(0.2535)   | 0.1396***<br>(0.0443)  | 0.6275**<br>(0.2749)   | 0.0977**<br>(0.0417)   |
| Committee Score             |                       |                       |                       |  |                         |                        | 0.0783<br>(0.0602)      | 0.0141<br>(0.0108)     | 0.0464<br>(0.0655)     | 0.0072<br>(0.0102)     |
| Year Transition More        | 0.0664<br>(0.1281)    | 0.0163<br>(0.0314)    | 0.1132<br>(0.1261)    |  | 0.9840***<br>(0.1472)   | 0.1915***<br>(0.0263)  | -0.9249***<br>(0.2512)  | -0.1669***<br>(0.0427) | -0.0237<br>(0.2763)    | -0.0037<br>(0.0430)    |
| Program Rank                |                       |                       |                       |  |                         |                        |                         |                        | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |
| $\sqrt{MSE}$                |                       |                       | 1.9949***<br>(0.0422) |  |                         |                        |                         |                        |                        |                        |
| Observations                | 1000                  |                       |                       |  |                         |                        |                         |                        |                        |                        |
| LL   Parameters   AIC       | -3738.4               | 37                    | 7550.9                |  |                         |                        |                         |                        |                        |                        |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

# A4: Unrestricted Dynamic Estimates (Suboptimal)

Table 19b: Unrestricted Dynamic Estimates (Suboptimal)

|                             | Advanced Math         |                       | Committee Score       | Transition              |                        | Accept                 |                        | Success                |                        |
|-----------------------------|-----------------------|-----------------------|-----------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient           | AME                   |                       | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -1.8990**<br>(0.7563) | -0.4654**<br>(0.1830) | 3.5753***<br>(0.7241) | -11.2291***<br>(1.0144) | -2.1800***<br>(0.1474) | -6.7296***<br>(1.7375) | -1.3296***<br>(0.3202) | -9.5854***<br>(1.9593) | -1.4958***<br>(0.2508) |
| GRE Quantitative Percentile | 0.0207**<br>(0.0090)  | 0.0051**<br>(0.0022)  | 0.0127<br>(0.0086)    | 0.1226***<br>(0.0117)   | 0.0238***<br>(0.0018)  | 0.0550***<br>(0.0191)  | 0.0109***<br>(0.0036)  | 0.1161***<br>(0.0216)  | 0.0181***<br>(0.0027)  |
| Gender                      | -0.2335*<br>(0.1309)  | -0.0572*<br>(0.0319)  | -0.0579<br>(0.1315)   | 0.2576*<br>(0.1468)     | 0.0500*<br>(0.0283)    | 0.0105<br>(0.2368)     | 0.0021<br>(0.0468)     | 0.0579<br>(0.2668)     | 0.0090<br>(0.0416)     |
| Demographic 1               | 0.3688**<br>(0.1732)  | 0.0904**<br>(0.0421)  | -0.3561**<br>(0.1714) | -0.8290***<br>(0.1966)  | -0.1609***<br>(0.0371) | -0.3340<br>(0.3394)    | -0.0660<br>(0.0669)    | -0.5160<br>(0.3647)    | -0.0805<br>(0.0563)    |
| Demographic 2               | 0.0920<br>(0.1911)    | 0.0226<br>(0.0468)    | 0.1205<br>(0.1890)    | 0.0801<br>(0.2135)      | 0.0155<br>(0.0414)     | 0.1765<br>(0.3345)     | 0.0349<br>(0.0659)     | -0.4751<br>(0.3855)    | -0.0741<br>(0.0598)    |
| Advanced Math               |                       |                       | 0.9752***<br>(0.1270) |                         |                        | 0.3579<br>(0.2375)     | 0.0707<br>(0.0465)     | 0.5367**<br>(0.2703)   | 0.0837**<br>(0.0416)   |
| Committee Score             |                       |                       |                       |                         |                        | 0.2961***<br>(0.0592)  | 0.0585***<br>(0.0106)  | 0.0614<br>(0.0642)     | 0.0096<br>(0.0100)     |
| Year Transition More        | 0.0663<br>(0.1281)    | 0.0163<br>(0.0314)    | 0.1135<br>(0.1262)    | 1.1105***<br>(0.1479)   | 0.2156***<br>(0.0257)  | -0.7183***<br>(0.2397) | -0.1419***<br>(0.0455) | 0.1308<br>(0.2794)     | 0.0204<br>(0.0434)     |
| Program Rank                |                       |                       |                       |                         |                        |                        |                        | -0.0279***<br>(0.0028) | -0.0044***<br>(0.0003) |
| $\sqrt{\text{MSE}}$         |                       |                       | 1.9949***<br>(0.0422) |                         |                        |                        |                        |                        |                        |
| Observations                | 1000                  |                       |                       |                         |                        |                        |                        |                        |                        |
| LL   Parameters   AIC       | -3765.0               |                       | 37                    | 7604.0                  |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Restricted Dynamic Estimates (Optimal)

Table 20a: Restricted Dynamic Estimates (Optimal)

|                             | Advanced Math          | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|------------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | -2.1172***<br>(0.7570) | 3.5091***<br>(0.7215) | -0.3361***<br>(0.0803) | 0.7146<br>(0.6105, 0.8364) | -0.0136<br>(0.2115) | 49.66%<br>(39.46%, 59.89%) | -9.4809***<br>(1.9406) |                       |
| GRE Quantitative Percentile | 0.0234***<br>(0.0090)  | 0.0136<br>(0.0086)    |                        |                            |                     |                            | 0.1127***<br>(0.0214)  |                       |
| Gender                      | -0.2143<br>(0.1304)    | -0.0515<br>(0.1314)   |                        |                            |                     |                            | 0.2244*<br>(0.1226)    |                       |
| Demographic 1               | 0.3587**<br>(0.1724)   | -0.3597**<br>(0.1719) |                        |                            |                     |                            | -0.3285*<br>(0.1835)   |                       |
| Demographic 2               | 0.0934<br>(0.1905)     | 0.1204<br>(0.1890)    |                        |                            |                     |                            | 0.0147<br>(0.1768)     |                       |
| Advanced Math               |                        | 0.9770***<br>(0.1271) |                        |                            |                     |                            | 0.5624***<br>(0.1710)  |                       |
| Committee Score             |                        |                       |                        |                            |                     |                            | 0.0529<br>(0.0377)     |                       |
| Year Transition More        | 0.0724<br>(0.1277)     | 0.1160<br>(0.1261)    | -0.0201<br>(0.0931)    | 0.7003<br>(0.5956, 0.8235) | 0.5863*<br>(0.3217) | 63.94%<br>(53.02%, 73.58%) | 0.0732<br>(0.2673)     |                       |
| Program Rank                |                        |                       |                        |                            |                     |                            | -0.0282***<br>(0.0027) |                       |
| $\sqrt{\text{MSE}}$         |                        | 1.9950***<br>(0.0422) |                        |                            |                     |                            |                        |                       |
| Sigma                       |                        |                       |                        |                            |                     |                            |                        | 0.1363***<br>(0.0247) |
| Observations                | 1000                   |                       |                        |                            |                     |                            |                        |                       |
| LL   Parameters   AIC       | -3765.3                | 28                    | 7586.6                 |                            |                     |                            |                        |                       |
| LR Statistic   DF   P-Value | 53.7172                | 9                     | 0.0000                 |                            |                     |                            |                        |                       |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01. Confidence intervals are 95%.

# A4: Restricted Dynamic Estimates (Suboptimal)

Table 20b: Restricted Dynamic Estimates (Suboptimal)

|                             | Advanced Math         | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|-----------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | -2.180***<br>(0.7588) | 3.2134***<br>(0.7270) | -0.3479***<br>(0.1032) | 0.7062<br>(0.5769, 0.8644) | -0.0962<br>(0.2228) | 47.60%<br>(36.98%, 58.43%) | -7.4740***<br>(2.3075) |                       |
| GRE Quantitative Percentile | 0.0243***<br>(0.0090) | 0.0173**<br>(0.0086)  |                        |                            |                     |                            | 0.0846***<br>(0.0246)  |                       |
| Gender                      | -0.2194*<br>(0.1307)  | -0.0391<br>(0.1310)   |                        |                            |                     |                            | 0.1565<br>(0.1219)     |                       |
| Demographic 1               | 0.3437**<br>(0.1729)  | -0.3903**<br>(0.1732) |                        |                            |                     |                            | -0.5023**<br>(0.2086)  |                       |
| Demographic 2               | 0.1004<br>(0.1906)    | 0.1302<br>(0.1892)    |                        |                            |                     |                            | -0.0312<br>(0.1793)    |                       |
| Advanced Math               |                       | 0.9783***<br>(0.1272) |                        |                            |                     |                            | 0.3499**<br>(0.1694)   |                       |
| Committee Score             |                       |                       |                        |                            |                     |                            | 0.1753***<br>(0.0367)  |                       |
| Year Transition More        | 0.0694<br>(0.1278)    | 0.1178<br>(0.1260)    | -0.0558<br>(0.0980)    | 0.6679<br>(0.5515, 0.8088) | 0.5471<br>(0.3423)  | 61.08%<br>(49.42%, 71.60%) | 0.1204<br>(0.2771)     |                       |
| Program Rank                |                       |                       |                        |                            |                     |                            | -0.0279***<br>(0.0027) |                       |
| $\sqrt{\text{MSE}}$         |                       | 1.9956***<br>(0.0423) |                        |                            |                     |                            |                        |                       |
| Sigma                       |                       |                       |                        |                            |                     |                            |                        | 0.1450***<br>(0.0396) |
| Observations                | 1000                  |                       |                        |                            |                     |                            |                        |                       |
| LL   Parameters   AIC       | -3808.9               | 28                    | 7673.9                 |                            |                     |                            |                        |                       |
| LR Statistic   DF   P-Value | 87.8382               | 9                     | 0.0000                 |                            |                     |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Unrestricted Dynamic Estimates, z Missing (Optimal)

Table 21a: Unrestricted Dynamic Estimates, z Missing (Optimal)

|                             | Selection              |                        | Advanced Math        |                     | Committee Score       | Transition              |                        | Accept                  |                        | Success                |                        |
|-----------------------------|------------------------|------------------------|----------------------|---------------------|-----------------------|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient            | AME                    | Coefficient          | AME                 |                       | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    |
| Intercept                   | -6.9053***<br>(0.5980) | -2.2429***<br>(0.1484) | -0.7532<br>(16.4490) | -0.1858<br>(4.0579) | 0.4540<br>(16.1199)   | -11.5912***<br>(1.0233) | -2.2563***<br>(0.1454) | -13.1834***<br>(2.0113) | -2.3773***<br>(0.2886) | -9.5112***<br>(2.0680) | -1.4815***<br>(0.2689) |
| GRE Quantitative Percentile | 0.0736***<br>(0.0070)  | 0.0239***<br>(0.0018)  | 0.0067<br>(0.1484)   | 0.0017<br>(0.0366)  | 0.0408<br>(0.1453)    | 0.1236***<br>(0.0118)   | 0.0241***<br>(0.0018)  | 0.1375***<br>(0.0223)   | 0.0248***<br>(0.0033)  | 0.1156***<br>(0.0227)  | 0.0180***<br>(0.0029)  |
| Gender                      | 0.1239<br>(0.0894)     | 0.0402<br>(0.0289)     | -0.0734<br>(0.3367)  | -0.0181<br>(0.0830) | -0.0113<br>(0.3305)   | 0.2260<br>(0.1464)      | 0.0440<br>(0.0283)     | 0.1303<br>(0.2504)      | 0.0235<br>(0.0451)     | 0.1258<br>(0.2697)     | 0.0196<br>(0.0419)     |
| Demographic 1               | -0.1824<br>(0.1205)    | -0.0593<br>(0.0390)    | 0.1516<br>(0.4510)   | 0.0374<br>(0.1112)  | -0.4348<br>(0.4207)   | -0.3017<br>(0.1988)     | -0.0587<br>(0.0386)    | -0.2665<br>(0.3745)     | -0.0480<br>(0.0672)    | -0.4607<br>(0.3930)    | -0.0718<br>(0.0608)    |
| Demographic 2               | 0.1058<br>(0.1373)     | 0.0344<br>(0.0446)     | -0.1438<br>(0.3776)  | -0.0355<br>(0.0931) | -0.1288<br>(0.3852)   | 0.1540<br>(0.2215)      | 0.0300<br>(0.0431)     | 0.1689<br>(0.3845)      | 0.0304<br>(0.0693)     | -0.2324<br>(0.4318)    | -0.0362<br>(0.0672)    |
| Advanced Math               |                        |                        |                      |                     | 0.6829***<br>(0.2024) |                         |                        | 0.7727***<br>(0.2536)   | 0.1393***<br>(0.0443)  | 0.6273**<br>(0.2747)   | 0.0977**<br>(0.0417)   |
| Committee Score             |                        |                        |                      |                     |                       |                         |                        | 0.0788<br>(0.0602)      | 0.0142<br>(0.0108)     | 0.0459<br>(0.0655)     | 0.0071<br>(0.0102)     |
| Year Transition More        | 0.5858***<br>(0.0953)  | 0.1903***<br>(0.0289)  | 0.3337<br>(1.1895)   | 0.0823<br>(0.2933)  | 0.4070<br>(1.1600)    | 0.9831***<br>(0.1472)   | 0.1914***<br>(0.0263)  | -0.9261***<br>(0.2514)  | -0.1670***<br>(0.0427) | -0.0242<br>(0.2761)    | -0.0038<br>(0.0430)    |
| Program Rank                |                        |                        |                      |                     |                       |                         |                        |                         |                        | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |
| Inverse Mills               |                        |                        | 0.0351<br>(3.0728)   | 0.0086<br>(0.7581)  | 0.8063<br>(2.9924)    |                         |                        |                         |                        |                        |                        |
| √MSE                        |                        |                        |                      |                     | 1.9477***<br>(0.0675) |                         |                        |                         |                        |                        |                        |
| Observations                | 1000                   |                        |                      |                     |                       |                         |                        |                         |                        |                        |                        |
| LL   Parameters   AIC       | -2554.0                | 45                     | 5197.9               |                     |                       |                         |                        |                         |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Unrestricted Dynamic Estimates, z Missing (Suboptimal)

Table 21b: Unrestricted Dynamic Estimates, z Missing (Suboptimal)

|                             | Selection              |                        | Advanced Math        |                     | Committee Score       |  | Transition              |                        | Accept                 |                        | Success                |                        |
|-----------------------------|------------------------|------------------------|----------------------|---------------------|-----------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient            | AME                    | Coefficient          | AME                 | Coefficient           |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -6.6165***<br>(0.5842) | -2.1481***<br>(0.1486) | -0.5421<br>(12.7975) | -0.1334<br>(3.1481) | 1.8701<br>(13.5038)   |  | -11.2042***<br>(1.0125) | -2.1767***<br>(0.1474) | -6.7415***<br>(1.7378) | -1.3322***<br>(0.3202) | -9.6658***<br>(1.9656) | -1.5071***<br>(0.2507) |
| GRE Quantitative Percentile | 0.0722***<br>(0.0069)  | 0.0234***<br>(0.0018)  | 0.0038<br>(0.1169)   | 0.0009<br>(0.0287)  | 0.0294<br>(0.1230)    |  | 0.1223***<br>(0.0117)   | 0.0238***<br>(0.0018)  | 0.0552***<br>(0.0191)  | 0.0109***<br>(0.0036)  | 0.1169***<br>(0.0216)  | 0.0182***<br>(0.0027)  |
| Gender                      | 0.1426<br>(0.0877)     | 0.0463<br>(0.0283)     | -0.1766<br>(0.3200)  | -0.0435<br>(0.0786) | -0.0351<br>(0.3308)   |  | 0.2581*<br>(0.1468)     | 0.0501*<br>(0.0283)    | 0.0114<br>(0.2367)     | 0.0023<br>(0.0468)     | 0.0600<br>(0.2670)     | 0.0094<br>(0.0416)     |
| Demographic 1               | -0.4934***<br>(0.1237) | -0.1602***<br>(0.0392) | 0.2524<br>(0.8210)   | 0.0621<br>(0.2019)  | -0.6904<br>(0.8275)   |  | -0.8236***<br>(0.1965)  | -0.1600***<br>(0.0371) | -0.3349<br>(0.3393)    | -0.0662<br>(0.0669)    | -0.5160<br>(0.3649)    | -0.0805<br>(0.0563)    |
| Demographic 2               | 0.0582<br>(0.1400)     | 0.0189<br>(0.0455)     | -0.0043<br>(0.3072)  | -0.0011<br>(0.0756) | -0.2509<br>(0.3042)   |  | 0.0826<br>(0.2134)      | 0.0160<br>(0.0415)     | 0.1764<br>(0.3345)     | 0.0349<br>(0.0659)     | -0.4738<br>(0.3859)    | -0.0739<br>(0.0598)    |
| Advanced Math               |                        |                        |                      |                     | 0.7188***<br>(0.1970) |  |                         |                        | 0.3552<br>(0.2375)     | 0.0702<br>(0.0465)     | 0.5347**<br>(0.2705)   | 0.0834**<br>(0.0416)   |
| Committee Score             |                        |                        |                      |                     |                       |  |                         |                        | 0.2960***<br>(0.0592)  | 0.0585***<br>(0.0106)  | 0.0624<br>(0.0643)     | 0.0097<br>(0.0100)     |
| Year Transition More        | 0.6602***<br>(0.0934)  | 0.2143***<br>(0.0278)  | 0.3658<br>(1.0892)   | 0.0900<br>(0.2678)  | 0.2878<br>(1.1431)    |  | 1.1100***<br>(0.1478)   | 0.2156***<br>(0.0257)  | -0.7155***<br>(0.2397) | -0.1414***<br>(0.0455) | 0.1338<br>(0.2796)     | 0.0209<br>(0.0434)     |
| Program Rank                |                        |                        |                      |                     |                       |  |                         |                        |                        |                        | -0.0279***<br>(0.0028) | -0.0043***<br>(0.0003) |
| Inverse Mills               |                        |                        | 0.0163<br>(2.5251)   | 0.0040<br>(0.6212)  | 0.6657<br>(2.6584)    |  |                         |                        |                        |                        |                        |                        |
| $\sqrt{\text{MSE}}$         |                        |                        |                      |                     | 1.9194***<br>(0.0682) |  |                         |                        |                        |                        |                        |                        |
| Observations                | 1000                   |                        |                      |                     |                       |  |                         |                        |                        |                        |                        |                        |
| LL   Parameters   AIC       | -2603.4                | 45                     | 5296.8               |                     |                       |  |                         |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Restricted Dynamic Estimates, z Missing (Optimal)

Table 22a: Restricted Dynamic Estimates, z Missing (Optimal)

|                             | Selection              | Advanced Math        | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|------------------------|----------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | -6.8792***<br>(0.5949) | -2.4193<br>(16.6178) | 1.4434<br>(16.1462)   | -0.3447***<br>(0.0812) | 0.7084<br>(0.6042, 0.8307) | -0.0214<br>(0.2119) | 49.46%<br>(39.25%, 59.72%) | -9.4493***<br>(1.9334) |                       |
| GRE Quantitative Percentile | 0.0733***<br>(0.0069)  | 0.0234<br>(0.1498)   | 0.0323<br>(0.1455)    |                        |                            |                     |                            | 0.1125***<br>(0.0213)  |                       |
| Gender                      | 0.1260<br>(0.0892)     | -0.0102<br>(0.3398)  | -0.0180<br>(0.3334)   |                        |                            |                     |                            | 0.2138*<br>(0.1218)    |                       |
| Demographic 1               | -0.1841<br>(0.1208)    | 0.1072<br>(0.4574)   | -0.4172<br>(0.4245)   |                        |                            |                     |                            | -0.3183*<br>(0.1815)   |                       |
| Demographic 2               | 0.1018<br>(0.1380)     | -0.1257<br>(0.3730)  | -0.1388<br>(0.3780)   |                        |                            |                     |                            | 0.0303<br>(0.1753)     |                       |
| Advanced Math               |                        |                      | 0.6924***<br>(0.2026) |                        |                            |                     |                            | 0.5558***<br>(0.1712)  |                       |
| Committee Score             |                        |                      |                       |                        |                            |                     |                            | 0.0495<br>(0.0374)     |                       |
| Year Transition More        | 0.5882***<br>(0.0958)  | 0.4228<br>(1.2131)   | 0.3258<br>(1.1708)    | -0.0114<br>(0.0939)    | 0.7004<br>(0.5959, 0.8233) | 0.5954*<br>(0.3209) | 63.97%<br>(53.08%, 73.59%) | 0.0819<br>(0.2670)     |                       |
| Program Rank                |                        |                      |                       |                        |                            |                     |                            | -0.0283***<br>(0.0027) |                       |
| Inverse Mills               |                        | 0.2346<br>(3.1142)   | 0.5910<br>(3.0031)    |                        |                            |                     |                            |                        |                       |
| $\sqrt{\text{MSE}}$         |                        |                      | 1.9477***<br>(0.0675) |                        |                            |                     |                            |                        |                       |
| Sigma                       |                        |                      |                       |                        |                            |                     |                            |                        | 0.1351***<br>(0.0245) |
| Observations                | 1000                   |                      |                       |                        |                            |                     |                            |                        |                       |
| LL   Parameters   AIC       | -2581.1                | 36                   | 5234.1                |                        |                            |                     |                            |                        |                       |
| LR Statistic   DF   P-Value | 54.2149                | 9                    | 0.0000                |                        |                            |                     |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.



# A4: Restricted Dynamic Estimates, z Missing (Suboptimal)

Table 22b: Restricted Dynamic Estimates, z Missing (Suboptimal)

|                             | Selection              | Advanced Math        | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|------------------------|----------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | -6.6868***<br>(0.5780) | -5.9722<br>(12.5445) | 6.0863<br>(12.6804)   | -0.3541***<br>(0.1030) | 0.7018<br>(0.5735, 0.8588) | -0.0872<br>(0.2230) | 47.82%<br>(37.18%, 58.66%) | -7.2568***<br>(2.2504) |                       |
| GRE Quantitative Percentile | 0.0730***<br>(0.0068)  | 0.0555<br>(0.1146)   | -0.0063<br>(0.1160)   |                        |                            |                     |                            | 0.0825***<br>(0.0240)  |                       |
| Gender                      | 0.1415<br>(0.0878)     | -0.0550<br>(0.3174)  | -0.0932<br>(0.3104)   |                        |                            |                     |                            | 0.1497<br>(0.1196)     |                       |
| Demographic 1               | -0.4971***<br>(0.1222) | -0.0903<br>(0.7976)  | -0.4590<br>(0.7904)   |                        |                            |                     |                            | -0.4767**<br>(0.2039)  |                       |
| Demographic 2               | 0.0466<br>(0.1367)     | 0.0249<br>(0.3185)   | -0.2698<br>(0.2805)   |                        |                            |                     |                            | -0.0081<br>(0.1739)    |                       |
| Advanced Math               |                        |                      | 0.7312***<br>(0.1979) |                        |                            |                     |                            | 0.3296**<br>(0.1679)   |                       |
| Committee Score             |                        |                      |                       |                        |                            |                     |                            | 0.1697***<br>(0.0362)  |                       |
| Year Transition More        | 0.6716***<br>(0.0901)  | 0.7657<br>(1.0909)   | -0.1418<br>(1.0900)   | -0.0559<br>(0.0984)    | 0.6637<br>(0.5483, 0.8032) | 0.5253<br>(0.3393)  | 60.78%<br>(49.25%, 71.22%) | 0.1089<br>(0.2762)     |                       |
| Program Rank                |                        |                      |                       |                        |                            |                     |                            | -0.0280***<br>(0.0027) |                       |
| Inverse Mills               |                        | 0.9443<br>(2.4602)   | -0.3480<br>(2.4678)   |                        |                            |                     |                            |                        |                       |
| $\sqrt{\text{MSE}}$         |                        |                      | 1.9214***<br>(0.0687) |                        |                            |                     |                            |                        |                       |
| Sigma                       |                        |                      |                       |                        |                            |                     |                            |                        | 0.1416***<br>(0.0389) |
| Observations                | 1000                   |                      |                       |                        |                            |                     |                            |                        |                       |
| LL   Parameters   AIC       | -2647.8                | 36                   | 5367.6                |                        |                            |                     |                            |                        |                       |
| LR Statistic   DF   P-Value | 88.8466                | 9                    | 0.0000                |                        |                            |                     |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Unrestricted Myopic Estimates (Optimal)

Table 23a: Unrestricted Myopic Estimates (Optimal)

|                             | Advanced Math      |                    | Committee Score       |  | Transition              |                        | Accept                  |                        | Success                |                        |
|-----------------------------|--------------------|--------------------|-----------------------|--|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient        | AME                | Coefficient           |  | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    |
| Intercept                   | 0.0000<br>(0.0894) | 0.0000<br>(0.0224) | 4.4770***<br>(0.1102) |  | -11.5716***<br>(1.0218) | -2.2538***<br>(0.1453) | -12.9691***<br>(1.9862) | -2.3484***<br>(0.2885) | -9.3783***<br>(2.0563) | -1.4632***<br>(0.2690) |
| GRE Quantitative Percentile |                    |                    |                       |  | 0.1234***<br>(0.0117)   | 0.0240***<br>(0.0018)  | 0.1352***<br>(0.0221)   | 0.0245***<br>(0.0033)  | 0.1144***<br>(0.0226)  | 0.0178***<br>(0.0029)  |
| Gender                      |                    |                    |                       |  | 0.2261<br>(0.1463)      | 0.0440<br>(0.0283)     | 0.1259<br>(0.2493)      | 0.0228<br>(0.0451)     | 0.1247<br>(0.2693)     | 0.0195<br>(0.0419)     |
| Demographic 1               |                    |                    |                       |  | -0.2971<br>(0.1987)     | -0.0579<br>(0.0386)    | -0.2548<br>(0.3731)     | -0.0461<br>(0.0673)    | -0.4727<br>(0.3922)    | -0.0738<br>(0.0608)    |
| Demographic 2               |                    |                    |                       |  | 0.1557<br>(0.2214)      | 0.0303<br>(0.0431)     | 0.1743<br>(0.3832)      | 0.0316<br>(0.0693)     | -0.2539<br>(0.4308)    | -0.0396<br>(0.0671)    |
| Advanced Math               |                    |                    | 0.9581***<br>(0.1269) |  |                         |                        | 0.7702***<br>(0.2525)   | 0.1395***<br>(0.0443)  | 0.6256**<br>(0.2743)   | 0.0976**<br>(0.0417)   |
| Committee Score             |                    |                    |                       |  |                         |                        | 0.0771<br>(0.0599)      | 0.0140<br>(0.0108)     | 0.0450<br>(0.0654)     | 0.0070<br>(0.0102)     |
| Year Transition More        | 0.0480<br>(0.1265) | 0.0120<br>(0.0316) | 0.1245<br>(0.1269)    |  | 0.9804***<br>(0.1471)   | 0.1910***<br>(0.0263)  | -0.9257***<br>(0.2503)  | -0.1676***<br>(0.0427) | -0.0277<br>(0.2757)    | -0.0043<br>(0.0430)    |
| Program Rank                |                    |                    |                       |  |                         |                        |                         |                        | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |
| $\sqrt{MSE}$                |                    |                    | 2.0063***<br>(0.0426) |  |                         |                        |                         |                        |                        |                        |
| Observations                | 1000               |                    |                       |  |                         |                        |                         |                        |                        |                        |
| LL   Parameters   AIC       | -3754.0            |                    | 29                    |  | 7566.0                  |                        |                         |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Unrestricted Myopic Estimates (Suboptimal)

Table 23b: Unrestricted Myopic Estimates (Suboptimal)

|                             | Advanced Math      |                    | Committee Score       | Transition              |                        | Accept                 |                        | Success                |                        |
|-----------------------------|--------------------|--------------------|-----------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient        | AME                | Coefficient           | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | 0.0000<br>(0.0894) | 0.0000<br>(0.0224) | 4.4770***<br>(0.1102) | -11.1839***<br>(1.0111) | -2.1737***<br>(0.1474) | -6.6736***<br>(1.7336) | -1.3193***<br>(0.3201) | -9.3520***<br>(1.9398) | -1.4646***<br>(0.2513) |
| GRE Quantitative Percentile |                    |                    |                       | 0.1221***<br>(0.0117)   | 0.0237***<br>(0.0018)  | 0.0545***<br>(0.0190)  | 0.0108***<br>(0.0036)  | 0.1137***<br>(0.0214)  | 0.0178***<br>(0.0027)  |
| Gender                      |                    |                    |                       | 0.2564*<br>(0.1467)     | 0.0498*<br>(0.0283)    | 0.0084<br>(0.2367)     | 0.0017<br>(0.0468)     | 0.0564<br>(0.2659)     | 0.0088<br>(0.0416)     |
| Demographic 1               |                    |                    |                       | -0.8277***<br>(0.1964)  | -0.1609***<br>(0.0371) | -0.3320<br>(0.3391)    | -0.0656<br>(0.0669)    | -0.5181<br>(0.3634)    | -0.0811<br>(0.0563)    |
| Demographic 2               |                    |                    |                       | 0.0807<br>(0.2132)      | 0.0157<br>(0.0414)     | 0.1761<br>(0.3343)     | 0.0348<br>(0.0659)     | -0.4851<br>(0.3840)    | -0.0760<br>(0.0597)    |
| Advanced Math               |                    |                    | 0.9581***<br>(0.1269) |                         |                        | 0.3556<br>(0.2374)     | 0.0703<br>(0.0465)     | 0.5345**<br>(0.2694)   | 0.0837**<br>(0.0416)   |
| Committee Score             |                    |                    |                       |                         |                        | 0.2959***<br>(0.0592)  | 0.0585***<br>(0.0106)  | 0.0594<br>(0.0640)     | 0.0093<br>(0.0100)     |
| Year Transition More        | 0.0480<br>(0.1265) | 0.0120<br>(0.0316) | 0.1245<br>(0.1269)    | 1.1073***<br>(0.1476)   | 0.2152***<br>(0.0257)  | -0.7208***<br>(0.2396) | -0.1425***<br>(0.0455) | 0.1235<br>(0.2783)     | 0.0193<br>(0.0434)     |
| Program Rank                |                    |                    |                       |                         |                        |                        |                        | -0.0279***<br>(0.0028) | -0.0044***<br>(0.0003) |
| √MSE                        |                    |                    | 2.0063***<br>(0.0426) |                         |                        |                        |                        |                        |                        |
| Observations                | 1000               |                    |                       |                         |                        |                        |                        |                        |                        |
| LL   Parameters   AIC       | -3780.5            | 29                 | 7619.1                |                         |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Restricted Myopic Estimates (Optimal)

Table 24a: Restricted Myopic Estimates (Optimal)

|                             | Advanced Math      | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|--------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | 0.0142<br>(0.0889) | 4.4801***<br>(0.1100) | -0.3358***<br>(0.0801) | 0.7148<br>(0.6110, 0.8363) | 0.0072<br>(0.2102)  | 50.18%<br>(40.02%, 60.33%) | -9.3825***<br>(1.9463) |                       |
| GRE Quantitative Percentile |                    |                       |                        |                            |                     |                            | 0.1123***<br>(0.0215)  |                       |
| Gender                      |                    |                       |                        |                            |                     |                            | 0.2094*<br>(0.1204)    |                       |
| Demographic 1               |                    |                       |                        |                            |                     |                            | -0.3133*<br>(0.1794)   |                       |
| Demographic 2               |                    |                       |                        |                            |                     |                            | 0.0257<br>(0.1730)     |                       |
| Advanced Math               |                    | 0.9603***<br>(0.1269) |                        |                            |                     |                            | 0.5383***<br>(0.1692)  |                       |
| Committee Score             |                    |                       |                        |                            |                     |                            | 0.0488<br>(0.0375)     |                       |
| Year Transition More        | 0.0536<br>(0.1261) | 0.1268<br>(0.1268)    | -0.0286<br>(0.0938)    | 0.6947<br>(0.5894, 0.8188) | 0.5497*<br>(0.3209) | 63.57%<br>(52.59%, 73.30%) | 0.0418<br>(0.2666)     |                       |
| Program Rank                |                    |                       |                        |                            |                     |                            | -0.0283***<br>(0.0027) |                       |
| $\sqrt{\text{MSE}}$         |                    | 2.0064***<br>(0.0426) |                        |                            |                     |                            |                        |                       |
| Sigma                       |                    |                       |                        |                            |                     |                            |                        | 0.1343***<br>(0.0246) |
| Observations                | 1000               |                       |                        |                            |                     |                            |                        |                       |
| LL   Parameters   AIC       | -3781.7            | 20                    | 7603.4                 |                            |                     |                            |                        |                       |
| LR Statistic   DF   P-Value | 55.4277            | 9                     | 0.0000                 |                            |                     |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Restricted Myopic Estimates (Suboptimal)

Table 24b: Restricted Myopic Estimates (Suboptimal)

|                             | Advanced Math      | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|--------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | 0.0128<br>(0.0891) | 4.4927***<br>(0.1102) | -0.3233***<br>(0.1044) | 0.7238<br>(0.5898, 0.8881) | -0.1041<br>(0.2279) | 47.40%<br>(36.57%, 58.48%) | -8.4360***<br>(2.3737) |                       |
| GRE Quantitative Percentile |                    |                       |                        |                            |                     |                            | 0.0955***<br>(0.0256)  |                       |
| Gender                      |                    |                       |                        |                            |                     |                            | 0.1572<br>(0.1307)     |                       |
| Demographic 1               |                    |                       |                        |                            |                     |                            | -0.5597**<br>(0.2190)  |                       |
| Demographic 2               |                    |                       |                        |                            |                     |                            | -0.0383<br>(0.1952)    |                       |
| Advanced Math               |                    | 0.9593***<br>(0.1270) |                        |                            |                     |                            | 0.3705**<br>(0.1785)   |                       |
| Committee Score             |                    |                       |                        |                            |                     |                            | 0.1748***<br>(0.0377)  |                       |
| Year Transition More        | 0.0496<br>(0.1263) | 0.1264<br>(0.1267)    | -0.0599<br>(0.0978)    | 0.6817<br>(0.5615, 0.8276) | 0.5985*<br>(0.3523) | 62.12%<br>(49.99%, 72.89%) | 0.1347<br>(0.2807)     |                       |
| Program Rank                |                    |                       |                        |                            |                     |                            | -0.0279***<br>(0.0028) |                       |
| $\sqrt{\text{MSE}}$         |                    | 2.0066***<br>(0.0427) |                        |                            |                     |                            |                        |                       |
| Sigma                       |                    |                       |                        |                            |                     |                            |                        | 0.1595***<br>(0.0412) |
| Observations                | 1000               |                       |                        |                            |                     |                            |                        |                       |
| LL   Parameters   AIC       | -3826.5            | 20                    | 7693.0                 |                            |                     |                            |                        |                       |
| LR Statistic   DF   P-Value | 91.9109            | 9                     | 0.0000                 |                            |                     |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Unrestricted Myopic Estimates, z Missing (Optimal)

Table 25a: Unrestricted Myopic Estimates, z Missing (Optimal)

|                             | Selection              |                        | Advanced Math       |                     | Committee Score       | Transition              |                        | Accept                  |                        | Success                |                        |
|-----------------------------|------------------------|------------------------|---------------------|---------------------|-----------------------|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient            | AME                    | Coefficient         | AME                 |                       | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    |
| Intercept                   | -6.8909***<br>(0.5823) | -2.2392***<br>(0.1446) | -0.0356<br>(0.3890) | -0.0088<br>(0.0965) | 4.6342***<br>(0.3853) | -11.5912***<br>(1.0234) | -2.2562***<br>(0.1454) | -13.0585***<br>(1.9964) | -2.3613***<br>(0.2886) | -9.4465***<br>(2.0619) | -1.4731***<br>(0.2690) |
| GRE Quantitative Percentile | 0.0735***<br>(0.0067)  | 0.0239***<br>(0.0018)  |                     |                     |                       | 0.1236***<br>(0.0118)   | 0.0241***<br>(0.0018)  | 0.1361***<br>(0.0222)   | 0.0246***<br>(0.0033)  | 0.1149***<br>(0.0226)  | 0.0179***<br>(0.0029)  |
| Gender                      | 0.1228<br>(0.0877)     | 0.0399<br>(0.0284)     |                     |                     |                       | 0.2268<br>(0.1464)      | 0.0441<br>(0.0283)     | 0.1296<br>(0.2497)      | 0.0234<br>(0.0451)     | 0.1250<br>(0.2694)     | 0.0195<br>(0.0419)     |
| Demographic 1               | -0.1830<br>(0.1209)    | -0.0595<br>(0.0392)    |                     |                     |                       | -0.3016<br>(0.1988)     | -0.0587<br>(0.0386)    | -0.2622<br>(0.3734)     | -0.0474<br>(0.0672)    | -0.4506<br>(0.3924)    | -0.0703<br>(0.0608)    |
| Demographic 2               | 0.0980<br>(0.1328)     | 0.0319<br>(0.0431)     |                     |                     |                       | 0.1543<br>(0.2215)      | 0.0300<br>(0.0431)     | 0.1694<br>(0.3835)      | 0.0306<br>(0.0693)     | -0.2284<br>(0.4313)    | -0.0356<br>(0.0672)    |
| Advanced Math               |                        |                        |                     |                     | 0.6765***<br>(0.2007) |                         |                        | 0.7693***<br>(0.2528)   | 0.1391***<br>(0.0443)  | 0.6250**<br>(0.2744)   | 0.0975**<br>(0.0417)   |
| Committee Score             |                        |                        |                     |                     |                       |                         |                        | 0.0784<br>(0.0600)      | 0.0142<br>(0.0108)     | 0.0456<br>(0.0654)     | 0.0071<br>(0.0102)     |
| Year Transition More        | 0.5880***<br>(0.0872)  | 0.1911***<br>(0.0264)  | 0.2996<br>(0.2346)  | 0.0743<br>(0.0577)  | 0.1113<br>(0.2267)    | 0.9841***<br>(0.1472)   | 0.1916***<br>(0.0263)  | -0.9229***<br>(0.2506)  | -0.1669***<br>(0.0427) | -0.0260<br>(0.2758)    | -0.0041<br>(0.0430)    |
| Program Rank                |                        |                        |                     |                     |                       |                         |                        |                         |                        | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |
| Inverse Mills               |                        |                        | -0.0743<br>(0.3370) | -0.0184<br>(0.0836) | 0.0311<br>(0.3087)    |                         |                        |                         |                        |                        |                        |
| √MSE                        |                        |                        |                     |                     | 1.9512***<br>(0.0675) |                         |                        |                         |                        |                        |                        |
| Observations                | 1000                   |                        |                     |                     |                       |                         |                        |                         |                        |                        |                        |
| LL   Parameters   AIC       | -2555.7                | 37                     | 5185.3              |                     |                       |                         |                        |                         |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Unrestricted Myopic Estimates, z Missing (Suboptimal)

Table 25b: Unrestricted Myopic Estimates, z Missing (Suboptimal)

|                             | Selection              |                        | Advanced Math       |                     | Committee Score       | Transition              |                        | Accept                 |                        | Success                |                        |
|-----------------------------|------------------------|------------------------|---------------------|---------------------|-----------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient            | AME                    | Coefficient         | AME                 |                       | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -6.6734***<br>(0.5784) | -2.1617***<br>(0.1467) | -0.2138<br>(0.3875) | -0.0529<br>(0.0958) | 4.7862***<br>(0.3887) | -11.2298***<br>(1.0144) | -2.1801***<br>(0.1474) | -6.7245***<br>(1.7375) | -1.3285***<br>(0.3202) | -9.4829***<br>(1.9504) | -1.4828***<br>(0.2511) |
| GRE Quantitative Percentile | 0.0729***<br>(0.0067)  | 0.0236***<br>(0.0018)  |                     |                     |                       | 0.1226***<br>(0.0117)   | 0.0238***<br>(0.0018)  | 0.0550***<br>(0.0191)  | 0.0109***<br>(0.0036)  | 0.1149***<br>(0.0215)  | 0.0180***<br>(0.0027)  |
| Gender                      | 0.1429<br>(0.0884)     | 0.0463<br>(0.0285)     |                     |                     |                       | 0.2571*<br>(0.1469)     | 0.0499*<br>(0.0283)    | 0.0097<br>(0.2368)     | 0.0019<br>(0.0468)     | 0.0577<br>(0.2663)     | 0.0090<br>(0.0416)     |
| Demographic 1               | -0.5017***<br>(0.1224) | -0.1625***<br>(0.0388) |                     |                     |                       | -0.8292***<br>(0.1966)  | -0.1610***<br>(0.0371) | -0.3330<br>(0.3393)    | -0.0658<br>(0.0669)    | -0.5073<br>(0.3638)    | -0.0793<br>(0.0563)    |
| Demographic 2               | 0.0514<br>(0.1303)     | 0.0166<br>(0.0422)     |                     |                     |                       | 0.0799<br>(0.2135)      | 0.0155<br>(0.0414)     | 0.1748<br>(0.3346)     | 0.0345<br>(0.0659)     | -0.4657<br>(0.3845)    | -0.0728<br>(0.0598)    |
| Advanced Math               |                        |                        |                     |                     | 0.7042***<br>(0.1959) |                         |                        | 0.3584<br>(0.2375)     | 0.0708<br>(0.0465)     | 0.5315**<br>(0.2697)   | 0.0831**<br>(0.0416)   |
| Committee Score             |                        |                        |                     |                     |                       |                         |                        | 0.2965***<br>(0.0592)  | 0.0586***<br>(0.0106)  | 0.0607<br>(0.0641)     | 0.0095<br>(0.0100)     |
| Year Transition More        | 0.6648***<br>(0.0874)  | 0.2153***<br>(0.0259)  | 0.3863<br>(0.2388)  | 0.0956<br>(0.0583)  | 0.0131<br>(0.2327)    | 1.1109***<br>(0.1479)   | 0.2157***<br>(0.0257)  | -0.7187***<br>(0.2397) | -0.1420***<br>(0.0455) | 0.1324<br>(0.2789)     | 0.0207<br>(0.0434)     |
| Program Rank                |                        |                        |                     |                     |                       |                         |                        |                        |                        | -0.0279***<br>(0.0028) | -0.0044***<br>(0.0003) |
| Inverse Mills               |                        |                        | 0.0677<br>(0.3344)  | 0.0168<br>(0.0828)  | -0.0261<br>(0.3195)   |                         |                        |                        |                        |                        |                        |
| √MSE                        |                        |                        |                     |                     | 1.9289***<br>(0.0683) |                         |                        |                        |                        |                        |                        |
| Observations                | 1000                   |                        |                     |                     |                       |                         |                        |                        |                        |                        |                        |
| LL   Parameters   AIC       | -2606.5                | 37                     | 5287.1              |                     |                       |                         |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Restricted Myopic Estimates, z Missing (Optimal)

Table 26a: Restricted Myopic Estimates, z Missing (Optimal)

|                             | Selection              | Advanced Math       | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|------------------------|---------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | -6.8904***<br>(0.5821) | 0.1647<br>(0.3873)  | 4.6943***<br>(0.3826) | -0.3407***<br>(0.0808) | 0.7113<br>(0.6071, 0.8334) | -0.0083<br>(0.2112) | 49.79%<br>(39.60%, 60.01%) | -9.4107***<br>(1.9311) |                       |
| GRE Quantitative Percentile | 0.0735***<br>(0.0067)  |                     |                       |                        |                            |                     |                            | 0.1123***<br>(0.0213)  |                       |
| Gender                      | 0.1228<br>(0.0879)     |                     |                       |                        |                            |                     |                            | 0.2102*<br>(0.1205)    |                       |
| Demographic 1               | -0.1830<br>(0.1212)    |                     |                       |                        |                            |                     |                            | -0.3135*<br>(0.1790)   |                       |
| Demographic 2               | 0.0948<br>(0.1331)     |                     |                       |                        |                            |                     |                            | 0.0201<br>(0.1734)     |                       |
| Advanced Math               |                        |                     | 0.6812***<br>(0.2010) |                        |                            |                     |                            | 0.5541***<br>(0.1704)  |                       |
| Committee Score             |                        |                     |                       |                        |                            |                     |                            | 0.0497<br>(0.0373)     |                       |
| Year Transition More        | 0.5864***<br>(0.0872)  | 0.2482<br>(0.2330)  | 0.0958<br>(0.2260)    | -0.0189<br>(0.0937)    | 0.6980<br>(0.5934, 0.8210) | 0.5724*<br>(0.3201) | 63.74%<br>(52.85%, 73.38%) | 0.0579<br>(0.2665)     |                       |
| Program Rank                |                        |                     |                       |                        |                            |                     |                            | -0.0282***<br>(0.0027) |                       |
| Inverse Mills               |                        | -0.2164<br>(0.3354) | -0.0135<br>(0.3083)   |                        |                            |                     |                            |                        |                       |
| $\sqrt{\text{MSE}}$         |                        |                     | 1.9515***<br>(0.0675) |                        |                            |                     |                            |                        |                       |
| Sigma                       |                        |                     |                       |                        |                            |                     |                            |                        | 0.1348***<br>(0.0245) |
| Observations                | 1000                   |                     |                       |                        |                            |                     |                            |                        |                       |
| LL   Parameters   AIC       | -2582.9                | 28                  | 5221.9                |                        |                            |                     |                            |                        |                       |
| LR Statistic   DF   P-Value | 54.5188                | 9                   | 0.0000                |                        |                            |                     |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.



# A4: Restricted Myopic Estimates, z Missing (Suboptimal)

Table 26b: Restricted Myopic Estimates, z Missing (Suboptimal)

|                             | Selection              | Advanced Math       | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|------------------------|---------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | -6.6682***<br>(0.5803) | 0.0078<br>(0.3895)  | 5.0936***<br>(0.3904) | -0.3395***<br>(0.1037) | 0.7121<br>(0.5811, 0.8727) | -0.0914<br>(0.2252) | 47.72%<br>(36.99%, 58.66%) | -7.7523***<br>(2.3404) |                       |
| GRE Quantitative Percentile | 0.0730***<br>(0.0067)  |                     |                       |                        |                            |                     |                            | 0.0879***<br>(0.0251)  |                       |
| Gender                      | 0.1385<br>(0.0887)     |                     |                       |                        |                            |                     |                            | 0.1506<br>(0.1232)     |                       |
| Demographic 1               | -0.5131***<br>(0.1232) |                     |                       |                        |                            |                     |                            | -0.5253**<br>(0.2119)  |                       |
| Demographic 2               | 0.0390<br>(0.1310)     |                     |                       |                        |                            |                     |                            | -0.0346<br>(0.1827)    |                       |
| Advanced Math               |                        |                     | 0.7137***<br>(0.1969) |                        |                            |                     |                            | 0.3526**<br>(0.1749)   |                       |
| Committee Score             |                        |                     |                       |                        |                            |                     |                            | 0.1753***<br>(0.0367)  |                       |
| Year Transition More        | 0.6644***<br>(0.0875)  | 0.3149<br>(0.2379)  | -0.0888<br>(0.2319)   | -0.0586<br>(0.0978)    | 0.6716<br>(0.5538, 0.8144) | 0.5511<br>(0.3453)  | 61.29%<br>(49.50%, 71.89%) | 0.1142<br>(0.2789)     |                       |
| Program Rank                |                        |                     |                       |                        |                            |                     |                            | -0.0279***<br>(0.0027) |                       |
| Inverse Mills               |                        | -0.1023<br>(0.3354) | -0.2670<br>(0.3202)   |                        |                            |                     |                            |                        |                       |
| $\sqrt{\text{MSE}}$         |                        |                     | 1.9315***<br>(0.0689) |                        |                            |                     |                            |                        |                       |
| Sigma                       |                        |                     |                       |                        |                            |                     |                            |                        | 0.1489***<br>(0.0403) |
| Observations                | 1000                   |                     |                       |                        |                            |                     |                            |                        |                       |
| LL   Parameters   AIC       | -2651.4                | 28                  | 5358.9                |                        |                            |                     |                            |                        |                       |
| LR Statistic   DF   P-Value | 89.7944                | 9                   | 0.0000                |                        |                            |                     |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Full Unrestricted Matriculation-Static Estimates (Optimal)

Table 27a: Unrestricted Matriculation-Static Estimates (Optimal)

|                                  | Transition              |                        | Accept                  |                        | Success 1              |                        | Success 2              |                        | Matriculate 1         |                       | Matriculate 2          |                        |
|----------------------------------|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|-----------------------|-----------------------|------------------------|------------------------|
|                                  | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient           | AME                   | Coefficient            | AME                    |
| Intercept                        | -11.5882***<br>(1.0232) | -2.2558***<br>(0.1454) | -13.1577***<br>(2.0081) | -2.3739***<br>(0.2886) | -8.1342***<br>(1.1379) | -1.1674***<br>(0.1344) | -9.5637***<br>(2.0728) | -1.4883***<br>(0.2688) | -9.1629**<br>(4.3152) | -1.4870**<br>(0.6431) | -4.5946<br>(4.6472)    | -0.6112<br>(0.6186)    |
| GRE Quantitative Percentile      | 0.1236***<br>(0.0118)   | 0.0241***<br>(0.0018)  | 0.1372***<br>(0.0223)   | 0.0248***<br>(0.0033)  | 0.1079***<br>(0.0133)  | 0.0155***<br>(0.0015)  | 0.1162***<br>(0.0227)  | 0.0181***<br>(0.0029)  | 0.0766*<br>(0.0463)   | 0.0124*<br>(0.0071)   | 0.0592<br>(0.0459)     | 0.0079<br>(0.0061)     |
| Gender                           | 0.2266<br>(0.1464)      | 0.0441<br>(0.0283)     | 0.1289<br>(0.2502)      | 0.0233<br>(0.0451)     | 0.1399<br>(0.1739)     | 0.0201<br>(0.0249)     | 0.1264<br>(0.2700)     | 0.0197<br>(0.0419)     | 0.4989<br>(0.4398)    | 0.0810<br>(0.0703)    | 0.4882<br>(0.5063)     | 0.0649<br>(0.0657)     |
| Demographic 1                    | -0.3013<br>(0.1988)     | -0.0587<br>(0.0386)    | -0.2639<br>(0.3744)     | -0.0476<br>(0.0672)    | -0.2532<br>(0.2359)    | -0.0363<br>(0.0338)    | -0.4607<br>(0.3933)    | -0.0717<br>(0.0608)    | 0.3408<br>(0.5565)    | 0.0553<br>(0.0907)    | 0.3585<br>(0.7459)     | 0.0477<br>(0.1001)     |
| Demographic 2                    | 0.1538<br>(0.2215)      | 0.0299<br>(0.0431)     | 0.1709<br>(0.3844)      | 0.0308<br>(0.0693)     | -0.1179<br>(0.2580)    | -0.0169<br>(0.0371)    | -0.2340<br>(0.4323)    | -0.0364<br>(0.0672)    | -0.1604<br>(0.6964)   | -0.0260<br>(0.1119)   | 0.1832<br>(0.7962)     | 0.0244<br>(0.1069)     |
| Advanced Math                    |                         |                        | 0.7737***<br>(0.2535)   | 0.1396***<br>(0.0443)  |                        |                        | 0.6276**<br>(0.2749)   | 0.0977**<br>(0.0417)   |                       |                       | 0.8105<br>(0.5772)     | 0.1078<br>(0.0733)     |
| Committee Score                  |                         |                        | 0.0783<br>(0.0602)      | 0.0141<br>(0.0108)     |                        |                        | 0.0465<br>(0.0655)     | 0.0072<br>(0.0102)     |                       |                       | -0.7403***<br>(0.2065) | -0.0985***<br>(0.0235) |
| Year Transition/Matriculate More | 0.9840***<br>(0.1472)   | 0.1915***<br>(0.0263)  | -0.9251***<br>(0.2513)  | -0.1669***<br>(0.0427) | -0.0587<br>(0.1690)    | -0.0084<br>(0.0242)    | -0.0234<br>(0.2763)    | -0.0036<br>(0.0430)    | 2.8450***<br>(0.5222) | 0.4617***<br>(0.0280) | 3.3277***<br>(0.5672)  | 0.4427***<br>(0.0331)  |
| Program Rank                     |                         |                        |                         |                        | -0.0310***<br>(0.0019) | -0.0045***<br>(0.0002) | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |                       |                       |                        |                        |
| Observations                     | 1000                    |                        |                         |                        |                        |                        |                        |                        |                       |                       |                        |                        |
| LL   Parameters   AIC            | -1491.1                 | 44                     | 3070.2                  |                        |                        |                        |                        |                        |                       |                       |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Full Unrestricted Matriculation-Static Estimates (Suboptimal)

Table 27b: Unrestricted Matriculation-Static Estimates (Suboptimal)

|                                  | Transition              |                        | Accept                 |                        | Success 1              |                        | Success 2              |                        | Matriculate 1         |                       | Matriculate 2          |                        |
|----------------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|-----------------------|-----------------------|------------------------|------------------------|
|                                  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient           | AME                   | Coefficient            | AME                    |
| Intercept                        | -11.2298***<br>(1.0144) | -2.1801***<br>(0.1474) | -6.7300***<br>(1.7376) | -1.3296***<br>(0.3202) | -8.2841***<br>(1.1393) | -1.1915***<br>(0.1341) | -9.6026***<br>(1.9608) | -1.4981***<br>(0.2508) | -6.1997*<br>(3.2762)  | -0.9843*<br>(0.5037)  | -1.0401<br>(3.6083)    | -0.1351<br>(0.4721)    |
| GRE Quantitative Percentile      | 0.1226***<br>(0.0117)   | 0.0238***<br>(0.0018)  | 0.0550***<br>(0.0191)  | 0.0109***<br>(0.0036)  | 0.1097***<br>(0.0133)  | 0.0158***<br>(0.0015)  | 0.1162***<br>(0.0216)  | 0.0181***<br>(0.0027)  | 0.0419<br>(0.0375)    | 0.0066<br>(0.0059)    | 0.0129<br>(0.0391)     | 0.0017<br>(0.0051)     |
| Gender                           | 0.2575*<br>(0.1469)     | 0.0500*<br>(0.0283)    | 0.0098<br>(0.2368)     | 0.0019<br>(0.0468)     | 0.1355<br>(0.1741)     | 0.0195<br>(0.0250)     | 0.0584<br>(0.2669)     | 0.0091<br>(0.0416)     | 0.4469<br>(0.4602)    | 0.0710<br>(0.0717)    | 0.1298<br>(0.4717)     | 0.0169<br>(0.0608)     |
| Demographic 1                    | -0.8293***<br>(0.1966)  | -0.1610***<br>(0.0371) | -0.3340<br>(0.3394)    | -0.0660<br>(0.0669)    | -0.2777<br>(0.2344)    | -0.0399<br>(0.0337)    | -0.5156<br>(0.3648)    | -0.0804<br>(0.0563)    | 0.2529<br>(0.5465)    | 0.0402<br>(0.0870)    | 0.6328<br>(0.6281)     | 0.0822<br>(0.0823)     |
| Demographic 2                    | 0.0797<br>(0.2135)      | 0.0155<br>(0.0414)     | 0.1764<br>(0.3346)     | 0.0348<br>(0.0659)     | -0.1809<br>(0.2566)    | -0.0260<br>(0.0369)    | -0.4743<br>(0.3856)    | -0.0740<br>(0.0598)    | -0.6596<br>(0.6474)   | -0.1047<br>(0.0966)   | -0.2222<br>(0.6864)    | -0.0289<br>(0.0880)    |
| Advanced Math                    |                         |                        | 0.3580<br>(0.2375)     | 0.0707<br>(0.0465)     |                        |                        | 0.5366**<br>(0.2704)   | 0.0837**<br>(0.0416)   |                       |                       | 0.5876<br>(0.5321)     | 0.0763<br>(0.0674)     |
| Committee Score                  |                         |                        | 0.2962***<br>(0.0592)  | 0.0585***<br>(0.0106)  |                        |                        | 0.0616<br>(0.0643)     | 0.0096<br>(0.0100)     |                       |                       | -0.6092***<br>(0.1637) | -0.0791***<br>(0.0176) |
| Year Transition/Matriculate More | 1.1108***<br>(0.1479)   | 0.2156***<br>(0.0257)  | -0.7185***<br>(0.2397) | -0.1420***<br>(0.0455) | -0.0766<br>(0.1688)    | -0.0110<br>(0.0243)    | 0.1315<br>(0.2794)     | 0.0205<br>(0.0434)     | 2.8800***<br>(0.5694) | 0.4572***<br>(0.0414) | 3.4846***<br>(0.6046)  | 0.4526***<br>(0.0349)  |
| Program Rank                     |                         |                        |                        |                        | -0.0304***<br>(0.0019) | -0.0044***<br>(0.0002) | -0.0279***<br>(0.0028) | -0.0044***<br>(0.0003) |                       |                       |                        |                        |
| Observations                     | 1000                    |                        |                        |                        |                        |                        |                        |                        |                       |                       |                        |                        |
| LL   Parameters   AIC            | -1521.6                 | 44                     | 3131.1                 |                        |                        |                        |                        |                        |                       |                       |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Full Restricted Matriculation-Static Estimates (Optimal)

Table 28a: Restricted Matriculation-Static Estimates (Optimal)

|                                  | Cutoff 1               | $P(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success 1              | Success 2              | Matriculate 1          | Matriculate 2          | Sigma                 |
|----------------------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|------------------------|------------------------|------------------------|-----------------------|
| Intercept                        | -1.0201***<br>(0.1697) | 26.50%<br>(20.54%, 33.46%) | 0.1479<br>(0.2334)  | 53.69%<br>(42.32%, 64.69%) | -8.0229***<br>(1.2042) | -8.3237***<br>(1.7684) | -8.2894***<br>(2.5032) | -4.6066<br>(4.6480)    |                       |
| GRE Quantitative Percentile      |                        |                            |                     |                            | 0.1071***<br>(0.0140)  | 0.1006***<br>(0.0194)  | 0.0788***<br>(0.0269)  | 0.0593<br>(0.0459)     |                       |
| Gender                           |                        |                            |                     |                            | 0.1039<br>(0.1522)     | 0.1026<br>(0.1470)     | 0.3893*<br>(0.2210)    | 0.4885<br>(0.5064)     |                       |
| Demographic 1                    |                        |                            |                     |                            | -0.2437<br>(0.2094)    | -0.2769<br>(0.2145)    | -0.0899<br>(0.3273)    | 0.3587<br>(0.7459)     |                       |
| Demographic 2                    |                        |                            |                     |                            | -0.0480<br>(0.2258)    | 0.0110<br>(0.2131)     | 0.0296<br>(0.3847)     | 0.1834<br>(0.7962)     |                       |
| Advanced Math                    |                        |                            |                     |                            |                        | 0.5492***<br>(0.1679)  |                        | 0.8106<br>(0.5773)     |                       |
| Committee Score                  |                        |                            |                     |                            |                        | 0.0492<br>(0.0368)     |                        | -0.7402***<br>(0.2064) |                       |
| Year Transition/Matriculate More | 0.1323<br>(0.2222)     | 29.16%<br>(21.96%, 37.57%) | 0.6306*<br>(0.3547) | 68.54%<br>(55.89%, 78.93%) | -0.1369<br>(0.1686)    | 0.0137<br>(0.2730)     | 1.8740***<br>(0.4069)  | 3.3280***<br>(0.5674)  |                       |
| Program Rank                     |                        |                            |                     |                            | -0.0311***<br>(0.0019) | -0.0280***<br>(0.0026) |                        |                        |                       |
| Sigma                            |                        |                            |                     |                            |                        |                        |                        |                        | 0.1554***<br>(0.0253) |
| Observations                     | 1000                   |                            |                     |                            |                        |                        |                        |                        |                       |
| LL   Parameters   AIC            | -1501.0                | 35                         | 3072.1              |                            |                        |                        |                        |                        |                       |
| LR Statistic   DF   P-Value      | 19.8266                | 9                          | 0.0190              |                            |                        |                        |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Full Restricted Matriculation-Static Estimates (Suboptimal)

Table 28b: Restricted Matriculation-Static Estimates (Suboptimal)

|                                  | Cutoff 1               | $P(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success 1              | Success 2              | Matriculate 1          | Matriculate 2          | Sigma                 |
|----------------------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|------------------------|------------------------|-----------------------|
| Intercept                        | -1.1917***<br>(0.2112) | 23.30%<br>(16.72%, 31.48%) | 0.1231<br>(0.2396) | 53.07%<br>(41.42%, 64.40%) | -8.0768***<br>(1.2382) | -3.7273**<br>(1.6988)  | -6.1199***<br>(2.2306) | -1.0400<br>(3.6084)    |                       |
| GRE Quantitative Percentile      |                        |                            |                    |                            | 0.1084***<br>(0.0144)  | 0.0455***<br>(0.0175)  | 0.0569**<br>(0.0248)   | 0.0129<br>(0.0391)     |                       |
| Gender                           |                        |                            |                    |                            | 0.1111<br>(0.1581)     | 0.0275<br>(0.1164)     | 0.3411<br>(0.2134)     | 0.1298<br>(0.4717)     |                       |
| Demographic 1                    |                        |                            |                    |                            | -0.3744*<br>(0.2131)   | -0.2437<br>(0.1781)    | -0.5399*<br>(0.2762)   | 0.6329<br>(0.6281)     |                       |
| Demographic 2                    |                        |                            |                    |                            | -0.0814<br>(0.2176)    | -0.0173<br>(0.1697)    | -0.0874<br>(0.3016)    | -0.2225<br>(0.6864)    |                       |
| Advanced Math                    |                        |                            |                    |                            |                        | 0.2559*<br>(0.1400)    |                        | 0.5874<br>(0.5320)     |                       |
| Committee Score                  |                        |                            |                    |                            |                        | 0.1375***<br>(0.0346)  |                        | -0.6092***<br>(0.1637) |                       |
| Year Transition/Matriculate More | -0.0312<br>(0.2089)    | 22.74%<br>(16.19%, 30.97%) | 0.3687<br>(0.3334) | 62.05%<br>(50.06%, 72.73%) | -0.1520<br>(0.1700)    | 0.0048<br>(0.2775)     | 1.4674***<br>(0.4102)  | 3.4846***<br>(0.6047)  |                       |
| Program Rank                     |                        |                            |                    |                            | -0.0305***<br>(0.0019) | -0.0281***<br>(0.0026) |                        |                        |                       |
| Sigma                            |                        |                            |                    |                            |                        |                        |                        |                        | 0.1279***<br>(0.0330) |
| Observations                     | 1000                   |                            |                    |                            |                        |                        |                        |                        |                       |
| LL   Parameters   AIC            | -1547.8                | 35                         | 3165.6             |                            |                        |                        |                        |                        |                       |
| LR Statistic   DF   P-Value      | 52.5157                | 9                          | 0.0000             |                            |                        |                        |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Unrestricted Matriculation-Dynamic Estimates (Optimal)

Table 29a: Unrestricted Matriculation-Dynamic Estimates (Optimal)

|                                  | Advanced Math         |                       | Committee Score       |  | Transition              |                        | Accept                  |                        | Success                |                        | Matriculate            |                        |
|----------------------------------|-----------------------|-----------------------|-----------------------|--|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                  | Coefficient           | AME                   | Coefficient           |  | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                        | -1.8996**<br>(0.7563) | -0.4655**<br>(0.1830) | 3.5766***<br>(0.7240) |  | -11.5894***<br>(1.0233) | -2.2560***<br>(0.1454) | -13.1546***<br>(2.0078) | -2.3735***<br>(0.2886) | -9.5555***<br>(2.0722) | -1.4871***<br>(0.2688) | -4.5520<br>(4.6442)    | -0.6056<br>(0.6185)    |
| GRE Quantitative Percentile      | 0.0207**<br>(0.0090)  | 0.0051**<br>(0.0022)  | 0.0127<br>(0.0086)    |  | 0.1236***<br>(0.0118)   | 0.0241***<br>(0.0018)  | 0.1372***<br>(0.0223)   | 0.0248***<br>(0.0033)  | 0.1161***<br>(0.0227)  | 0.0181***<br>(0.0029)  | 0.0588<br>(0.0458)     | 0.0078<br>(0.0061)     |
| Gender                           | -0.2335*<br>(0.1309)  | -0.0572*<br>(0.0319)  | -0.0579<br>(0.1315)   |  | 0.2266<br>(0.1464)      | 0.0441<br>(0.0283)     | 0.1289<br>(0.2502)      | 0.0233<br>(0.0451)     | 0.1263<br>(0.2699)     | 0.0197<br>(0.0419)     | 0.4868<br>(0.5061)     | 0.0648<br>(0.0657)     |
| Demographic 1                    | 0.3691**<br>(0.1732)  | 0.0905**<br>(0.0421)  | -0.3564**<br>(0.1714) |  | -0.3014<br>(0.1988)     | -0.0587<br>(0.0386)    | -0.2641<br>(0.3743)     | -0.0477<br>(0.0672)    | -0.4598<br>(0.3933)    | -0.0716<br>(0.0608)    | 0.3575<br>(0.7459)     | 0.0476<br>(0.1001)     |
| Demographic 2                    | 0.0924<br>(0.1911)    | 0.0226<br>(0.0468)    | 0.1202<br>(0.1890)    |  | 0.1537<br>(0.2215)      | 0.0299<br>(0.0431)     | 0.1707<br>(0.3844)      | 0.0308<br>(0.0693)     | -0.2336<br>(0.4322)    | -0.0363<br>(0.0672)    | 0.1821<br>(0.7962)     | 0.0242<br>(0.1069)     |
| Advanced Math                    |                       |                       | 0.9753***<br>(0.1270) |  |                         |                        | 0.7738***<br>(0.2535)   | 0.1396***<br>(0.0443)  | 0.6274**<br>(0.2749)   | 0.0976**<br>(0.0417)   | 0.8104<br>(0.5770)     | 0.1078<br>(0.0733)     |
| Committee Score                  |                       |                       |                       |  |                         |                        | 0.0783<br>(0.0602)      | 0.0141<br>(0.0108)     | 0.0464<br>(0.0655)     | 0.0072<br>(0.0102)     | -0.7407***<br>(0.2066) | -0.0985***<br>(0.0235) |
| Year Transition/Matriculate More | 0.0663<br>(0.1281)    | 0.0163<br>(0.0314)    | 0.1134<br>(0.1262)    |  | 0.9841***<br>(0.1472)   | 0.1916***<br>(0.0263)  | -0.9251***<br>(0.2513)  | -0.1669***<br>(0.0427) | -0.0235<br>(0.2763)    | -0.0037<br>(0.0430)    | 3.3263***<br>(0.5667)  | 0.4425***<br>(0.0331)  |
| Program Rank                     |                       |                       |                       |  |                         |                        |                         |                        | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |                        |                        |
| $\sqrt{\text{MSE}}$              |                       |                       | 1.9949***<br>(0.0422) |  |                         |                        |                         |                        |                        |                        |                        |                        |
| Observations                     | 1000                  |                       |                       |  |                         |                        |                         |                        |                        |                        |                        |                        |
| LL   Parameters   AIC            | -3790.8      45       |                       | 7671.7                |  |                         |                        |                         |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

Table 29b: Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

|                                  | Advanced Math         |                       | Committee Score       | Transition              |                        | Accept                 |                        | Success                |                        | Matriculate            |                        |
|----------------------------------|-----------------------|-----------------------|-----------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                  | Coefficient           | AME                   |                       | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                        | -1.9040**<br>(0.7564) | -0.4666**<br>(0.1830) | 3.5860***<br>(0.7240) | -11.2294***<br>(1.0144) | -2.1800***<br>(0.1474) | -6.7296***<br>(1.7376) | -1.3296***<br>(0.3202) | -9.5956***<br>(1.9602) | -1.4971***<br>(0.2508) | -1.0400<br>(3.6083)    | -0.1351<br>(0.4721)    |
| GRE Quantitative Percentile      | 0.0208**<br>(0.0090)  | 0.0051**<br>(0.0022)  | 0.0126<br>(0.0086)    | 0.1226***<br>(0.0117)   | 0.0238***<br>(0.0018)  | 0.0550***<br>(0.0191)  | 0.0109***<br>(0.0036)  | 0.1162***<br>(0.0216)  | 0.0181***<br>(0.0027)  | 0.0129<br>(0.0391)     | 0.0017<br>(0.0051)     |
| Gender                           | -0.2334*<br>(0.1309)  | -0.0572*<br>(0.0319)  | -0.0581<br>(0.1315)   | 0.2575*<br>(0.1469)     | 0.0500*<br>(0.0283)    | 0.0099<br>(0.2368)     | 0.0020<br>(0.0468)     | 0.0583<br>(0.2669)     | 0.0091<br>(0.0416)     | 0.1298<br>(0.4717)     | 0.0169<br>(0.0608)     |
| Demographic 1                    | 0.3691**<br>(0.1732)  | 0.0904**<br>(0.0421)  | -0.3561**<br>(0.1714) | -0.8293***<br>(0.1966)  | -0.1610***<br>(0.0371) | -0.3340<br>(0.3394)    | -0.0660<br>(0.0669)    | -0.5154<br>(0.3647)    | -0.0804<br>(0.0563)    | 0.6329<br>(0.6282)     | 0.0822<br>(0.0823)     |
| Demographic 2                    | 0.0925<br>(0.1911)    | 0.0227<br>(0.0468)    | 0.1200<br>(0.1890)    | 0.0798<br>(0.2135)      | 0.0155<br>(0.0414)     | 0.1764<br>(0.3346)     | 0.0348<br>(0.0659)     | -0.4742<br>(0.3856)    | -0.0740<br>(0.0598)    | -0.2221<br>(0.6864)    | -0.0288<br>(0.0880)    |
| Advanced Math                    |                       |                       | 0.9753***<br>(0.1270) |                         |                        | 0.3580<br>(0.2375)     | 0.0707<br>(0.0465)     | 0.5367**<br>(0.2704)   | 0.0837**<br>(0.0416)   | 0.5876<br>(0.5321)     | 0.0763<br>(0.0674)     |
| Committee Score                  |                       |                       |                       |                         |                        | 0.2962***<br>(0.0592)  | 0.0585***<br>(0.0106)  | 0.0616<br>(0.0642)     | 0.0096<br>(0.0100)     | -0.6093***<br>(0.1637) | -0.0791***<br>(0.0176) |
| Year Transition/Matriculate More | 0.0664<br>(0.1281)    | 0.0163<br>(0.0314)    | 0.1132<br>(0.1261)    | 1.1108***<br>(0.1479)   | 0.2157***<br>(0.0257)  | -0.7186***<br>(0.2397) | -0.1420***<br>(0.0455) | 0.1312<br>(0.2794)     | 0.0205<br>(0.0434)     | 3.4846***<br>(0.6046)  | 0.4526***<br>(0.0349)  |
| Program Rank                     |                       |                       |                       |                         |                        |                        |                        | -0.0279***<br>(0.0028) | -0.0044***<br>(0.0003) |                        |                        |
| $\sqrt{\text{MSE}}$              |                       |                       | 1.9949***<br>(0.0422) |                         |                        |                        |                        |                        |                        |                        |                        |
| Observations                     | 1000                  |                       |                       |                         |                        |                        |                        |                        |                        |                        |                        |
| LL   Parameters   AIC            | -3818.8               |                       | 45                    | 7727.6                  |                        |                        |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

►► Unrestricted Matriculation-Static AMEs (Suboptimal)

# A4: Restricted Matriculation-Dynamic Estimates (Optimal)

Table 30a: Restricted Matriculation-Dynamic Estimates (Optimal)

|                                  | Advanced Math          | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Matriculate            | Sigma                 |
|----------------------------------|------------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                        | -2.1669***<br>(0.7595) | 3.8799***<br>(0.7336) | -1.8456***<br>(0.2148) | 0.1579<br>(0.1037, 0.2406) | -0.0356<br>(0.2309) | 49.11%<br>(38.03%, 60.27%) | -9.5935***<br>(1.9857) | -9.8293**<br>(3.9945)  |                       |
| GRE Quantitative Percentile      | 0.0240***<br>(0.0090)  | 0.0089<br>(0.0087)    |                        |                            |                     |                            | 0.1148***<br>(0.0219)  | 0.1167***<br>(0.0390)  |                       |
| Gender                           | -0.2221*<br>(0.1308)   | -0.0705<br>(0.1313)   |                        |                            |                     |                            | 0.1531<br>(0.1426)     | 0.7455*<br>(0.4502)    |                       |
| Demographic 1                    | 0.3610**<br>(0.1730)   | -0.3432**<br>(0.1710) |                        |                            |                     |                            | -0.3117<br>(0.2128)    | 0.1035<br>(0.7071)     |                       |
| Demographic 2                    | 0.0936<br>(0.1908)     | 0.1159<br>(0.1888)    |                        |                            |                     |                            | 0.0162<br>(0.2072)     | 0.0750<br>(0.7667)     |                       |
| Advanced Math                    |                        | 0.9646***<br>(0.1272) |                        |                            |                     |                            | 0.6048***<br>(0.1759)  | 1.0214*<br>(0.5809)    |                       |
| Committee Score                  |                        |                       |                        |                            |                     |                            | 0.0301<br>(0.0398)     | -0.6438***<br>(0.1969) |                       |
| Year Transition/Matriculate More | 0.0723<br>(0.1280)     | 0.1304<br>(0.1260)    | -1.1129***<br>(0.2818) | 0.0519<br>(0.0278, 0.0970) | 0.6391*<br>(0.3414) | 64.65%<br>(53.26%, 74.59%) | 0.0969<br>(0.2727)     | 3.0092***<br>(0.5385)  |                       |
| Program Rank                     |                        |                       |                        |                            |                     |                            | -0.0281***<br>(0.0027) |                        |                       |
| $\sqrt{\text{MSE}}$              |                        | 1.9943***<br>(0.0422) |                        |                            |                     |                            |                        |                        |                       |
| Sigma                            |                        |                       |                        |                            |                     |                            |                        |                        | 0.1442***<br>(0.0289) |
| Observations                     | 1000                   |                       |                        |                            |                     |                            |                        |                        |                       |
| LL   Parameters   AIC            | -3838.7                | 36                    | 7749.4                 |                            |                     |                            |                        |                        |                       |
| LR Statistic   DF   P-Value      | 95.7273                | 9                     | 0.0000                 |                            |                     |                            |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.



# A4: Restricted Matriculation-Dynamic Estimates (Suboptimal)

Table 30b: Restricted Matriculation-Dynamic Estimates (Suboptimal)

|                                  | Advanced Math          | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success                | Matriculate            | Sigma                 |
|----------------------------------|------------------------|-----------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                        | -2.0386***<br>(0.7576) | 3.6309***<br>(0.7286) | -1.5730***<br>(0.3128) | 0.2074<br>(0.1124, 0.3829) | 0.0671<br>(0.2646) | 51.68%<br>(38.90%, 64.24%) | -8.2302***<br>(2.4820) | -6.3541*<br>(3.3941)   |                       |
| GRE Quantitative Percentile      | 0.0225**<br>(0.0090)   | 0.0120<br>(0.0087)    |                        |                            |                    |                            | 0.0939***<br>(0.0270)  | 0.0749**<br>(0.0366)   |                       |
| Gender                           | -0.2288*<br>(0.1310)   | -0.0613<br>(0.1309)   |                        |                            |                    |                            | 0.0806<br>(0.1654)     | 0.3939<br>(0.4605)     |                       |
| Demographic 1                    | 0.3557**<br>(0.1730)   | -0.3494**<br>(0.1709) |                        |                            |                    |                            | -0.5220**<br>(0.2624)  | 0.0541<br>(0.6458)     |                       |
| Demographic 2                    | 0.0919<br>(0.1908)     | 0.1134<br>(0.1884)    |                        |                            |                    |                            | -0.1330<br>(0.2603)    | -0.4344<br>(0.6927)    |                       |
| Advanced Math                    |                        | 0.9720***<br>(0.1269) |                        |                            |                    |                            | 0.4381**<br>(0.1981)   | 0.7196<br>(0.5458)     |                       |
| Committee Score                  |                        |                       |                        |                            |                    |                            | 0.1748***<br>(0.0395)  | -0.5253***<br>(0.1603) |                       |
| Year Transition/Matriculate More | 0.0721<br>(0.1280)     | 0.1248<br>(0.1260)    | -1.3756***<br>(0.3777) | 0.0524<br>(0.0210, 0.1309) | 0.5763<br>(0.4136) | 65.55%<br>(50.60%, 77.95%) | 0.0809<br>(0.2936)     | 3.1389***<br>(0.5995)  |                       |
| Program Rank                     |                        |                       |                        |                            |                    |                            | -0.0281***<br>(0.0027) |                        |                       |
| $\sqrt{\text{MSE}}$              |                        | 1.9930***<br>(0.0421) |                        |                            |                    |                            |                        |                        |                       |
| Sigma                            |                        |                       |                        |                            |                    |                            |                        |                        | 0.2012***<br>(0.0603) |
| Observations                     | 1000                   |                       |                        |                            |                    |                            |                        |                        |                       |
| LL   Parameters   AIC            | -3886.7                | 36                    | 7845.5                 |                            |                    |                            |                        |                        |                       |
| LR Statistic   DF   P-Value      | 135.8481               | 9                     | 0.0000                 |                            |                    |                            |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Unrestricted Matriculation-Dynamic Estimates, z Missing (Optimal)

Table 31a: Unrestricted Matriculation-Dynamic Estimates, z Missing (Optimal)

|                                  | Selection              |                        | Advanced Math        |                     | Committee Score       | Transition              |                        | Accept                  |                        | Success                |                        | Matriculate            |                        |
|----------------------------------|------------------------|------------------------|----------------------|---------------------|-----------------------|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                  | Coefficient            | AME                    | Coefficient          | AME                 |                       | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                        | -6.9439***<br>(0.5964) | -2.2520***<br>(0.1474) | -1.3596<br>(16.2376) | -0.3354<br>(4.0058) | 0.3551<br>(15.9808)   | -11.5954***<br>(1.0237) | -2.2568***<br>(0.1454) | -13.1564***<br>(2.0079) | -2.3738***<br>(0.2886) | -9.5573***<br>(2.0723) | -1.4875***<br>(0.2688) | -4.6257<br>(4.6540)    | -0.6149<br>(0.6190)    |
| GRE Quantitative Percentile      | 0.0741***<br>(0.0070)  | 0.0240***<br>(0.0018)  | 0.0121<br>(0.1466)   | 0.0030<br>(0.0362)  | 0.0417<br>(0.1442)    | 0.1236***<br>(0.0118)   | 0.0241***<br>(0.0018)  | 0.1372***<br>(0.0223)   | 0.0248***<br>(0.0033)  | 0.1161***<br>(0.0227)  | 0.0181***<br>(0.0029)  | 0.0594<br>(0.0459)     | 0.0079<br>(0.0061)     |
| Gender                           | 0.1245<br>(0.0893)     | 0.0404<br>(0.0288)     | -0.0615<br>(0.3334)  | -0.0152<br>(0.0822) | -0.0053<br>(0.3305)   | 0.2269<br>(0.1464)      | 0.0442<br>(0.0283)     | 0.1280<br>(0.2502)      | 0.0231<br>(0.0451)     | 0.1265<br>(0.2699)     | 0.0197<br>(0.0419)     | 0.4876<br>(0.5068)     | 0.0648<br>(0.0657)     |
| Demographic 1                    | -0.1831<br>(0.1209)    | -0.0594<br>(0.0391)    | 0.1381<br>(0.4473)   | 0.0341<br>(0.1103)  | -0.4335<br>(0.4176)   | -0.3015<br>(0.1988)     | -0.0587<br>(0.0386)    | -0.2634<br>(0.3743)     | -0.0475<br>(0.0672)    | -0.4615<br>(0.3933)    | -0.0718<br>(0.0608)    | 0.3718<br>(0.7461)     | 0.0494<br>(0.1001)     |
| Demographic 2                    | 0.1054<br>(0.1376)     | 0.0342<br>(0.0446)     | -0.1379<br>(0.3751)  | -0.0340<br>(0.0925) | -0.1277<br>(0.3833)   | 0.1535<br>(0.2215)      | 0.0299<br>(0.0431)     | 0.1702<br>(0.3844)      | 0.0307<br>(0.0693)     | -0.2311<br>(0.4322)    | -0.0360<br>(0.0672)    | 0.1948<br>(0.7963)     | 0.0259<br>(0.1069)     |
| Advanced Math                    |                        |                        |                      |                     | 0.6834***<br>(0.2024) |                         |                        | 0.7734***<br>(0.2535)   | 0.1395***<br>(0.0443)  | 0.6272**<br>(0.2749)   | 0.0976**<br>(0.0417)   | 0.8104<br>(0.5776)     | 0.1077<br>(0.0733)     |
| Committee Score                  |                        |                        |                      |                     |                       |                         |                        | 0.0778<br>(0.0602)      | 0.0140<br>(0.0108)     | 0.0465<br>(0.0655)     | 0.0072<br>(0.0102)     | -0.7405***<br>(0.2065) | -0.0984***<br>(0.0235) |
| Year Transition/Matriculate More | 0.5891***<br>(0.0952)  | 0.1911***<br>(0.0289)  | 0.3781<br>(1.1778)   | 0.0933<br>(0.2904)  | 0.4111<br>(1.1496)    | 0.9842***<br>(0.1473)   | 0.1916***<br>(0.0263)  | -0.9253***<br>(0.2513)  | -0.1669***<br>(0.0427) | -0.0234<br>(0.2763)    | -0.0036<br>(0.0430)    | 3.3325***<br>(0.5685)  | 0.4430***<br>(0.0331)  |
| Program Rank                     |                        |                        |                      |                     |                       |                         |                        |                         |                        | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |                        |                        |
| Inverse Mills                    |                        |                        | 0.1516<br>(3.0200)   | 0.0374<br>(0.7450)  | 0.8167<br>(2.9514)    |                         |                        |                         |                        |                        |                        |                        |                        |
| √MSE                             |                        |                        |                      |                     | 1.9477***<br>(0.0675) |                         |                        |                         |                        |                        |                        |                        |                        |
| Observations                     | 1000                   |                        |                      |                     |                       |                         |                        |                         |                        |                        |                        |                        |                        |
| LL   Parameters   AIC            | -2606.3                |                        | 53                   |                     | 5318.6                |                         |                        |                         |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

» Unrestricted Matriculation-Static AMEs (Optimal)

# A4: Unrestricted Matriculation-Dynamic Estimates, z Missing (Suboptimal)

Table 31b: Unrestricted Matriculation-Dynamic Estimates, z Missing (Suboptimal)

|                                  | Selection              |                        | Advanced Math        |                     | Committee Score       | Transition              |                        | Accept                 |                        | Success                |                        | Matriculate            |                        |
|----------------------------------|------------------------|------------------------|----------------------|---------------------|-----------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                  | Coefficient            | AME                    | Coefficient          | AME                 | Coefficient           | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                        | -6.7276***<br>(0.5845) | -2.1758***<br>(0.1470) | -0.4236<br>(12.5129) | -0.1042<br>(3.0787) | -0.0354<br>(12.9588)  | -11.3115***<br>(1.0200) | -2.1922***<br>(0.1474) | -6.7975***<br>(1.7427) | -1.3417***<br>(0.3203) | -9.5522***<br>(1.9577) | -1.4907***<br>(0.2509) | -1.0583<br>(3.6036)    | -0.1376<br>(0.4721)    |
| GRE Quantitative Percentile      | 0.0735***<br>(0.0068)  | 0.0238***<br>(0.0018)  | 0.0029<br>(0.1146)   | 0.0007<br>(0.0282)  | 0.0472<br>(0.1184)    | 0.1236***<br>(0.0118)   | 0.0239***<br>(0.0018)  | 0.0557***<br>(0.0191)  | 0.0110***<br>(0.0036)  | 0.1157***<br>(0.0216)  | 0.0181***<br>(0.0027)  | 0.0131<br>(0.0391)     | 0.0017<br>(0.0051)     |
| Gender                           | 0.1451*<br>(0.0879)    | 0.0469*<br>(0.0283)    | -0.1798<br>(0.3165)  | -0.0442<br>(0.0777) | 0.0024<br>(0.3266)    | 0.2595*<br>(0.1471)     | 0.0503*<br>(0.0283)    | 0.0113<br>(0.2370)     | 0.0022<br>(0.0468)     | 0.0580<br>(0.2669)     | 0.0091<br>(0.0416)     | 0.1301<br>(0.4711)     | 0.0169<br>(0.0608)     |
| Demographic 1                    | -0.4965***<br>(0.1220) | -0.1606***<br>(0.0385) | 0.2589<br>(0.7992)   | 0.0637<br>(0.1965)  | -0.8130<br>(0.7898)   | -0.8329***<br>(0.1971)  | -0.1614***<br>(0.0371) | -0.3369<br>(0.3398)    | -0.0665<br>(0.0669)    | -0.5150<br>(0.3647)    | -0.0804<br>(0.0564)    | 0.6291<br>(0.6275)     | 0.0818<br>(0.0823)     |
| Demographic 2                    | 0.0600<br>(0.1363)     | 0.0194<br>(0.0441)     | -0.0052<br>(0.3070)  | -0.0013<br>(0.0755) | -0.2543<br>(0.3120)   | 0.0794<br>(0.2139)      | 0.0154<br>(0.0414)     | 0.1774<br>(0.3349)     | 0.0350<br>(0.0659)     | -0.4688<br>(0.3854)    | -0.0732<br>(0.0598)    | -0.2206<br>(0.6858)    | -0.0287<br>(0.0880)    |
| Advanced Math                    |                        |                        |                      |                     | 0.7199***<br>(0.1971) |                         |                        | 0.3587<br>(0.2378)     | 0.0708<br>(0.0465)     | 0.5346**<br>(0.2703)   | 0.0834**<br>(0.0416)   | 0.5865<br>(0.5314)     | 0.0763<br>(0.0674)     |
| Committee Score                  |                        |                        |                      |                     |                       |                         |                        | 0.2970***<br>(0.0593)  | 0.0586***<br>(0.0106)  | 0.0611<br>(0.0642)     | 0.0095<br>(0.0100)     | -0.6086***<br>(0.1636) | -0.0791***<br>(0.0176) |
| Year Transition/Matriculate More | 0.6628***<br>(0.0932)  | 0.2144***<br>(0.0276)  | 0.3500<br>(1.0548)   | 0.0861<br>(0.2594)  | 0.4393<br>(1.0838)    | 1.1112***<br>(0.1482)   | 0.2153***<br>(0.0257)  | -0.7175***<br>(0.2399) | -0.1416***<br>(0.0455) | 0.1308<br>(0.2794)     | 0.0204<br>(0.0434)     | 3.4787***<br>(0.6027)  | 0.4524***<br>(0.0350)  |
| Program Rank                     |                        |                        |                      |                     |                       |                         |                        |                        |                        | -0.0280***<br>(0.0028) | -0.0044***<br>(0.0003) |                        |                        |
| Inverse Mills                    |                        |                        | -0.0105<br>(2.4357)  | -0.0026<br>(0.5993) | 1.0077<br>(2.5170)    |                         |                        |                        |                        |                        |                        |                        |                        |
| √MSE                             |                        |                        |                      |                     | 1.9201***<br>(0.0683) |                         |                        |                        |                        |                        |                        |                        |                        |
| Observations                     | 1000                   |                        |                      |                     |                       |                         |                        |                        |                        |                        |                        |                        |                        |
| LL   Parameters   AIC            | -2657.3                |                        | 53                   |                     | 5420.7                |                         |                        |                        |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

» Unrestricted Matriculation-Static AMEs (Suboptimal)

# A4: Restricted Matriculation-Dynamic Estimates, z Missing (Optimal)

Table 32a: Restricted Matriculation-Dynamic Estimates, z Missing (Optimal)

|                                  | Selection              | Advanced Math        | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success                | Matriculate            | Sigma                 |
|----------------------------------|------------------------|----------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                        | -6.8724***<br>(0.5876) | -2.3905<br>(16.2524) | 0.8934<br>(15.7133)   | -1.8424***<br>(0.2144) | 0.1584<br>(0.1041, 0.2412) | -0.0195<br>(0.2312) | 49.51%<br>(38.40%, 60.67%) | -9.6049***<br>(1.9966) | -9.7043**<br>(4.1058)  |                       |
| GRE Quantitative Percentile      | 0.0732***<br>(0.0069)  | 0.0233<br>(0.1464)   | 0.0345<br>(0.1415)    |                        |                            |                     |                            | 0.1154***<br>(0.0220)  | 0.1170***<br>(0.0399)  |                       |
| Gender                           | 0.1227<br>(0.0892)     | -0.0345<br>(0.3311)  | -0.0217<br>(0.3238)   |                        |                            |                     |                            | 0.1550<br>(0.1427)     | 0.7422<br>(0.4541)     |                       |
| Demographic 1                    | -0.1805<br>(0.1206)    | 0.1132<br>(0.4463)   | -0.4055<br>(0.4096)   |                        |                            |                     |                            | -0.3161<br>(0.2126)    | 0.1389<br>(0.7180)     |                       |
| Demographic 2                    | 0.1054<br>(0.1372)     | -0.1272<br>(0.3751)  | -0.1289<br>(0.3788)   |                        |                            |                     |                            | 0.0125<br>(0.2066)     | 0.0858<br>(0.7721)     |                       |
| Advanced Math                    |                        |                      | 0.6547***<br>(0.2032) |                        |                            |                     |                            | 0.6020***<br>(0.1767)  | 0.9693<br>(0.5942)     |                       |
| Committee Score                  |                        |                      |                       |                        |                            |                     |                            | 0.0261<br>(0.0398)     | -0.6798***<br>(0.1997) |                       |
| Year Transition/Matriculate More | 0.5886***<br>(0.0928)  | 0.4054<br>(1.1910)   | 0.4817<br>(1.1421)    | -1.0996***<br>(0.2789) | 0.0528<br>(0.0283, 0.0983) | 0.6125*<br>(0.3393) | 64.41%<br>(53.12%, 74.29%) | 0.0734<br>(0.2725)     | 3.0554***<br>(0.5403)  |                       |
| Program Rank                     |                        |                      |                       |                        |                            |                     |                            | -0.0281***<br>(0.0027) |                        |                       |
| Inverse Mills                    |                        | 0.2334<br>(3.0552)   | 0.8380<br>(2.9269)    |                        |                            |                     |                            |                        |                        |                       |
| $\sqrt{\text{MSE}}$              |                        |                      | 1.9466***<br>(0.0673) |                        |                            |                     |                            |                        |                        |                       |
| Sigma                            |                        |                      |                       |                        |                            |                     |                            |                        |                        | 0.1445***<br>(0.0289) |
| Observations                     | 1000                   |                      |                       |                        |                            |                     |                            |                        |                        |                       |
| LL   Parameters   AIC            | -2653.6                | 44                   | 5395.3                |                        |                            |                     |                            |                        |                        |                       |
| LR Statistic   DF   P-Value      | 94.6726                | 9                    | 0.0000                |                        |                            |                     |                            |                        |                        |                       |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01. Confidence intervals are 95%.

# A4: Restricted Matriculation-Dynamic Estimates, z Missing (Suboptimal)

**Table 32b: Restricted Matriculation-Dynamic Estimates, z Missing (Suboptimal)**

|                                  | Selection              | Advanced Math        | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success                | Matriculate            | Sigma                 |
|----------------------------------|------------------------|----------------------|-----------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                        | -6.6775***<br>(0.5790) | -5.4847<br>(12.5476) | 2.9925<br>(12.8141)   | -1.5898***<br>(0.3132) | 0.2040<br>(0.1104, 0.3768) | 0.0720<br>(0.2631) | 51.80%<br>(39.09%, 64.28%) | -8.0004***<br>(2.4225) | -6.4228*<br>(3.4043)   |                       |
| GRE Quantitative Percentile      | 0.0728***<br>(0.0068)  | 0.0503<br>(0.1144)   | 0.0185<br>(0.1168)    |                        |                            |                    |                            | 0.0917***<br>(0.0263)  | 0.0769**<br>(0.0367)   |                       |
| Gender                           | 0.1426<br>(0.0879)     | -0.0799<br>(0.3179)  | -0.0622<br>(0.3173)   |                        |                            |                    |                            | 0.0784<br>(0.1634)     | 0.3801<br>(0.4593)     |                       |
| Demographic 1                    | -0.4914***<br>(0.1229) | -0.0561<br>(0.7898)  | -0.6127<br>(0.7796)   |                        |                            |                    |                            | -0.5131**<br>(0.2579)  | 0.0626<br>(0.6460)     |                       |
| Demographic 2                    | 0.0593<br>(0.1385)     | 0.0129<br>(0.3191)   | -0.2632<br>(0.2950)   |                        |                            |                    |                            | -0.1242<br>(0.2556)    | -0.4402<br>(0.6987)    |                       |
| Advanced Math                    |                        |                      | 0.7126***<br>(0.1964) |                        |                            |                    |                            | 0.4313**<br>(0.1974)   | 0.6735<br>(0.5543)     |                       |
| Committee Score                  |                        |                      |                       |                        |                            |                    |                            | 0.1701***<br>(0.0391)  | -0.5440***<br>(0.1605) |                       |
| Year Transition/Matriculate More | 0.6697***<br>(0.0904)  | 0.7533<br>(1.0931)   | 0.2395<br>(1.0955)    | -1.3565***<br>(0.3734) | 0.0525<br>(0.0211, 0.1307) | 0.5585<br>(0.4094) | 65.26%<br>(50.59%, 77.51%) | 0.0706<br>(0.2920)     | 3.1731***<br>(0.5913)  |                       |
| Program Rank                     |                        |                      |                       |                        |                            |                    |                            | -0.0281***<br>(0.0027) |                        |                       |
| Inverse Mills                    |                        | 0.8987<br>(2.4674)   | 0.4744<br>(2.5094)    |                        |                            |                    |                            |                        |                        |                       |
| $\sqrt{\text{MSE}}$              |                        |                      | 1.9147***<br>(0.0676) |                        |                            |                    |                            |                        |                        |                       |
| Sigma                            |                        |                      |                       |                        |                            |                    |                            |                        |                        | 0.1986***<br>(0.0595) |
| Observations                     | 1000                   |                      |                       |                        |                            |                    |                            |                        |                        |                       |
| LL   Parameters   AIC            | -2724.6                | 44                   | 5537.2                |                        |                            |                    |                            |                        |                        |                       |
| LR Statistic   DF   P-Value      | 134.4685               | 9                    | 0.0000                |                        |                            |                    |                            |                        |                        |                       |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01. Confidence intervals are 95%.

# A4: Full Unrestricted Waitlist-Hybrid Estimates (Optimal)

Table 33a: Unrestricted Waitlist-Hybrid Estimates (Optimal)

|                                  | $\overline{m}_{2A}$<br>Coefficient | Transition<br>Coefficient | Transition<br>AME      | Accept 2<br>Coefficient | Accept 2<br>AME        | Accept 3<br>Coefficient | Accept 3<br>AME        | Success 1<br>Coefficient | Success 1<br>AME       | Success 3<br>Coefficient | Success 3<br>AME       | Matriculate<br>Coefficient | Matriculate<br>AME     |
|----------------------------------|------------------------------------|---------------------------|------------------------|-------------------------|------------------------|-------------------------|------------------------|--------------------------|------------------------|--------------------------|------------------------|----------------------------|------------------------|
| Intercept                        | -0.0252<br>(0.0518)                | -11.5905***<br>(1.0234)   | -2.2561***<br>(0.1454) | -11.9661***<br>(2.5509) | -1.6170***<br>(0.3154) | -13.2739***<br>(2.4117) | -1.8259***<br>(0.3091) | -8.1384***<br>(1.1381)   | -1.2642***<br>(0.1516) | -9.5601***<br>(2.0725)   | -1.4878***<br>(0.2688) | -5.8313<br>(6.4970)        | -0.6741<br>(0.7706)    |
| GRE Quantitative Percentile      |                                    | 0.1236***<br>(0.0118)     | 0.0241***<br>(0.0018)  | 0.1097***<br>(0.0271)   | 0.0148***<br>(0.0034)  | 0.1387***<br>(0.0264)   | 0.0191***<br>(0.0034)  | 0.1079***<br>(0.0133)    | 0.0168***<br>(0.0017)  | 0.1161***<br>(0.0227)    | 0.0181***<br>(0.0029)  | 0.0834<br>(0.0644)         | 0.0096<br>(0.0075)     |
| Gender                           |                                    | 0.2266<br>(0.1464)        | 0.0441<br>(0.0283)     | 0.0233<br>(0.2899)      | 0.0031<br>(0.0392)     | 0.3039<br>(0.3287)      | 0.0418<br>(0.0451)     | 0.1399<br>(0.1739)       | 0.0217<br>(0.0270)     | 0.1264<br>(0.2699)       | 0.0197<br>(0.0419)     | 0.4650<br>(0.9003)         | 0.0538<br>(0.1052)     |
| Demographic 1                    |                                    | -0.3014<br>(0.1988)       | -0.0587<br>(0.0386)    | 0.5530<br>(0.4940)      | 0.0747<br>(0.0665)     | -0.8834*<br>(0.4622)    | -0.1215**<br>(0.0619)  | -0.2535<br>(0.2359)      | -0.0394<br>(0.0366)    | -0.4606<br>(0.3933)      | -0.0717<br>(0.0608)    | -0.2627<br>(0.7744)        | -0.0304<br>(0.0880)    |
| Demographic 2                    |                                    | 0.1538<br>(0.2215)        | 0.0299<br>(0.0431)     | 0.5128<br>(0.5204)      | 0.0693<br>(0.0699)     | -0.0526<br>(0.4528)     | -0.0072<br>(0.0623)    | -0.1180<br>(0.2581)      | -0.0183<br>(0.0401)    | -0.2340<br>(0.4322)      | -0.0364<br>(0.0672)    | -0.3966<br>(1.2105)        | -0.0459<br>(0.1393)    |
| Advanced Math                    |                                    |                           |                        | 0.5065*<br>(0.2907)     | 0.0684*<br>(0.0390)    | 0.8621***<br>(0.3306)   | 0.1186***<br>(0.0451)  |                          |                        | 0.6273**<br>(0.2749)     | 0.0976**<br>(0.0417)   | 1.1399<br>(0.9839)         | 0.1318<br>(0.1080)     |
| Committee Score                  |                                    |                           |                        | 0.1061<br>(0.0689)      | 0.0143<br>(0.0093)     | 0.0378<br>(0.0672)      | 0.0052<br>(0.0092)     |                          |                        | 0.0464<br>(0.0655)       | 0.0072<br>(0.0102)     | -0.9123**<br>(0.3787)      | -0.1055***<br>(0.0317) |
| Year Transition/Matriculate More |                                    | 0.9842***<br>(0.1472)     | 0.1916***<br>(0.0263)  |                         |                        |                         |                        | -0.0586<br>(0.1690)      | -0.0091<br>(0.0262)    | -0.0233<br>(0.2763)      | -0.0036<br>(0.0430)    | 3.3933***<br>(0.7682)      | 0.3922***<br>(0.0697)  |
| Program Rank                     |                                    |                           |                        |                         |                        |                         |                        | -0.0310***<br>(0.0019)   | -0.0048***<br>(0.0002) | -0.0282***<br>(0.0027)   | -0.0044***<br>(0.0003) |                            |                        |
| $\overline{m}_2$                 | 1.0892***<br>(0.0851)              |                           |                        | -1.5983**<br>(0.7869)   | -0.2160**<br>(0.1049)  |                         |                        |                          |                        |                          |                        |                            |                        |
| $\overline{m}_{2A}$              |                                    |                           |                        |                         |                        | -2.0170***<br>(0.7696)  | -0.2774***<br>(0.1033) |                          |                        |                          |                        |                            |                        |
| $\sqrt{\text{MSE}}$              | 0.0015***<br>(0.0005)              |                           |                        |                         |                        |                         |                        |                          |                        |                          |                        |                            |                        |
| Observations                     | 1000                               |                           |                        |                         |                        |                         |                        |                          |                        |                          |                        |                            |                        |
| LL   Parameters   AIC            | 438.6                              | 41                        | -795.1                 |                         |                        |                         |                        |                          |                        |                          |                        |                            |                        |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Full Unrestricted Waitlist-Hybrid Estimates (Suboptimal)

Table 33b: Unrestricted Waitlist-Hybrid Estimates (Suboptimal)

|                                  | $\bar{m}_{2A}$<br>Coefficient | Transition<br>Coefficient | Transition<br>AME      | Accept 2<br>Coefficient | Accept 2<br>AME        | Accept 3<br>Coefficient | Accept 3<br>AME        | Success 1<br>Coefficient | Success 1<br>AME       | Success 3<br>Coefficient | Success 3<br>AME       | Matriculate<br>Coefficient | Matriculate<br>AME     |
|----------------------------------|-------------------------------|---------------------------|------------------------|-------------------------|------------------------|-------------------------|------------------------|--------------------------|------------------------|--------------------------|------------------------|----------------------------|------------------------|
| Intercept                        | -0.0498*<br>(0.0260)          | -11.2292***<br>(1.0143)   | -2.1801***<br>(0.1474) | -6.7111***<br>(2.0006)  | -1.0638***<br>(0.3043) | -7.1640***<br>(2.5069)  | -0.8321***<br>(0.2952) | -8.2823***<br>(1.1392)   | -1.2871***<br>(0.1493) | -9.6011***<br>(1.9606)   | -1.4979***<br>(0.2508) | -1.4373<br>(3.8091)        | -0.2080<br>(0.5533)    |
| GRE Quantitative Percentile      |                               | 0.1226***<br>(0.0117)     | 0.0238***<br>(0.0018)  | 0.0387*<br>(0.0215)     | 0.0061*<br>(0.0034)    | 0.0717***<br>(0.0278)   | 0.0083**<br>(0.0033)   | 0.1097***<br>(0.0133)    | 0.0170***<br>(0.0017)  | 0.1162***<br>(0.0216)    | 0.0181***<br>(0.0027)  | 0.0355<br>(0.0435)         | 0.0051<br>(0.0063)     |
| Gender                           |                               | 0.2576*<br>(0.1468)       | 0.0500*<br>(0.0283)    | -0.2763<br>(0.2746)     | -0.0438<br>(0.0431)    | 0.4099<br>(0.3417)      | 0.0476<br>(0.0394)     | 0.1354<br>(0.1741)       | 0.0210<br>(0.0270)     | 0.0584<br>(0.2669)       | 0.0091<br>(0.0416)     | -0.3544<br>(0.5436)        | -0.0513<br>(0.0788)    |
| Demographic 1                    |                               | -0.8291***<br>(0.1966)    | -0.1610***<br>(0.0371) | 0.2570<br>(0.3773)      | 0.0407<br>(0.0594)     | -1.1547**<br>(0.5110)   | -0.1341**<br>(0.0581)  | -0.2774<br>(0.2344)      | -0.0431<br>(0.0364)    | -0.5155<br>(0.3648)      | -0.0804<br>(0.0563)    | 0.3334<br>(0.7088)         | 0.0482<br>(0.1030)     |
| Demographic 2                    |                               | 0.0799<br>(0.2135)        | 0.0155<br>(0.0414)     | 0.5360<br>(0.3885)      | 0.0850<br>(0.0605)     | -0.3696<br>(0.4652)     | -0.0429<br>(0.0539)    | -0.1806<br>(0.2566)      | -0.0281<br>(0.0399)    | -0.4743<br>(0.3856)      | -0.0740<br>(0.0598)    | -0.7602<br>(0.7927)        | -0.1100<br>(0.1089)    |
| Advanced Math                    |                               |                           |                        | 0.2361<br>(0.2639)      | 0.0374<br>(0.0417)     | 0.4867<br>(0.3566)      | 0.0565<br>(0.0417)     |                          |                        | 0.5367**<br>(0.2704)     | 0.0837**<br>(0.0416)   | 0.3982<br>(0.6445)         | 0.0576<br>(0.0922)     |
| Committee Score                  |                               |                           |                        | 0.3736***<br>(0.0674)   | 0.0592***<br>(0.0096)  | 0.0653<br>(0.0837)      | 0.0076<br>(0.0098)     |                          |                        | 0.0617<br>(0.0643)       | 0.0096<br>(0.0100)     | -0.6725***<br>(0.1981)     | -0.0973***<br>(0.0217) |
| Year Transition/Matriculate More |                               | 1.1108***<br>(0.1479)     | 0.2157***<br>(0.0257)  |                         |                        |                         |                        | -0.0767<br>(0.1688)      | -0.0119<br>(0.0262)    | 0.1318<br>(0.2794)       | 0.0206<br>(0.0434)     | 2.9410***<br>(0.6722)      | 0.4256***<br>(0.0578)  |
| Program Rank                     |                               |                           |                        |                         |                        |                         |                        | -0.0304***<br>(0.0019)   | -0.0047***<br>(0.0002) | -0.0279***<br>(0.0028)   | -0.0044***<br>(0.0003) |                            |                        |
| $\bar{m}_2$                      | 0.8883***<br>(0.0489)         |                           |                        | -0.5486<br>(0.5705)     | -0.0870<br>(0.0900)    |                         |                        |                          |                        |                          |                        |                            |                        |
| $\bar{m}_{2A}$                   |                               |                           |                        |                         |                        | -2.9216***<br>(0.9934)  | -0.3393***<br>(0.1141) |                          |                        |                          |                        |                            |                        |
| $\sqrt{\text{MSE}}$              | 0.0010***<br>(0.0002)         |                           |                        |                         |                        |                         |                        |                          |                        |                          |                        |                            |                        |
| Observations                     | 1000                          |                           |                        |                         |                        |                         |                        |                          |                        |                          |                        |                            |                        |
| LL   Parameters   AIC            | 618.0                         | 41                        | -1153.9                |                         |                        |                         |                        |                          |                        |                          |                        |                            |                        |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Full Restricted Waitlist-Hybrid Estimates (Optimal)

Table 34a: Restricted Waitlist-Hybrid Estimates (Optimal)

|                                               | $\bar{m}_{2A}$        | Cutoff 1               | $P(\text{Cutoff})_1$       | Cutoff 3            | $P(\text{Cutoff})_3$       | Success 1              | Success 3              | Matriculate           | Beta                  | Sigma                 |
|-----------------------------------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|------------------------|-----------------------|-----------------------|-----------------------|
| Year Transition/Matriculate Fewer   Intercept | -0.0252<br>(0.0518)   | 0.6891***<br>(0.1830)  | 66.58%<br>(58.19%, 74.03%) | 0.4185<br>(0.4184)  | 69.29%<br>(54.94%, 80.68%) | -8.0491***<br>(1.1651) | -9.6599***<br>(1.7228) | -5.8313<br>(6.4970)   |                       |                       |
| GRE Quantitative Percentile                   |                       |                        |                            |                     |                            | 0.1060***<br>(0.0135)  | 0.1151***<br>(0.0191)  | 0.0834<br>(0.0644)    |                       |                       |
| Gender                                        |                       |                        |                            |                     |                            | 0.1997**<br>(0.1017)   | 0.1149<br>(0.1664)     | 0.4650<br>(0.9003)    |                       |                       |
| Demographic 1                                 |                       |                        |                            |                     |                            | -0.2477*<br>(0.1349)   | -0.3286<br>(0.2390)    | -0.2627<br>(0.7744)   |                       |                       |
| Demographic 2                                 |                       |                        |                            |                     |                            | 0.0405<br>(0.1507)     | -0.0036<br>(0.2410)    | -0.3966<br>(1.2105)   |                       |                       |
| Advanced Math                                 |                       |                        |                            |                     |                            |                        | 0.6284***<br>(0.1758)  | 1.1399<br>(0.9839)    |                       |                       |
| Committee Score                               |                       |                        |                            |                     |                            |                        | 0.0548<br>(0.0419)     | -0.9123**<br>(0.3787) |                       |                       |
| Year Transition/Matriculate More              |                       | -0.8464***<br>(0.2185) | 46.08%<br>(39.62%, 52.67%) |                     | 87.77%<br>(63.97%, 96.67%) | -0.0609<br>(0.1668)    | 0.0204<br>(0.2685)     | 3.3933***<br>(0.7682) |                       |                       |
| Program Rank                                  |                       |                        |                            |                     |                            | -0.0310***<br>(0.0019) | -0.0283***<br>(0.0027) |                       |                       |                       |
| $\bar{m}_2$                                   | 1.0892***<br>(0.0851) |                        |                            |                     |                            |                        |                        |                       |                       |                       |
| $\bar{m}_{2A}$                                |                       |                        |                            | 2.4392*<br>(1.4478) |                            |                        |                        |                       |                       |                       |
| $\sqrt{\text{MSE}}$                           | 0.0015***<br>(0.0005) |                        |                            |                     |                            |                        |                        |                       |                       |                       |
| Beta                                          |                       |                        |                            |                     |                            |                        |                        |                       | 0.9493***<br>(0.0396) |                       |
| Sigma                                         |                       |                        |                            |                     |                            |                        |                        |                       |                       | 0.1917***<br>(0.0251) |
| Observations                                  | 1000                  |                        |                            |                     |                            |                        |                        |                       |                       |                       |
| LL   Parameters   AIC                         | 434.1                 | 25                     | -818.1                     |                     |                            |                        |                        |                       |                       |                       |
| LR Statistic   DF   P-Value                   | 8.9955                | 16                     | 0.9136                     |                     |                            |                        |                        |                       |                       |                       |

Murphy-Topel robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.



# A4: Full Restricted Waitlist-Hybrid Estimates (Suboptimal)

Table 34b: Restricted Waitlist-Hybrid Estimates (Suboptimal)

|                                               | $\overline{m}_{2A}$   | Cutoff 1               | $P(\text{Cutoff})_1$       | Cutoff 3           | $P(\text{Cutoff})_3$       | Success 1              | Success 3              | Matriculate            | Beta                  | Sigma                 |
|-----------------------------------------------|-----------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|------------------------|-----------------------|-----------------------|
| Year Transition/Matriculate Fewer   Intercept | -0.0498*<br>(0.0260)  | 0.7428***<br>(0.2082)  | 67.76%<br>(58.29%, 75.97%) | 0.5736<br>(0.5852) | 74.61%<br>(55.80%, 87.24%) | -8.0749***<br>(1.2691) | -6.3299***<br>(1.7373) | -1.4373<br>(3.8091)    |                       |                       |
| GRE Quantitative Percentile                   |                       |                        |                            |                    |                            | 0.1082***<br>(0.0147)  | 0.0718***<br>(0.0186)  | 0.0355<br>(0.0435)     |                       |                       |
| Gender                                        |                       |                        |                            |                    |                            | 0.2189**<br>(0.1045)   | 0.0369<br>(0.1594)     | -0.3545<br>(0.5436)    |                       |                       |
| Demographic 1                                 |                       |                        |                            |                    |                            | -0.5509***<br>(0.1397) | -0.3579<br>(0.2337)    | 0.3334<br>(0.7088)     |                       |                       |
| Demographic 2                                 |                       |                        |                            |                    |                            | -0.0156<br>(0.1510)    | -0.0774<br>(0.2321)    | -0.7602<br>(0.7927)    |                       |                       |
| Advanced Math                                 |                       |                        |                            |                    |                            |                        | 0.3857**<br>(0.1669)   | 0.3982<br>(0.6445)     |                       |                       |
| Committee Score                               |                       |                        |                            |                    |                            |                        | 0.1748***<br>(0.0374)  | -0.6725***<br>(0.1981) |                       |                       |
| Year Transition/Matriculate More              |                       | -1.0055***<br>(0.2419) | 43.47%<br>(37.12%, 50.04%) |                    | 93.27%<br>(41.58%, 99.63%) | -0.0797<br>(0.1668)    | 0.1064<br>(0.2766)     | 2.9410***<br>(0.6722)  |                       |                       |
| Program Rank                                  |                       |                        |                            |                    |                            | -0.0306***<br>(0.0019) | -0.0281***<br>(0.0027) |                        |                       |                       |
| $\overline{m}_2$                              | 0.8883***<br>(0.0489) |                        |                            |                    |                            |                        |                        |                        |                       |                       |
| $\overline{m}_{2A}$                           |                       |                        |                            | 3.6299<br>(3.2728) |                            |                        |                        |                        |                       |                       |
| $\sqrt{\text{MSE}}$                           | 0.0010***<br>(0.0002) |                        |                            |                    |                            |                        |                        |                        |                       |                       |
| Beta                                          |                       |                        |                            |                    |                            |                        |                        |                        | 0.8280***<br>(0.0422) |                       |
| Sigma                                         |                       |                        |                            |                    |                            |                        |                        |                        |                       | 0.2012***<br>(0.0325) |
| Observations                                  | 1000                  |                        |                            |                    |                            |                        |                        |                        |                       |                       |
| LL   Parameters   AIC                         | 596.3                 | 25                     | -1142.6                    |                    |                            |                        |                        |                        |                       |                       |
| LR Statistic   DF   P-Value                   | 43.3371               | 16                     | 0.0002                     |                    |                            |                        |                        |                        |                       |                       |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01. Confidence intervals are 95%.

# A4: Unrestricted Waitlist-Dynamic Estimates (Optimal)

Table 35a: Unrestricted Waitlist-Dynamic Estimates (Optimal)

|                                  | Advanced Math         |                       | Committee Score       | $\overline{m}_{2A}$    | Transition              |                        | Accept 2                |                        | Accept 3                |                        | Success                |                        | Matriculate           |                        |
|----------------------------------|-----------------------|-----------------------|-----------------------|------------------------|-------------------------|------------------------|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|-----------------------|------------------------|
|                                  | Coefficient           | AME                   | Coefficient           |                        | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient           | AME                    |
| Intercept                        | -1.9043**<br>(0.7564) | -0.4667**<br>(0.1830) | 3.5868***<br>(0.7240) | -0.0252<br>(0.0518)    | -11.5883***<br>(1.0232) | -2.2558***<br>(0.1454) | -11.9537***<br>(2.5492) | -1.6156***<br>(0.3153) | -13.2624***<br>(2.4102) | -1.8245***<br>(0.3091) | -9.5491***<br>(2.0715) | -1.4864***<br>(0.2688) | -5.8313<br>(6.4970)   | -0.6741<br>(0.7706)    |
| GRE Quantative Percentile        | 0.0208**<br>(0.0090)  | 0.0051**<br>(0.0022)  | 0.0126<br>(0.0086)    |                        | 0.1236***<br>(0.0118)   | 0.0241***<br>(0.0018)  | 0.1095***<br>(0.0271)   | 0.0148***<br>(0.0034)  | 0.1386***<br>(0.0264)   | 0.0191***<br>(0.0034)  | 0.1160***<br>(0.0227)  | 0.0181***<br>(0.0029)  | 0.0834<br>(0.0644)    | 0.0096<br>(0.0075)     |
| Gender                           | -0.2334*<br>(0.1309)  | -0.0572*<br>(0.0319)  | -0.0581<br>(0.1315)   |                        | 0.2266<br>(0.1464)      | 0.0441<br>(0.0283)     | 0.0232<br>(0.2898)      | 0.0031<br>(0.0392)     | 0.3035<br>(0.3287)      | 0.0418<br>(0.0451)     | 0.1262<br>(0.2699)     | 0.0196<br>(0.0419)     | 0.4650<br>(0.9003)    | 0.0538<br>(0.1052)     |
| Demographic 1                    | 0.3690**<br>(0.1732)  | 0.0904**<br>(0.0421)  | -0.3559**<br>(0.1714) |                        | -0.3012<br>(0.1988)     | -0.0586<br>(0.0386)    | 0.5532<br>(0.4939)      | 0.0748<br>(0.0665)     | -0.8833*<br>(0.4621)    | -0.1215**<br>(0.0619)  | -0.4603<br>(0.3932)    | -0.0717<br>(0.0608)    | -0.2627<br>(0.7744)   | -0.0304<br>(0.0880)    |
| Demographic 2                    | 0.0924<br>(0.1911)    | 0.0227<br>(0.0468)    | 0.1203<br>(0.1890)    |                        | 0.1537<br>(0.2215)      | 0.0299<br>(0.0431)     | 0.5133<br>(0.5202)      | 0.0694<br>(0.0699)     | -0.0529<br>(0.4527)     | -0.0073<br>(0.0623)    | -0.2342<br>(0.4322)    | -0.0365<br>(0.0672)    | -0.3966<br>(1.2105)   | -0.0459<br>(0.1393)    |
| Advanced Math                    |                       |                       | 0.9753***<br>(0.1270) |                        |                         |                        | 0.5066*<br>(0.2906)     | 0.0685*<br>(0.0390)    | 0.8620***<br>(0.3305)   | 0.1186***<br>(0.0451)  | 0.6275**<br>(0.2749)   | 0.0977**<br>(0.0417)   | 1.1399<br>(0.9839)    | 0.1318<br>(0.1080)     |
| Committee Score                  |                       |                       |                       |                        |                         |                        | 0.1060<br>(0.0689)      | 0.0143<br>(0.0093)     | 0.0378<br>(0.0672)      | 0.0052<br>(0.0092)     | 0.0464<br>(0.0655)     | 0.0072<br>(0.0102)     | -0.9123**<br>(0.3787) | -0.1055***<br>(0.0317) |
| Year Transition/Matriculate More | 0.0664<br>(0.1281)    | 0.0163<br>(0.0314)    | 0.1132<br>(0.1261)    |                        | 0.9840***<br>(0.1472)   | 0.1915***<br>(0.0263)  |                         |                        |                         |                        | -0.0238<br>(0.2763)    | -0.0037<br>(0.0430)    | 3.3933***<br>(0.7682) | 0.3922***<br>(0.0697)  |
| Program Rank                     |                       |                       |                       |                        |                         |                        |                         |                        |                         |                        | -0.0282***<br>(0.0027) | -0.0044***<br>(0.0003) |                       |                        |
| $\overline{m}_2$                 |                       |                       |                       | 1.0892***<br>(0.0851)  |                         |                        | -1.5989**<br>(0.7870)   | -0.2161**<br>(0.1049)  |                         |                        |                        |                        |                       |                        |
| $\overline{m}_{2A}$              |                       |                       |                       |                        |                         |                        |                         |                        | -2.0177***<br>(0.7696)  | -0.2776***<br>(0.1033) |                        |                        |                       |                        |
| $\sqrt{\text{MSE}}$              |                       |                       | 1.9949***<br>(0.0422) | -0.0015***<br>(0.0005) |                         |                        |                         |                        |                         |                        |                        |                        |                       |                        |
| Observations                     | 1000                  |                       |                       |                        |                         |                        |                         |                        |                         |                        |                        |                        |                       |                        |
| LL   Parameters   AIC            | -1924.0               | 48                    | 3944.1                |                        |                         |                        |                         |                        |                         |                        |                        |                        |                       |                        |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

►► Unrestricted Waitlist-Hybrid AMEs (Optimal)

# A4: Unrestricted Waitlist-Dynamic Estimates (Suboptimal)

Table 35b: Unrestricted Waitlist-Dynamic Estimates (Suboptimal)

|                                  | Advanced Math         |                       | Committee Score       |  | $\overline{m}_{2A}$   |  | Transition              |                        | Accept 2               |                        | Accept 3               |                        | Success                |                        | Matriculate            |                        |
|----------------------------------|-----------------------|-----------------------|-----------------------|--|-----------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                  | Coefficient           | AME                   | Coefficient           |  | Coefficient           |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                        | -1.9037**<br>(0.7564) | -0.4665**<br>(0.1830) | 3.5854***<br>(0.7240) |  | -0.0498*<br>(0.0260)  |  | -11.2294***<br>(1.0144) | -2.1801***<br>(0.1474) | -6.7112***<br>(2.0007) | -1.0637***<br>(0.3043) | -7.1638***<br>(2.5069) | -0.8321***<br>(0.2952) | -9.5963***<br>(1.9602) | -1.4972***<br>(0.2508) | -1.4373<br>(3.8091)    | -0.2080<br>(0.5533)    |
| GRE Quantative Percentile        | 0.0208**<br>(0.0090)  | 0.0051**<br>(0.0022)  | 0.0126<br>(0.0086)    |  |                       |  | 0.1226***<br>(0.0117)   | 0.0238***<br>(0.0018)  | 0.0387*<br>(0.0215)    | 0.0061*<br>(0.0034)    | 0.0717***<br>(0.0278)  | 0.0083**<br>(0.0033)   | 0.1162***<br>(0.0216)  | 0.0181***<br>(0.0027)  | 0.0355<br>(0.0435)     | 0.0051<br>(0.0063)     |
| Gender                           | -0.2334*<br>(0.1309)  | -0.0572*<br>(0.0319)  | -0.0580<br>(0.1315)   |  |                       |  | 0.2575*<br>(0.1469)     | 0.0500*<br>(0.0283)    | -0.2767<br>(0.2746)    | -0.0439<br>(0.0431)    | 0.4102<br>(0.3417)     | 0.0476<br>(0.0394)     | 0.0583<br>(0.2669)     | 0.0091<br>(0.0416)     | -0.3544<br>(0.5436)    | -0.0513<br>(0.0788)    |
| Demographic 1                    | 0.3691**<br>(0.1732)  | 0.0905**<br>(0.0421)  | -0.3562**<br>(0.1714) |  |                       |  | -0.8293***<br>(0.1966)  | -0.1610***<br>(0.0371) | 0.2577<br>(0.3774)     | 0.0408<br>(0.0594)     | -1.1546**<br>(0.5110)  | -0.1341**<br>(0.0581)  | -0.5155<br>(0.3647)    | -0.0804<br>(0.0563)    | 0.3334<br>(0.7088)     | 0.0482<br>(0.1030)     |
| Demographic 2                    | 0.0925<br>(0.1911)    | 0.0227<br>(0.0468)    | 0.1201<br>(0.1890)    |  |                       |  | 0.0797<br>(0.2135)      | 0.0155<br>(0.0414)     | 0.5370<br>(0.3886)     | 0.0851<br>(0.0605)     | -0.3694<br>(0.4652)    | -0.0429<br>(0.0539)    | -0.4743<br>(0.3856)    | -0.0740<br>(0.0598)    | -0.7602<br>(0.7927)    | -0.1100<br>(0.1089)    |
| Advanced Math                    |                       |                       | 0.9753***<br>(0.1270) |  |                       |  |                         |                        | 0.2364<br>(0.2639)     | 0.0375<br>(0.0417)     | 0.4866<br>(0.3566)     | 0.0565<br>(0.0417)     | 0.5366**<br>(0.2704)   | 0.0837**<br>(0.0416)   | 0.3982<br>(0.6445)     | 0.0576<br>(0.0922)     |
| Committee Score                  |                       |                       |                       |  |                       |  |                         |                        | 0.3737***<br>(0.0674)  | 0.0592***<br>(0.0096)  | 0.0653<br>(0.0837)     | 0.0076<br>(0.0098)     | 0.0616<br>(0.0642)     | 0.0096<br>(0.0100)     | -0.6725***<br>(0.1981) | -0.0973***<br>(0.0217) |
| Year Transition/Matriculate More | 0.0664<br>(0.1281)    | 0.0163<br>(0.0314)    | 0.1131<br>(0.1261)    |  |                       |  | 1.1108***<br>(0.1479)   | 0.2156***<br>(0.0257)  |                        |                        |                        |                        | 0.1313<br>(0.2794)     | 0.0205<br>(0.0434)     | 2.9410***<br>(0.6722)  | 0.4256***<br>(0.0578)  |
| Program Rank                     |                       |                       |                       |  |                       |  |                         |                        |                        |                        |                        |                        | -0.0279***<br>(0.0028) | -0.0044***<br>(0.0003) |                        |                        |
| $\overline{m}_2$                 |                       |                       |                       |  | 0.8883***<br>(0.0489) |  |                         |                        | -0.5496<br>(0.5706)    | -0.0871<br>(0.0900)    |                        |                        |                        |                        |                        |                        |
| $\overline{m}_{2A}$              |                       |                       |                       |  |                       |  |                         |                        |                        |                        | -2.9214***<br>(0.9934) | -0.3393***<br>(0.1141) |                        |                        |                        |                        |
| $\sqrt{MSE}$                     |                       |                       | 1.9949***<br>(0.0422) |  | 0.0010***<br>(0.0002) |  |                         |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| Observations                     | 1000                  |                       |                       |  |                       |  |                         |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| LL   Parameters   AIC            | -1743.0               | 48                    | 3582.0                |  |                       |  |                         |                        |                        |                        |                        |                        |                        |                        |                        |                        |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

» Unrestricted Waitlist-Hybrid AMEs (Suboptimal)

# A4: Restricted Waitlist-Dynamic Estimates (Optimal)

Table 36a: Restricted Waitlist-Dynamic Estimates (Optimal)

|                                    | Advanced Math          | Committee Score       | $\bar{m}_{2A}$         | Cutoff 1              | $L(\text{Cutoff})_1$       | Cutoff 3            | $P(\text{Cutoff})_3$       | Success                | Matriculate           | Beta                  | Sigma                 |
|------------------------------------|------------------------|-----------------------|------------------------|-----------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|-----------------------|-----------------------|
| Intercept                          | -2.1340***<br>(0.7572) | 3.5031***<br>(0.7215) | -0.0252<br>(0.0518)    | -0.2196**<br>(0.0861) | 0.8028<br>(0.6781, 0.9505) | 0.1887<br>(0.3071)  | 60.13%<br>(49.30%, 70.05%) | -9.5181***<br>(1.9761) | -5.8313<br>(6.4970)   |                       |                       |
| GRE Quantative Percentile          | 0.0236***<br>(0.0090)  | 0.0136<br>(0.0086)    |                        |                       |                            |                     |                            | 0.1131***<br>(0.0218)  | 0.0834<br>(0.0644)    |                       |                       |
| Gender                             | -0.2139<br>(0.1304)    | -0.0513<br>(0.1314)   |                        |                       |                            |                     |                            | 0.2235*<br>(0.1232)    | 0.4650<br>(0.9003)    |                       |                       |
| Demographic 1                      | 0.3573**<br>(0.1724)   | -0.3604**<br>(0.1719) |                        |                       |                            |                     |                            | -0.3224*<br>(0.1848)   | -0.2627<br>(0.7744)   |                       |                       |
| Demographic 2                      | 0.0939<br>(0.1905)     | 0.1211<br>(0.1890)    |                        |                       |                            |                     |                            | 0.0204<br>(0.1776)     | -0.3966<br>(1.2105)   |                       |                       |
| Advanced Math                      |                        | 0.9768***<br>(0.1271) |                        |                       |                            |                     |                            | 0.5615***<br>(0.1718)  | 1.1399<br>(0.9839)    |                       |                       |
| Committee Score                    |                        |                       |                        |                       |                            |                     |                            | 0.0539<br>(0.0377)     | -0.9123**<br>(0.3787) |                       |                       |
| Year Transition / Matriculate More | 0.0731<br>(0.1277)     | 0.1160<br>(0.1261)    |                        | -0.0227<br>(0.0823)   | 0.7848<br>(0.6569, 0.9377) |                     | 74.29%<br>(59.59%, 84.99%) | 0.0621<br>(0.2683)     | 3.3933***<br>(0.7682) |                       |                       |
| Program Rank                       |                        |                       |                        |                       |                            |                     |                            | -0.0282***<br>(0.0027) |                       |                       |                       |
| $\bar{m}_2$                        |                        |                       | 1.0892***<br>(0.0851)  |                       |                            |                     |                            |                        |                       |                       |                       |
| $\bar{m}_{2A}$                     |                        |                       |                        |                       |                            | 1.3706*<br>(0.8077) |                            |                        |                       |                       |                       |
| $\sqrt{\text{MSE}}$                |                        | 1.9950***<br>(0.0422) | -0.0015***<br>(0.0005) |                       |                            |                     |                            |                        |                       |                       |                       |
| Beta                               |                        |                       |                        |                       |                            |                     |                            |                        |                       | 0.9751***<br>(0.0345) |                       |
| Sigma                              |                        |                       |                        |                       |                            |                     |                            |                        |                       |                       | 0.1362***<br>(0.0249) |
| Observations                       | 1000                   |                       |                        |                       |                            |                     |                            |                        |                       |                       |                       |
| LL   Parameters   AIC              | -1955.3                | 32                    | 3974.6                 |                       |                            |                     |                            |                        |                       |                       |                       |
| LR Statistic   DF   P-Value        | 62.5662                | 16                    | 0.0000                 |                       |                            |                     |                            |                        |                       |                       |                       |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01. Confidence intervals are 95%.

# A4: Restricted Waitlist-Dynamic Estimates (Suboptimal)

Table 36b: Restricted Waitlist-Dynamic Estimates (Suboptimal)

|                                  | Advanced Math          | Committee Score       | $\bar{m}_{2A}$        | Cutoff 1              | $L(\text{Cutoff})_1$       | Cutoff 3           | $P(\text{Cutoff})_3$       | Success                | Matriculate            | Beta                  | Sigma                 |
|----------------------------------|------------------------|-----------------------|-----------------------|-----------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|-----------------------|-----------------------|
| Intercept                        | -2.1873***<br>(0.7590) | 3.1964***<br>(0.7272) | -0.0498*<br>(0.0260)  | -0.2951**<br>(0.1211) | 0.7445<br>(0.5872, 0.9439) | 0.2281<br>(0.3497) | 61.38%<br>(47.25%, 73.81%) | -6.7283***<br>(2.2718) | -1.4373<br>(3.8091)    |                       |                       |
| GRE Quantative Percentile        | 0.0243***<br>(0.0090)  | 0.0175**<br>(0.0087)  |                       |                       |                            |                    |                            | 0.0765***<br>(0.0241)  | 0.0355<br>(0.0435)     |                       |                       |
| Gender                           | -0.2178*<br>(0.1307)   | -0.0367<br>(0.1310)   |                       |                       |                            |                    |                            | 0.1394<br>(0.1150)     | -0.3545<br>(0.5436)    |                       |                       |
| Demographic 1                    | 0.3435**<br>(0.1728)   | -0.3913**<br>(0.1732) |                       |                       |                            |                    |                            | -0.4496**<br>(0.1976)  | 0.3334<br>(0.7088)     |                       |                       |
| Demographic 2                    | 0.0987<br>(0.1906)     | 0.1280<br>(0.1892)    |                       |                       |                            |                    |                            | -0.0106<br>(0.1662)    | -0.7602<br>(0.7927)    |                       |                       |
| Advanced Math                    |                        | 0.9787***<br>(0.1272) |                       |                       |                            |                    |                            | 0.3227**<br>(0.1629)   | 0.3982<br>(0.6445)     |                       |                       |
| Committee Score                  |                        |                       |                       |                       |                            |                    |                            | 0.1659***<br>(0.0368)  | -0.6725***<br>(0.1981) |                       |                       |
| Year Transition/Matriculate More | 0.0686<br>(0.1278)     | 0.1168<br>(0.1260)    |                       | -0.0354<br>(0.0874)   | 0.7186<br>(0.5826, 0.8863) |                    | 76.61%<br>(53.83%, 90.19%) | 0.1630<br>(0.2705)     | 2.9410***<br>(0.6722)  |                       |                       |
| Program Rank                     |                        |                       |                       |                       |                            |                    |                            | -0.0281***<br>(0.0027) |                        |                       |                       |
| $\bar{m}_2$                      |                        |                       | 0.8883***<br>(0.0489) |                       |                            |                    |                            |                        |                        |                       |                       |
| $\bar{m}_{2A}$                   |                        |                       |                       |                       |                            | 1.6924<br>(1.1232) |                            |                        |                        |                       |                       |
| $\sqrt{\text{MSE}}$              |                        | 1.9958***<br>(0.0424) | 0.0010***<br>(0.0002) |                       |                            |                    |                            |                        |                        |                       |                       |
| Beta                             |                        |                       |                       |                       |                            |                    |                            |                        |                        | 0.8727***<br>(0.0426) |                       |
| Sigma                            |                        |                       |                       |                       |                            |                    |                            |                        |                        |                       | 0.1345***<br>(0.0400) |
| Observations                     | 1000                   |                       |                       |                       |                            |                    |                            |                        |                        |                       |                       |
| LL   Parameters   AIC            | -1797.6                | 32                    | 3659.2                |                       |                            |                    |                            |                        |                        |                       |                       |
| LR Statistic   DF   P-Value      | 109.2603               | 16                    | 0.0000                |                       |                            |                    |                            |                        |                        |                       |                       |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01. Confidence intervals are 95%.

# A4: Unrestricted Waitlist-Dynamic Estimates, z Missing (Optimal)

Table 37a: Unrestricted Waitlist-Dynamic Estimates, z Missing (Optimal)

|                                  | Selection              |                        | Advanced Math        |                     | Committee Score       |                                 | Transition              |                        | Accept 2                |                        | Accept 3                |                        | Success                |                        | Matriculate           |                        |
|----------------------------------|------------------------|------------------------|----------------------|---------------------|-----------------------|---------------------------------|-------------------------|------------------------|-------------------------|------------------------|-------------------------|------------------------|------------------------|------------------------|-----------------------|------------------------|
|                                  | Coefficient            | AME                    | Coefficient          | AME                 | Coefficient           | $\overline{m}_2$<br>Coefficient | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient           | AME                    |
| Intercept                        | -6.8078***<br>(0.5924) | -2.2186***<br>(0.1485) | -0.6710<br>(16.7178) | -0.1655<br>(4.1242) | 0.1245<br>(16.2870)   | -0.0252<br>(0.0518)             | -11.5865***<br>(1.0231) | -2.2556***<br>(0.1454) | -11.9790***<br>(2.5521) | -1.6185***<br>(0.3154) | -13.3009***<br>(2.4144) | -1.8295***<br>(0.3092) | -9.5718***<br>(2.0725) | -1.4900***<br>(0.2688) | -5.8313<br>(6.4970)   | -0.6741<br>(0.7706)    |
| GRE Quantative Percentile        | 0.0725***<br>(0.0069)  | 0.0236***<br>(0.0018)  | 0.0060<br>(0.1504)   | 0.0015<br>(0.0371)  | 0.0437<br>(0.1464)    |                                 | 0.1235***<br>(0.0118)   | 0.0241***<br>(0.0018)  | 0.1099***<br>(0.0272)   | 0.0148***<br>(0.0034)  | 0.1390***<br>(0.0265)   | 0.0191***<br>(0.0034)  | 0.1162***<br>(0.0227)  | 0.0181***<br>(0.0029)  | 0.0834<br>(0.0644)    | 0.0096<br>(0.0075)     |
| Gender                           | 0.1226<br>(0.0890)     | 0.0399<br>(0.0289)     | -0.0741<br>(0.3403)  | -0.0183<br>(0.0839) | -0.0025<br>(0.3342)   |                                 | 0.2266<br>(0.1464)      | 0.0441<br>(0.0283)     | 0.0225<br>(0.2900)      | 0.0030<br>(0.0392)     | 0.3026<br>(0.3288)      | 0.0416<br>(0.0451)     | 0.1256<br>(0.2699)     | 0.0196<br>(0.0419)     | 0.4650<br>(0.9003)    | 0.0538<br>(0.1052)     |
| Demographic 1                    | -0.1796<br>(0.1196)    | -0.0585<br>(0.0389)    | 0.1539<br>(0.4546)   | 0.0380<br>(0.1121)  | -0.4463<br>(0.4241)   |                                 | -0.3021<br>(0.1988)     | -0.0588<br>(0.0386)    | 0.5445<br>(0.4924)      | 0.0736<br>(0.0663)     | -0.8825*<br>(0.4624)    | -0.1214**<br>(0.0619)  | -0.4599<br>(0.3932)    | -0.0716<br>(0.0608)    | -0.2627<br>(0.7744)   | -0.0304<br>(0.0880)    |
| Demographic 2                    | 0.1051<br>(0.1361)     | 0.0342<br>(0.0444)     | -0.1447<br>(0.3802)  | -0.0357<br>(0.0937) | -0.1301<br>(0.3884)   |                                 | 0.1535<br>(0.2214)      | 0.0299<br>(0.0431)     | 0.5056<br>(0.5188)      | 0.0683<br>(0.0697)     | -0.0507<br>(0.4530)     | -0.0070<br>(0.0623)    | -0.2334<br>(0.4322)    | -0.0363<br>(0.0672)    | -0.3966<br>(1.2105)   | -0.0459<br>(0.1393)    |
| Advanced Math                    |                        |                        |                      |                     | 0.6835***<br>(0.2024) |                                 |                         |                        | 0.5077*<br>(0.2908)     | 0.0686*<br>(0.3306)    | 0.8623***<br>(0.0451)   | 0.1186***<br>(0.2749)  | 0.6299**<br>(0.0417)   | 0.0980**<br>(0.0839)   | 1.1399<br>(0.9839)    | 0.1318<br>(0.1080)     |
| Committee Score                  |                        |                        |                      |                     |                       |                                 |                         |                        | 0.1058<br>(0.0689)      | 0.0143<br>(0.0093)     | 0.0379<br>(0.0673)      | 0.0052<br>(0.0092)     | 0.0465<br>(0.0655)     | 0.0072<br>(0.0102)     | -0.9123**<br>(0.3787) | -0.1055***<br>(0.0317) |
| Year Transition/Matriculate More | 0.5839***<br>(0.0955)  | 0.1903***<br>(0.0291)  | 0.3266<br>(1.2191)   | 0.0806<br>(0.3006)  | 0.4315<br>(1.1808)    |                                 | 0.9837***<br>(0.1472)   | 0.1915***<br>(0.0263)  |                         |                        |                         |                        | -0.0260<br>(0.2762)    | -0.0040<br>(0.0430)    | 3.3933***<br>(0.7682) | 0.3922***<br>(0.0697)  |
| Program Rank                     |                        |                        |                      |                     |                       |                                 |                         |                        |                         |                        |                         |                        | -0.0281***<br>(0.0027) | -0.0044***<br>(0.0003) |                       |                        |
| $m_2$                            |                        |                        |                      |                     |                       | 1.0892***<br>(0.0851)           |                         |                        | -1.5959**<br>(0.7865)   | -0.2156**<br>(0.1048)  |                         |                        |                        |                        |                       |                        |
| $\overline{m}_2$                 |                        |                        |                      |                     |                       |                                 |                         |                        |                         |                        | -2.0111***<br>(0.7688)  | -0.2766***<br>(0.1032) |                        |                        |                       |                        |
| Inverse Mills                    |                        |                        | 0.0155<br>(3.1591)   | 0.0038<br>(0.7793)  | 0.8758<br>(3.0588)    |                                 |                         |                        |                         |                        |                         |                        |                        |                        |                       |                        |
| $\sqrt{MSE}$                     |                        |                        |                      |                     | 1.9478***<br>(0.0675) | 0.0015***<br>(0.0005)           |                         |                        |                         |                        |                         |                        |                        |                        |                       |                        |
| Observations                     | 1000                   |                        |                      |                     |                       |                                 |                         |                        |                         |                        |                         |                        |                        |                        |                       |                        |
| LL   Parameters   AIC            | -739.6                 | 56                     | 1591.2               |                     |                       |                                 |                         |                        |                         |                        |                         |                        |                        |                        |                       |                        |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Unrestricted Waitlist-Dynamic Estimates, z Missing (Suboptimal)

Table 37b: Unrestricted Waitlist-Dynamic Estimates, z Missing (Suboptimal)

|                                  | Selection              |                        | Advanced Math        |                     | Committee Score       |  | $\overline{m}_{2A}$    |  | Transition              |                        | Accept 2               |                        | Accept 3               |                        | Success                |                        | Matriculate            |                        |
|----------------------------------|------------------------|------------------------|----------------------|---------------------|-----------------------|--|------------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                  | Coefficient            | AME                    | Coefficient          | AME                 | Coefficient           |  | Coefficient            |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                        | -6.5993***<br>(0.5812) | -2.1437***<br>(0.1480) | -0.7285<br>(12.7225) | -0.1792<br>(3.1302) | 0.2835<br>(13.5178)   |  | -0.0498*<br>(0.0260)   |  | -11.2416***<br>(1.0154) | -2.1816***<br>(0.1474) | -6.7325***<br>(2.0024) | -1.0671***<br>(0.3044) | -7.2150***<br>(2.5119) | -0.8378***<br>(0.2955) | -9.8173***<br>(1.9798) | -1.5260***<br>(0.2504) | -1.4373<br>(3.8091)    | -0.2080<br>(0.5533)    |
| GRE Quantative Percentile        | 0.0720***<br>(0.0068)  | 0.0234***<br>(0.0018)  | 0.0057<br>(0.1161)   | 0.0014<br>(0.0286)  | 0.0439<br>(0.1229)    |  |                        |  | 0.1227***<br>(0.0117)   | 0.0238***<br>(0.0018)  | 0.0389*<br>(0.0215)    | 0.0062*<br>(0.0034)    | 0.0722***<br>(0.0279)  | 0.0084***<br>(0.0033)  | 0.1185***<br>(0.0218)  | 0.0184***<br>(0.0027)  | 0.0355<br>(0.0435)     | 0.0051<br>(0.0063)     |
| Gender                           | 0.1419<br>(0.0876)     | 0.0461<br>(0.0283)     | -0.1727<br>(0.3182)  | -0.0425<br>(0.0782) | -0.0053<br>(0.3325)   |  |                        |  | 0.2580*<br>(0.1469)     | 0.0501*<br>(0.0283)    | -0.2746<br>(0.2746)    | -0.0435<br>(0.0431)    | 0.4111<br>(0.3418)     | 0.0477<br>(0.0394)     | 0.0607<br>(0.2678)     | 0.0094<br>(0.0416)     | -0.3544<br>(0.5436)    | -0.0513<br>(0.0788)    |
| Demographic 1                    | -0.4885***<br>(0.1235) | -0.1587***<br>(0.0391) | 0.2364<br>(0.8110)   | 0.0582<br>(0.1995)  | -0.7814<br>(0.8178)   |  |                        |  | -0.8289***<br>(0.1967)  | -0.1609***<br>(0.0371) | 0.2571<br>(0.3774)     | 0.0407<br>(0.0594)     | -1.1557**<br>(0.5112)  | -0.1342**<br>(0.0581)  | -0.5223<br>(0.3662)    | -0.0812<br>(0.0563)    | 0.3334<br>(0.7088)     | 0.0482<br>(0.1030)     |
| Demographic 2                    | 0.0631<br>(0.1391)     | 0.0205<br>(0.0452)     | -0.0077<br>(0.3091)  | -0.0019<br>(0.0761) | -0.2392<br>(0.3175)   |  |                        |  | 0.0804<br>(0.2136)      | 0.0156<br>(0.0414)     | 0.5369<br>(0.3887)     | 0.0851<br>(0.0606)     | -0.3689<br>(0.4654)    | -0.0428<br>(0.0539)    | -0.4756<br>(0.3873)    | -0.0739<br>(0.0598)    | -0.7602<br>(0.7927)    | -0.1100<br>(0.1089)    |
| Advanced Math                    |                        |                        |                      |                     | 0.7195***<br>(0.1971) |  |                        |  |                         |                        | 0.2358<br>(0.2639)     | 0.0374<br>(0.0417)     | 0.4872<br>(0.3568)     | 0.0566<br>(0.0417)     | 0.5389**<br>(0.0645)   | 0.0838**<br>(0.0100)   | 0.3982<br>(0.6445)     | 0.0576<br>(0.0922)     |
| Committee Score                  |                        |                        |                      |                     |                       |  |                        |  |                         |                        | 0.3737***<br>(0.0674)  | 0.0592***<br>(0.0096)  | 0.0662<br>(0.0837)     | 0.0077<br>(0.0098)     | 0.0632<br>(0.0645)     | 0.0098<br>(0.0100)     | -0.6725***<br>(0.1981) | -0.0973***<br>(0.0217) |
| Year Transition/Matriculate More | 0.6595***<br>(0.0933)  | 0.2142***<br>(0.0278)  | 0.3761<br>(1.0851)   | 0.0925<br>(0.2668)  | 0.4166<br>(1.1430)    |  |                        |  | 1.1125***<br>(0.1480)   | 0.2159***<br>(0.0257)  |                        |                        |                        |                        | 0.1385<br>(0.2805)     | 0.0215<br>(0.0434)     | 2.9410***<br>(0.6722)  | 0.4256***<br>(0.0578)  |
| Program Rank                     |                        |                        |                      |                     |                       |  |                        |  |                         |                        |                        |                        |                        |                        | -0.0279***<br>(0.0028) | -0.0043***<br>(0.0003) |                        |                        |
| $m_2$                            |                        |                        |                      |                     |                       |  | 0.8883***<br>(0.0489)  |  |                         |                        | -0.5439<br>(0.5704)    | -0.0862<br>(0.0900)    |                        |                        |                        |                        |                        |                        |
| $\overline{m}_{2A}$              |                        |                        |                      |                     |                       |  |                        |  |                         |                        |                        |                        | -2.9170***<br>(0.9928) | -0.3387***<br>(0.1141) |                        |                        |                        |                        |
| Inverse Mills                    |                        |                        | 0.0468<br>(2.5150)   | 0.0115<br>(0.6188)  | 0.9690<br>(2.6691)    |  |                        |  |                         |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| $\sqrt{MSE}$                     |                        |                        |                      |                     | 1.9199***<br>(0.0682) |  | -0.0010***<br>(0.0002) |  |                         |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| Observations                     | 1000                   |                        |                      |                     |                       |  |                        |  |                         |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| LL   Parameters   AIC            | -581.5                 | 56                     | 1274.9               |                     |                       |  |                        |  |                         |                        |                        |                        |                        |                        |                        |                        |                        |                        |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Restricted Waitlist-Dynamic Estimates, z Missing (Optimal)

Table 38a: Restricted Waitlist-Dynamic Estimates, z Missing (Optimal)

|                                  | Selection              | Advanced Math         | Committee Score       | $\bar{m}_{2A}$        | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 3            | $P(\text{Cutoff})_3$       | Success                | Matriculate           | Beta                  | Sigma                 |
|----------------------------------|------------------------|-----------------------|-----------------------|-----------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|-----------------------|-----------------------|-----------------------|
| Intercept                        | -6.8776***<br>(0.5895) | -16.4264<br>(16.7220) | 11.8785<br>(16.0244)  | -0.0252<br>(0.0518)   | -0.2299***<br>(0.0868) | 0.7946<br>(0.6703, 0.9419) | 0.1790<br>(0.3056)  | 59.88%<br>(49.06%, 69.82%) | -9.3510***<br>(1.9497) | -5.8313<br>(6.4970)   |                       |                       |
| GRE Quantative Percentile        | 0.0734***<br>(0.0068)  | 0.1497<br>(0.1505)    | -0.0618<br>(0.1451)   |                       |                        |                            |                     |                            | 0.1116***<br>(0.0215)  | 0.0834<br>(0.0644)    |                       |                       |
| Gender                           | 0.1092<br>(0.0903)     | 0.1881<br>(0.3337)    | -0.1699<br>(0.3107)   |                       |                        |                            |                     |                            | 0.2108*<br>(0.1214)    | 0.4650<br>(0.9003)    |                       |                       |
| Demographic 1                    | -0.1858<br>(0.1197)    | -0.1931<br>(0.5149)   | -0.2012<br>(0.4491)   |                       |                        |                            |                     |                            | -0.3113*<br>(0.1814)   | -0.2627<br>(0.7744)   |                       |                       |
| Demographic 2                    | 0.0780<br>(0.1399)     | -0.0169<br>(0.4383)   | -0.2417<br>(0.3391)   |                       |                        |                            |                     |                            | 0.0285<br>(0.1748)     | -0.3966<br>(1.2105)   |                       |                       |
| Advanced Math                    |                        |                       | 0.6994***<br>(0.2027) |                       |                        |                            |                     |                            | 0.5496***<br>(0.1708)  | 1.1399<br>(0.9839)    |                       |                       |
| Committee Score                  |                        |                       |                       |                       |                        |                            |                     |                            | 0.0494<br>(0.0370)     | -0.9123**<br>(0.3787) |                       |                       |
| Year Transition/Matriculate More | 0.6046***<br>(0.0873)  | 1.4541<br>(1.3071)    | -0.4384<br>(1.2031)   |                       | -0.0159<br>(0.0827)    | 0.7821<br>(0.6552, 0.9335) |                     | 74.06%<br>(59.56%, 84.70%) | 0.0632<br>(0.2657)     | 3.3933***<br>(0.7682) |                       |                       |
| Program Rank                     |                        |                       |                       |                       |                        |                            |                     |                            | -0.0282***<br>(0.0027) |                       |                       |                       |
| $\bar{m}_2$                      |                        |                       |                       | 1.0892***<br>(0.0851) |                        |                            |                     |                            |                        |                       |                       |                       |
| $\bar{m}_{2A}$                   |                        |                       |                       |                       |                        |                            | 1.3671*<br>(0.7957) |                            |                        |                       |                       |                       |
| Inverse Mills                    |                        | 2.8596<br>(3.1805)    | -1.3585<br>(2.9670)   |                       |                        |                            |                     |                            |                        |                       |                       |                       |
| $\sqrt{\text{MSE}}$              |                        |                       | 1.9458***<br>(0.0673) | 0.0015***<br>(0.0005) |                        |                            |                     |                            |                        |                       |                       |                       |
| Beta                             |                        |                       |                       |                       |                        |                            |                     |                            |                        |                       | 0.9756***<br>(0.0340) |                       |
| Sigma                            |                        |                       |                       |                       |                        |                            |                     |                            |                        |                       |                       | 0.1338***<br>(0.0247) |
| Observations                     | 1000                   |                       |                       |                       |                        |                            |                     |                            |                        |                       |                       |                       |
| LL   Parameters   AIC            | -769.9                 | 40                    | 1619.7                |                       |                        |                            |                     |                            |                        |                       |                       |                       |
| LR Statistic   DF   P-Value      | 60.5412                | 16                    | 0.0000                |                       |                        |                            |                     |                            |                        |                       |                       |                       |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01. Confidence intervals are 95%.



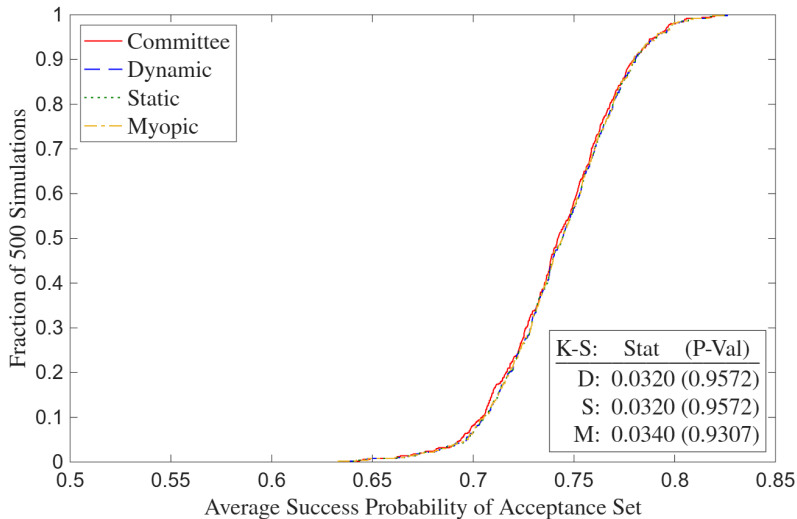
# A4: Restricted Waitlist-Dynamic Estimates, z Missing (Suboptimal)

Table 38b: Restricted Waitlist-Dynamic Estimates, z Missing (Suboptimal)

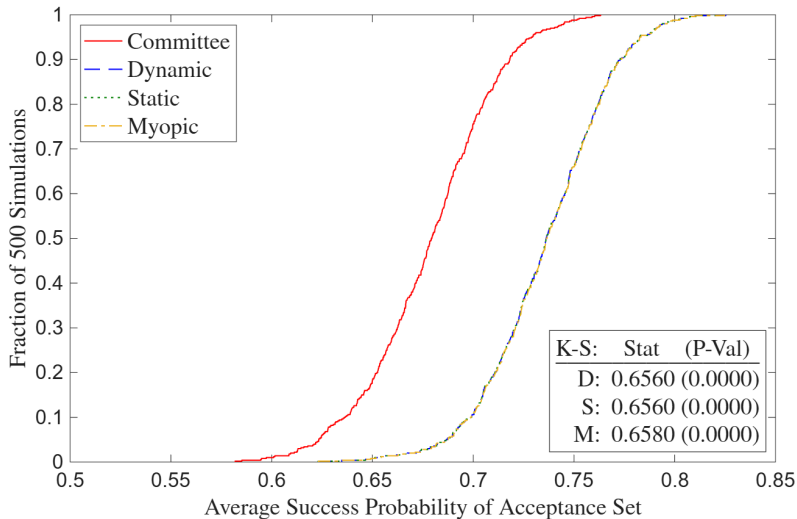
|                                  | Selection              | Advanced Math        | Committee Score       | $\bar{m}_{2A}$         | Cutoff 1              | $L(\text{Cutoff})_1$       | Cutoff 3           | $P(\text{Cutoff})_3$       | Success                | Matriculate            | Beta                  | Sigma                 |
|----------------------------------|------------------------|----------------------|-----------------------|------------------------|-----------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|-----------------------|-----------------------|
| Intercept                        | -6.681***<br>(0.5782)  | -9.6293<br>(12.8771) | 8.8365<br>(12.8190)   | -0.0498*<br>(0.0260)   | -0.2897**<br>(0.1214) | 0.7485<br>(0.5900, 0.9495) | 0.2333<br>(0.3559) | 61.64%<br>(47.26%, 74.23%) | -6.9197***<br>(2.2866) | -1.4373<br>(3.8091)    |                       |                       |
| GRE Quantative Percentile        | 0.0730***<br>(0.0068)  | 0.0890<br>(0.1175)   | -0.0314<br>(0.1173)   |                        |                       |                            |                    |                            | 0.0786***<br>(0.0243)  | 0.0355<br>(0.0435)     |                       |                       |
| Gender                           | 0.1395<br>(0.0878)     | 0.0114<br>(0.3276)   | -0.1444<br>(0.3104)   |                        |                       |                            |                    |                            | 0.1387<br>(0.1170)     | -0.3545<br>(0.5436)    |                       |                       |
| Demographic 1                    | -0.4950***<br>(0.1221) | -0.3103<br>(0.8106)  | -0.2947<br>(0.8024)   |                        |                       |                            |                    |                            | -0.4461**<br>(0.1992)  | 0.3334<br>(0.7088)     |                       |                       |
| Demographic 2                    | 0.0401<br>(0.1371)     | 0.0311<br>(0.3409)   | -0.2785<br>(0.2791)   |                        |                       |                            |                    |                            | 0.0037<br>(0.1686)     | -0.7602<br>(0.7927)    |                       |                       |
| Advanced Math                    |                        |                      | 0.7345***<br>(0.1979) |                        |                       |                            |                    |                            | 0.3150*<br>(0.1653)    | 0.3982<br>(0.6445)     |                       |                       |
| Committee Score                  |                        |                      |                       |                        |                       |                            |                    |                            | 0.1654***<br>(0.0367)  | -0.6725***<br>(0.1981) |                       |                       |
| Year Transition/Matriculate More | 0.6754***<br>(0.0884)  | 1.0752<br>(1.1364)   | -0.3792<br>(1.1126)   |                        | -0.0349<br>(0.0873)   | 0.7228<br>(0.5857, 0.8920) |                    | 77.11%<br>(53.97%, 90.64%) | 0.1681<br>(0.2720)     | 2.9410***<br>(0.6722)  |                       |                       |
| Program Rank                     |                        |                      |                       |                        |                       |                            |                    |                            | -0.0281***<br>(0.0027) |                        |                       |                       |
| $\bar{m}_2$                      |                        |                      |                       | 0.8883***<br>(0.0489)  |                       |                            |                    |                            |                        |                        |                       |                       |
| $\bar{m}_{2A}$                   |                        |                      |                       |                        |                       |                            | 1.7336<br>(1.1467) |                            |                        |                        |                       |                       |
| Inverse Mills                    |                        | 1.6566<br>(2.5358)   | -0.8873<br>(2.4938)   |                        |                       |                            |                    |                            |                        |                        |                       |                       |
| $\sqrt{\text{MSE}}$              |                        |                      | 1.9214***<br>(0.0688) | -0.0010***<br>(0.0002) |                       |                            |                    |                            |                        |                        |                       |                       |
| Beta                             |                        |                      |                       |                        |                       |                            |                    |                            |                        |                        | 0.8712***<br>(0.0429) |                       |
| Sigma                            |                        |                      |                       |                        |                       |                            |                    |                            |                        |                        |                       | 0.1370***<br>(0.0404) |
| Observations                     | 1000                   |                      |                       |                        |                       |                            |                    |                            |                        |                        |                       |                       |
| LL   Parameters   AIC            | -636.4                 | 40                   | 1352.8                |                        |                       |                            |                    |                            |                        |                        |                       |                       |
| LR Statistic   DF   P-Value      | 109.8389               | 16                   | 0.0000                |                        |                       |                            |                    |                            |                        |                        |                       |                       |

Murphy-Topel robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01. Confidence intervals are 95%.

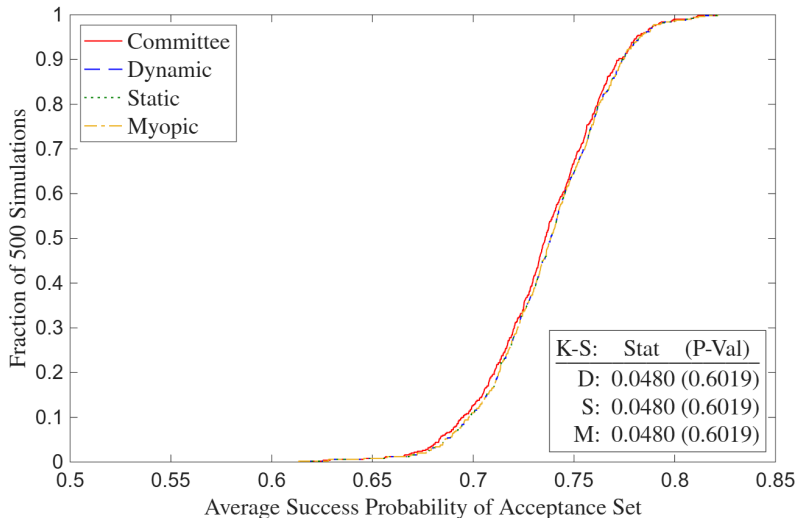
# A4: Full Deviation Gains (Optimal)



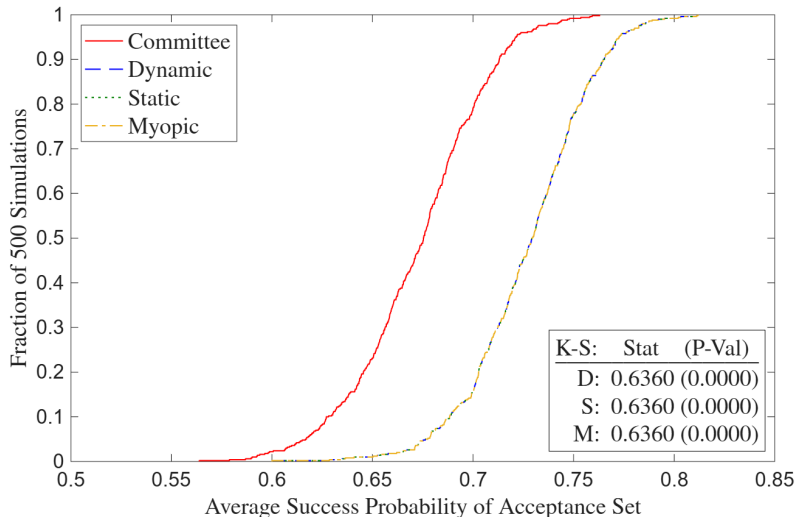
## A4: Full Deviation Gains (Suboptimal)



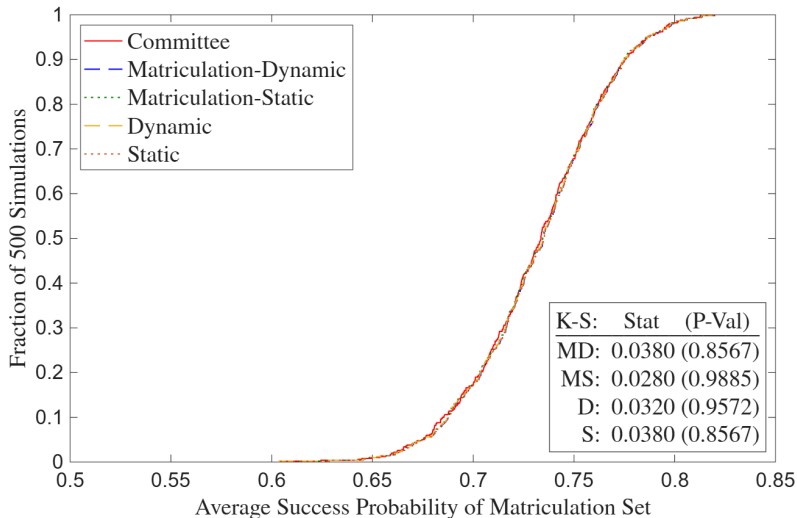
## A4: Deviation Gains, $z$ Missing (Optimal)



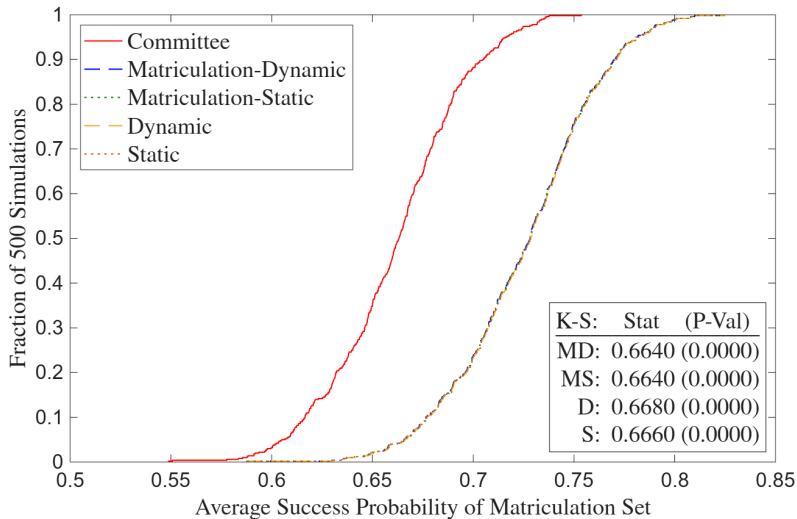
## A4: Deviation Gains, $z$ Missing (Suboptimal)



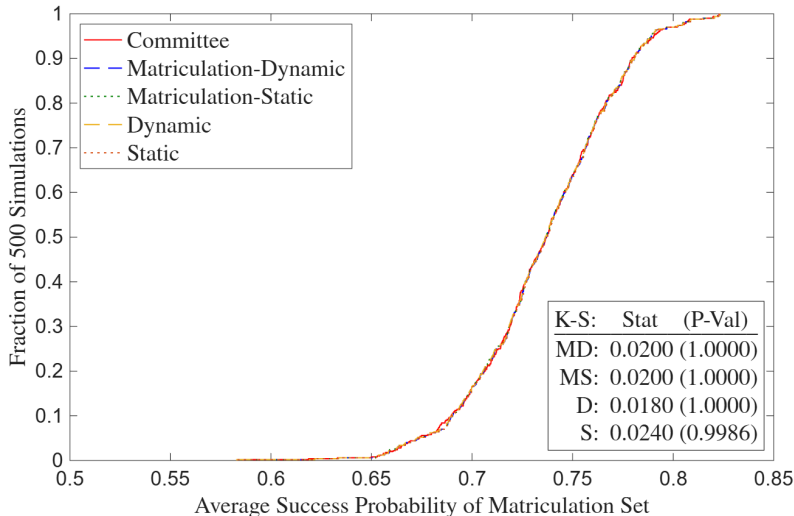
# A4: Full Matriculation Deviation Gains (Optimal)



# A4: Full Matriculation Deviation Gains (Suboptimal)

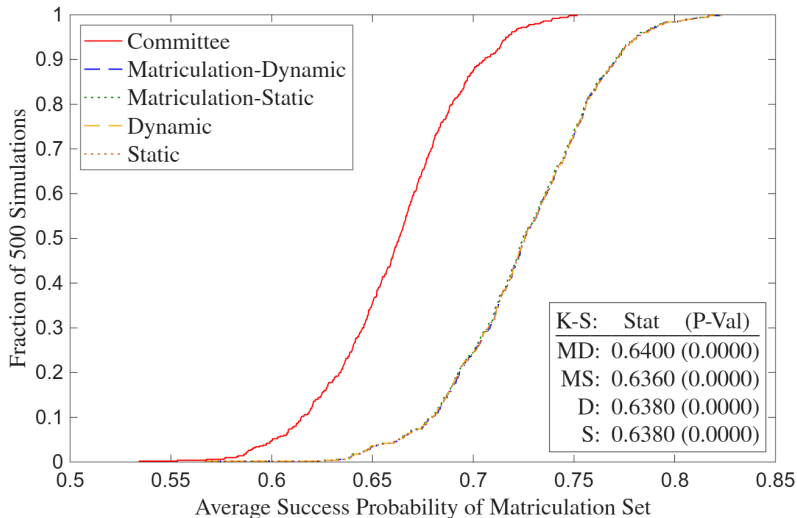


## A4: Matriculation Deviation Gains, $z$ Missing (Optimal)

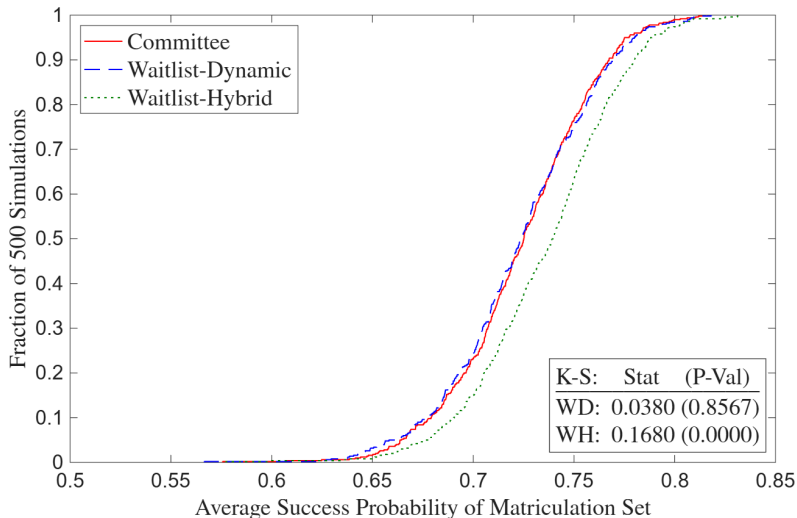




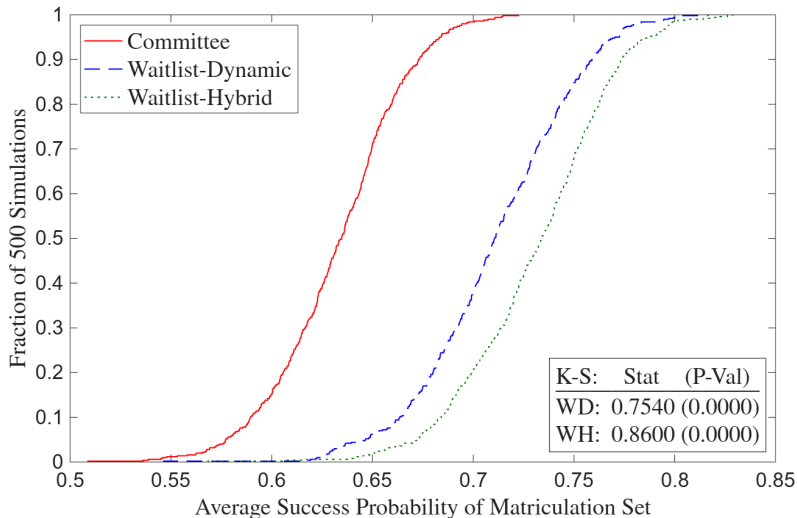
# A4: Matriculation Deviation Gains, $z$ Missing (Suboptimal)



# A4: Waitlist Deviation Gains, $z$ Missing (Optimal)



## A4: Waitlist Deviation Gains, $z$ Missing (Suboptimal)



# References I



Athey, S., L. F. Katz, A. B. Krueger, S. Levitt & J. Poterba (2007). "What Does Performance in Graduate School Predict? Graduate Economics Education and Student Outcomes". In: *American Economic Review* 97.2, pp. 512–520.



Athey, S. & M. Luca (2019). "Economists (and Economics) in Tech Companies". In: *Journal of Economic Perspectives* 33.1, pp. 209–230.



Attiyeh, G. & R. Attiyeh (1997). "Testing for Bias in Graduate School Admissions". In: *The Journal of Human Resources* 32.3, pp. 524–548.



Bai, J., M. Esche, W. B. MacLeod & Y. Shi (2022). "Subjective Evaluations and Stratification in Graduate Education". In: *NBER Working Paper* 30677.



Caplin, A. & M. Dean (2015). "Revealed Preference, Rational Inattention, and Costly Information Acquisition". In: *American Economic Review* 105.7, pp. 2183–2203.



Grove, W. A. & S. Wu (2007). "The Search for Economics Talent: Doctoral Completion and Research Productivity". In: *American Economic Review* 97.2, pp. 506–511.



Heckman, J. J. (1979). "Sample Selection Bias as a Specification Error". In: *Econometrica* 47.1, pp. 153–161.



Krueger, A. B. & S. Wu (2000). "Forecasting Job Placements of Economics Graduate Students". In: *The Journal of Economic Education* 31.1, pp. 81–94.



Lewis, M. (2004). *Moneyball: The Art of Winning an Unfair Game*. W.W. Norton & Company.



Madera, J. M., M. R. Hebl & R. C. Martin (2009). "Gender and Letters of Recommendation for Academia: Agentic and Communal Differences". In: *Journal of Applied Psychology* 94.6, pp. 1591–1599.



Matějka, F. & A. McKay (2015). "Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model". In: *American Economic Review* 105.1, pp. 272–298.

# References II



McCall, J. (1970). "Economics of Information and Job Search". In: *The Quarterly Journal of Economics* 84.1, pp. 113–126.



Mood, C. (2010). "Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do About It". In: *European Sociological Review* 26.1, pp. 67–82.



Mortensen, D. T. (1970). "Job Search, the Duration of Unemployment, and the Phillips Curve". In: *American Economic Review* 60.5, pp. 847–862.



Mullainathan, S. & J. Spiess (2017). "Machine Learning: An Applied Econometric Approach". In: *Journal of Economic Perspectives* 31.2, pp. 87–106.



Patki, N., R. Wedge & K. Veeramachaneni (2016). "The Synthetic data vault". In: *IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 399–410.



Simon, H. A. (1956). "Rational choice and the structure of the environment". In: *Psychological Review* 63.2, pp. 129–138.



Sims, C. A. (2003). "Implications of rational inattention". In: *Journal of Monetary Economics* 50.3, pp. 665–690.



Stock, W. A., T. A. Finegan & J. J. Siegfried (2006). "Attrition in Economics Ph.D. Programs". In: *American Economic Review* 96.2, pp. 458–466.



Stock, W. A. & J. J. Siegfried (2015). "The Undergraduate Origins of PhD Economics Revisited". In: *The Journal of Economic Education* 46.2, pp. 150–165.



Vuong, Q. H. (1989). "Likelihood Ratio Tests of Model Selection and Non-Nested Hypotheses". In: *Econometrica* 57.2, pp. 307–333.



Weitzman, M. L. (1979). "Optimal Search for the Best Alternative". In: *Econometrica* 47.3, pp. 641–654.



Young, L. & S. Soroka (2012). "Affective News: The Automated Coding of Sentiment in Political Texts". In: *Political Communications* 29.2, pp. 205–231.